GENERIC FINITENESS OF PLANAR SIX-BODY CENTRAL CONFIGURATIONS: PART I, EXCEPTIONAL-MASS FIRST LIMITS

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Abstract. We separate the generic positive-mass finiteness problem for planar six-body central configurations from finiteness on exceptional mass loci. The Chang–Chen diagram enumeration has 117 graphs. The arguments below confine 40 first-limit diagram obligations to proper necessary mass loci. In particular, a direct function-field argument establishes the No. 66 mass condition for both zero and nonzero drift. We establish the algebraic reduction from an infinite positive family to a germ with an unbounded seven-product, and use the stated diagram rules and computational certificates to confine its mass vector to the union of these loci. Finiteness on the exceptional positive mass loci, the proposed subject of Part II, will be discussed in another paper.

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1. Introduction

Let $m_1, \dots, m_6 > 0$ be masses at distinct positions $q_1, \dots, q_6 \in \mathbb{R}^2$, and let $c = (\sum_i m_i q_i)/(\sum_i m_i)$ be their centre of mass. The configuration is *central* if some $\lambda \in \mathbb{R}$ satisfies

$$\lambda(q_i - c) = \sum_{j \neq i} m_j \frac{q_j - q_i}{|q_j - q_i|^3}, \qquad i = 1, \dots, 6. \tag{1}$$

Translation, rotation and positive scaling preserve this condition. Smale's sixth problem asks whether, for each fixed positive mass vector, there are only finitely many such configurations modulo these symmetries.

For five positive masses, finiteness is known [1, 2]; the corresponding question for six positive masses remains open. Following the normalized algebraic formulation, set $z_i = x_i + iy_i$, $w_i = x_i - iy_i$ on real configurations and regard z_i, w_i as independent complex coordinates. Put $z_{ij} = z_i - z_j$, $w_{ij} = w_i - w_j$, and use signed inverse distances $\delta_{ij} = \delta_{ji}$. With the rotational gauge $z_{12} = w_{12}$,

the normalized system is

$$
\begin{aligned}
z_i &= \sum_{j\neq i} m_j \delta_{ij}^3 (z_i - z_j), \quad w_i = \sum_{j\neq i} m_j \delta_{ij}^3 (w_i - w_j), \\
\delta_{ij}^2 z_{ij} w_{ij} &= 1 \qquad\qquad (1 \le i < j \le 6), \qquad z_{12} = w_{12}.
\end{aligned} \tag{2}
$$

For fixed $m = (m_1, \dots, m_6) \in \mathbb{R}^6_{>0}$, let $\mathcal{A}_m$ denote its complex zero set. A point of $\mathcal{A}_m$ is *positive* when $w_i = \overline{z_i}$ and every $\delta_{ij} > 0$. The finiteness of these normalized positive points is equivalent to finiteness modulo the symmetries of (1).

Put $r_{ij}^2 = z_{ij} w_{ij}$ and, for each pair ij, define

$$
P_{ij} = r_{ij}^2 \prod_{\substack{k<\ell \\ \{k,\ell\}\cap\{i,j\}=\varnothing}} r_{k\ell}^2 \qquad (1 \le i < j \le 6). \tag{3}
$$

These are the 15 *seven-products* of squared distances. A *positive-point first limit* is a Puiseux branch at infinity of an irreducible curve in $\mathcal{A}_m$ containing a nonconstant arc of positive points, along which some P_{ij} has a pole. This is a condition on a curve that passes through physical configurations, not an assertion that the singular limit itself is positive real.

The normalized-curve approach for five bodies [1] motivates our reduction to such first limits. Chang and Chen's Part I [5] enumerates 117 six-body diagrams and eliminates 31 by Algorithm II. Their Part II [6, Theorem 6.1, Sections 6.2–6.12 and Table 6.1] excludes No. 116 for positive masses and confines 61 further diagrams to proper algebraic mass loci, leaving 24 diagrams historically open. This use of cited results does not independently reproduce their symbolic computations. By the local arguments below, 28 of the 61 reported-relation diagrams and 17 historically open diagrams are excluded at the positive-point first-limit tier, leaving 33 published and 7 derived relation obligations. Let $\mathcal{D}_{\mathrm{CC}}$ denote those 33 surviving reported labels and put $\mathcal{D}_{\mathrm{new}} = \{8, 63, 66, 88, 93, 94, 98\}$ and $\mathcal{D}_{40} = \mathcal{D}_{\mathrm{CC}} \cup \mathcal{D}_{\mathrm{new}}$.

On curves through positive configurations, all 28 additional reported-relation diagrams are excluded as first limits by the local arguments below. No. 4 follows from Lemma 4.5, and No. 20 from Section 4.7. No. 2 and 22 follow from Lemma 4.2; No. 3 from Lemma 4.3, and No. 78 from Lemma 4.4. No. 40, 44 and 45 follow from Sections 4.10, 4.12 and 4.14; No. 86 and 95 from Sections 4.20 and 4.21; No. 99 and 100 from Sections 4.25 and 4.24; No. 109/112 from Sections 4.27 and 4.26, and No. 111 from Section 4.28; No. 51/55 from Sections 4.9 and 4.11; No. 105/77 from Sections 4.6 and 4.22; No. 31/32/36 from Sections 4.15, 4.17 and 4.19; No. 49/50 from Sections 4.13 and 4.16; No. 57/61/84 from Sections 4.8, 4.23 and 4.18.

The five historical connectivity cases follow from Lemma 5.1; No. 6 follows from Section 5.2 and No. 71 from Section 5.7. No. 74 and 75 follow from Lemma 5.2; No. 82, 89 and 90 from Sections 5.13, 5.14 and 5.15; No. 81 from Section 5.12; No. 64 and 65 from Sections 5.11 and 5.8. No. 79 follows from Section 5.16. No. 52 is excluded by Proposition 5.8.

Using the published relation loci for $r \in \mathcal{D}_{\mathrm{CC}}$ and Propositions 5.10, 5.12, 5.13, 5.14, 5.16, 5.20 and 5.21 for the seven derived classes, choose an orbit-completed nonzero necessary polynomial $P_r(m)$ for each $r \in \mathcal{D}_{40}$. Every first limit of type r forces $P_r(m) = 0$; the converse is not claimed. Define

$$
E_{40} = \left\{ m \in \mathbb{R}^6_{>0} : \prod_{r\in\mathcal{D}_{40}} P_r(m) = 0 \right\}. \tag{4}
$$

The product is nonzero, so E_{40} is a proper algebraic subset of the positive mass cone. The number 40 counts diagram obligations, not distinct irreducible varieties.

Theorem 1.1 (First-limit confinement)**.** *Using Chang–Chen's cited classification and the specified computational certificates, if the planar six-body problem has infinitely many positive normalized central configurations at $m \in \mathbb{R}^6_{>0}$, then $m \in E_{40}$.*

Corollary 1.2 (Generic finiteness off E_{40})**.** *Positive normalized central configurations are finite for every $m \in \mathbb{R}^6_{>0} \setminus E_{40}$.*

This deduction uses the published Chang–Chen theorem; its computations and the local certificates still require independent publication-level audit. All-positive-mass finiteness, including E_{40}, is a distinct and unfinished Part II problem.

2. The normalized six-body system

In the normalized system (2), no δ_{ij} or $z_{ij}w_{ij}$ vanishes on $\mathcal{A}_m$. Summing mass-weighted coordinate rows gives $\sum_i m_i z_i = \sum_i m_i w_i = 0$. The finiteness question concerns the positive points of $\mathcal{A}_m$, not the entire complex zero set.

Relabelling the bodies or exchanging z and w preserves the ungauged equations and all squared-distance and seven-product orders. Since $z_{ij}w_{ij} \neq 0$, after a relabelling π choose $\lambda^2 = w_{\pi(1)\pi(2)}/z_{\pi(1)\pi(2)}$; the complex rotation $(z, w) \mapsto (\lambda z, \lambda^{-1} w)$ restores the gauge $z_{12} = w_{12}$. Diagram conclusions below therefore apply to the full labelled and colour-swapped orbit, with the masses permuted along with the body labels.

When a necessary mass factor has been obtained for only a canonical labelling, we replace it by the finite product of its pullbacks under all body permutations. We likewise include the factor obtained by the $z \leftrightarrow w$ involution if its canonical proof is colour-specific. Each factor is nonzero as a polynomial, so the orbit product remains nonzero and vanishes on every labelled occurrence of the diagram. In (4), P_r denotes this orbit-completed necessary polynomial, not just the expression in one choice of body labels.

Set $U = \sum_{i<j} m_i m_j \delta_{ij}$ and $I = \sum_i m_i z_i w_i$.

Lemma 2.1 (Constant positive energy on a positive-point curve)**.** *If an irreducible curve $\Gamma \subset \mathcal{A}_m$ contains a positive real-analytic arc, then $U = I = c > 0$ on Γ for a constant c.*

Proof. On the smooth distance-tie variety, differentiate $2U + I$ along coordinate directions. The derivatives are the mass-weighted coordinate residuals in (2), so they vanish along an algebraic curve of solutions. The function is constant on that irreducible curve. Simultaneously scaling (z, w, δ) by (s, s, s^{-1}) and differentiating at $s = 1$ gives $U = I$ at each solution. On the positive arc, $U > 0$, fixing the constant sign. The gauge is not used in this differential argument. □

Lemma 2.2 (Positive arc and bounded distance fibres)**.** *Infinitely many positive points of $\mathcal{A}_m$ yield an irreducible algebraic curve containing a nonconstant positive real-analytic arc. Fixing all 15 nonzero squared distances $z_{ij}w_{ij}$ on $\mathcal{A}_m$ leaves only finitely many normalized configurations.*

Proof. The positive locus is semialgebraic. If it is infinite, a positive-dimensional semialgebraic component admits a Nash arc; its complex Zariski closure contains an irreducible curve and retains that arc. For fixed nonzero squared distances the δ_{ij} take finitely many signs. Fix one choice. The pairwise Gram data of the six vectors $(z_i - z_1, w_i - w_1)$ are fixed. Because $z_{12}w_{12} \neq 0$, their Gram form is nondegenerate for the bilinear form $(a, b) \cdot (c, d) = (ad + bc)/2$. Two realizations with the same Gram data differ by an isometry of this complex plane; the mass centre fixes translation and $z_{12} = w_{12}$ leaves only finitely many scaling/reflection choices. This is the six-body version of the fixed-distance argument in the five-body proof. □

The 15 seven-products P_{ij} were defined in (3).

Lemma 2.3 (The six-body complement-product matrix)**.** *On every irreducible positive-dimensional curve in $\mathcal{A}_m$ at least one P_{ij} is nonconstant; on some branch at infinity it has a pole.*

Proof. The exponent matrix of these monomials in the 15 squared distances is $E = I_{15} + A(K(6, 2))$, where two pair-indices in the Kneser graph are adjacent exactly when disjoint. Its eigenvalues are 7 once, -2 five times and 2 nine times; thus $\det E = -114688 \neq 0$. If all P_{ij}

were constant, powers of all squared distances would be constant on the curve. Lemma 2.2 would make that curve finite, a contradiction. A nonconstant regular function on an affine irreducible algebraic curve attains arbitrarily large complex values. At a point at infinity of its normalization it has a pole, giving the required Puiseux branch. This argument does not assume that the branch itself is positive real. □

3. Diagram inputs and exceptional-mass accounting

The cited Chang–Chen classification and the local proofs separate excluded positive-point first limits from necessary mass relations. The disjoint diagram accounting is as follows. The

Chang–Chen category	Number	Positive-tier disposition
Algorithm II eliminated	31	no target diagram
Published positive-mass exclusion	1	No. 116 excluded in Chang–Chen Part II
Published relation diagrams	61	28 additionally excluded; 33 relation obligations
Historically open	24	17 excluded first limits; 7 derived relations
Total	117	40 relation obligations; no unrestricted positive-tier diagram

Table 1. Bookkeeping of diagrams, not of distinct mass varieties.

reported-relation and historically open cases are treated separately below.

4. Previously classified relation diagrams

Remark 4.1 (Disjoint membership of the excluded classes). The 31 Algorithm II labels are

$$1, 10, 21, 23, 24, 25, 26, 33, 34, 35, 37,$$
$$38, 39, 41, 42, 53, 54, 56, 59, 60, 68,$$
$$70, 72, 83, 85, 87, 91, 92, 96, 97, 113.$$

The 28 reported-relation diagrams whose positive first limits are additionally excluded are

$$2, 3, 4, 20, 22, 31, 32, 36, 40, 44,$$
$$45, 49, 50, 51, 55, 57, 61, 77, 78, 84,$$
$$86, 95, 99, 100, 105, 109, 111, 112.$$

The 17 historically open diagrams whose positive first limits are excluded are

$$5, 6, 52, 64, 65, 67, 69, 71, 73,$$
$$74, 75, 79, 80, 81, 82, 89, 90.$$

Together with positive-empty No. 116, the 33 published relation labels below and the 7 derived labels, these are disjoint and cover exactly $1, \ldots, 117$.

The 31 Algorithm II exclusions come from the cited Chang–Chen theorem. Among the additional 28 reported-relation positive first-limit exclusions, No. 2 and 22 follow from Lemma 4.2; No. 3 and 78 follow from Lemmas 4.3 and 4.4; No. 40, 44 and 45 follow from Sections 4.10, 4.12 and 4.14; No. 86 and 95 follow from Sections 4.20 and 4.21; No. 99 and 100 follow from Sections 4.25 and 4.24; No. 109 and 112 follow from Sections 4.27 and 4.26, and No. 111 from Section 4.28.

No. 51 and 55 follow from Sections 4.9 and 4.11; No. 105 and 77 follow from Sections 4.6 and 4.22; No. 31, 32 and 36 follow from Sections 4.15, 4.17 and 4.19; No. 49 and 50 follow from Sections 4.13 and 4.16; No. 57, 61 and 84 follow from Sections 4.8, 4.23 and 4.18. No. 4 and 20 follow from Lemma 4.5 and Section 4.7.

Among the historical diagrams, No. 5, 67, 69, 73 and 80 follow from Lemma 5.1; No. 6 from Section 5.2; No. 71 from Section 5.7; No. 74 and 75 from Lemma 5.2; No. 82, 89 and 90 from their torque subsections; No. 81 from Section 5.12; No. 64 and 65 from Sections 5.11 and 5.8; No. 79 from Section 5.16; No. 52 is treated separately below. Chang–Chen [6, Table 6.1] already excludes No. 116 for positive masses. This membership list records coverage; it is not a proof of any individual exclusion.

shared first-limit tests. The reduction from the 61 published relation diagrams to 33 still-relevant ones uses 28 additional positive-point first-limit exclusions. For each of these 28 graphs, the separate local argument in the subsections below prevents it from being a *first* limit on a curve through a positive configuration. The detailed original-row and branch-coverage arguments are scoped to such first limits. In particular No. 4 follows from Lemma 4.5 and No. 20 from section 4.7; neither is assumed.

Lemma 4.2 (Force transfer at an uncircled zw triangle)**.** *At positive masses, the signed diagrams No. 2 and No. 22 cannot occur as singular germs.*

Proof. Both contain an uncircled zw triangle $T = \{a, b, c\}$. The vertex a has no external stroke, while b has exactly one external stroke bd in the z colour and none in the w colour (interchange the colours for No. 22). Since a is uncircled, its two leading coordinate rows require $F^z_{T,a} = F^w_{T,a} = 0$. Both separations ab, ac have the same signed distance order. Solving these two nontrivial force equations gives $z_{ab}/w_{ab} = z_{ac}/w_{ac} = \rho$ at leading order. The triangle difference identity transfers the same ratio to bc, hence $F^z_{T,b} = \rho F^w_{T,b}$. With no external w stroke at b, its original w row requires $F^w_{T,b} = 0$. Its original z row then requires $F^z_{T,b} + m_d \delta^3_{bd}(z_d - z_b)^* = 0$. The first term is zero and the second is nonzero by the external stroke and $m_d > 0$, a contradiction.

For No. 2 the labelled triple and external edge are $(a, b, c; bd) = (1, 3, 4; 35)$; for No. 22, after the colour exchange, they are $(4, 6, 5; 16)$. This uses their circle patterns; the same stroke skeleton with additional circles need not obey the lemma. □

4.1. **No. 2: force transfer across triangle 134.** The vertices $(a, b, c; bd) = (1, 3, 4; 35)$ satisfy the hypotheses of Lemma 4.2: the triangle is uncircled, body 1 has no external stroke, and body 3 has exactly one external z stroke. Its w row makes the internal w-force zero; the common shape ratio then makes the internal z-force zero as well, contrary to the nonzero leading 35 force.

4.2. **No. 22: the colour-dual force transfer.** Interchange the two colours in Lemma 4.2 and take $(a, b, c; bd) = (4, 6, 5; 16)$. Bodies 4, 5 and 6 are uncircled, and body 6 has only the 16 exterior w stroke. Its internal z-force vanishes, so the transferred internal w-force vanishes, contradicting the nonzero signed 16 source in the original w row.

4.3. **No. 3: bounded seven-products.**

Lemma 4.3 (A bounded reported-relation diagram)**.** *No. 3 cannot be an unbounded-seven-product first limit.*

Proof. For a pair ij put $L_{ij} = 2$ on a zw edge, 0 on a single-colour stroke, -1 on an unmarked pair joined by a stroke path in at least one colour, and -4 on an unmarked pair joined by no such path in either colour. Stroke-path separation bounds give

$$\operatorname{ord}(z_{ij} w_{ij}) \geq L_{ij}.$$

The seven-product P_{ij} uses ij and its six disjoint pairs, so

$$\operatorname{ord} P_{ij} \geq L_{ij} + \sum_{\{k,\ell\} \cap \{i,j\} = \varnothing} L_{k\ell}. \tag{5}$$

For the pinned No. 3 graph these fifteen lower bounds are 12 for 12, 3 for the eight pairs $1j, 2j$ with $j \in \{3,4,5,6\}$, 2 for $34, 35, 45$, and 0 for $36, 46, 56$. The minimum is 0, so no seven-product can have a pole. This does not assert that No. 3 cannot occur elsewhere on a complex curve. □

4.4. **No. 78: a single-bridge energy contradiction.**

Lemma 4.4 (The single bridge of No. 78)**.** *No. 78 is not a first limit on a curve through a positive configuration, at any strictly positive mass.*

Proof. Partition the bodies into its two circled zw triangles $A = 123$ and $B = 456$, with cluster masses M_A, M_B, $M = M_A + M_B$, and put $\mu = m_3 m_6$. The original mass-weighted w rows give distinct leading cluster centres and leave 36 as the sole finite cross inverse distance $D_{36} \to d \neq 0$. Their relative equation implies $\mu d^3 = M_A M_B / M$. Contract the original z rows against the relative w coordinates of each triangle. Internal inertia tends to zero; for every crossed tie $\delta_{ij}^3 z_{ij} = \delta_{ij}/w_{ij}$, so the remaining terms fade. Both internal signed potentials tend to zero, and constancy of $U = I = c > 0$ therefore gives $c = \mu d$ and

$$Mc^3 = M_A M_B \mu^2. \tag{6}$$

At the positive point on that same curve, Hölder requires $\sigma^3 \leq Mc^3$, where $\sigma = \sum_{i<j} m_i m_j$. But $\sigma > M_A M_B > \mu$ for strictly positive masses; hence $\sigma^3 > M_A M_B \mu^2 = Mc^3$, a contradiction. This excludes the physical first limit, not an arbitrary complex-only No. 78 curve. □

4.5. **No. 4: vanishing cluster potentials.**

Lemma 4.5 (The two vanishing clusters of No. 4)**.** *No. 4 is not a positive-point first limit, including on its reported mass-relation zero sets.*

Proof. Its negative-seven-product window has $-2 < \alpha < -13/8$, where α is the common z-separation order on the z-only K_4 with vertex set $Q = \{3,4,5,6\}$. Put $T = \{1,2,6\}$ and $M = \sum_{i=1}^{6} m_i$. The zw edges of T have z- and w-orders both equal to 1. On each edge of Q, the signed tie and the marked z-force order -2 give

$$\operatorname{ord} z_{ij} = \alpha, \qquad \operatorname{ord} w_{ij} = \beta := \frac{4-\alpha}{3} > 1.$$

For $i \in \{1,2\}$ and $j \in \{3,4,5\}$, the separation identities through body 6 therefore give

$$\operatorname{ord} z_{ij} = \alpha, \qquad \operatorname{ord} w_{ij} = 1, \qquad \operatorname{ord} \delta_{ij} = -\frac{\alpha+1}{2} > 0.$$

The original mass centres yield

$$Mz_6 = -\sum_{i\neq 6} m_i z_{i6}, \qquad Mw_6 = -\sum_{i\neq 6} m_i w_{i6},$$

so $z_6 = O(\varepsilon^\alpha)$ and $w_6 = O(\varepsilon)$.

We contract the original w rows at bodies $1, 2$ against $m_i z_{i6}$, retaining the exterior forces. The internal pairs combine by their exact signed ties, giving

$$\begin{aligned} U_T = w_6 \sum_{i=1,2} m_i z_{i6} + \sum_{i=1,2} m_i z_{i6} w_{i6} \\ - \sum_{i=1,2} \sum_{j=3,4,5} m_i m_j \delta_{ij}^3 w_{ij} z_{i6}. \end{aligned}$$

Consequently,

$$U_T = O(\varepsilon^2) + O\left(\varepsilon^{(1-3\alpha)/2}\right).$$

Similarly, contracting the original z rows at bodies $3,4,5$ against $m_i w_{i6}$ gives

$$U_Q = z_6 \sum_{i=3,4,5} m_i w_{i6} + \sum_{i=3,4,5} m_i z_{i6} w_{i6} - \sum_{i=3,4,5} \sum_{j=1,2} m_i m_j \delta_{ij}^3 z_{ij} w_{i6},$$

and hence

$$U_Q = O\Big(\varepsilon^{2(\alpha+2)/3}\Big) + O\Big(\varepsilon^{-(5\alpha+1)/6}\Big). \tag{7}$$

All four displayed exponents are strictly positive throughout the stated window. The remaining six pairs have total signed potential

$$\sum_{i=1,2} \sum_{j=3,4,5} m_i m_j \delta_{ij} = O\Big(\varepsilon^{-(\alpha+1)/2}\Big) \longrightarrow 0.$$

These pairs, the pairs in T, and the pairs in Q partition all fifteen pairs. Thus $U \to 0$, contrary to the constant $U = c > 0$ on a curve through a positive configuration. □

Lemma 4.6 (Reference-body potential identity)**.** *Let $T \subset \{1,\dots,6\}$ have at least two elements, let $r \in T$, and put $U_T = \sum_{i<j,\ i,j\in T} m_i m_j \delta_{ij}$. On the original normalized six-body equations,*

$$U_T = z_r \sum_{i\in T\setminus\{r\}} m_i w_{ir} + \sum_{i\in T\setminus\{r\}} m_i z_{ir} w_{ir} - \sum_{i\in T\setminus\{r\}} \sum_{j\notin T} m_i m_j \delta_{ij}^3 z_{ij} w_{ir}. \tag{8}$$

The colour-dual identity is obtained by exchanging z and w. No row at the reference body is treated as an independent subsystem equation.

Proof. Multiply the original z_i row by $m_i w_{ir}$ and sum over $i \in T \setminus \{r\}$. For an internal pair i,j with neither endpoint equal to r, its two contributions combine as

$$m_i m_j \delta_{ij}^3 z_{ij}(w_{ir} - w_{jr}) = m_i m_j \delta_{ij}^3 z_{ij} w_{ij} = m_i m_j \delta_{ij}.$$

For the internal pair ir, its single contribution has the same value by the signed tie. Thus the internal terms sum to U_T, while the exterior terms are exactly the double sum in (8). Expanding $z_i = z_r + z_{ir}$ on the left and moving the exterior terms to the other side proves the identity. The original w_i rows give its colour-dual form. □

Lemma 4.7 (Internal potential with complete external forcing)**.** *For a set T of bodies, put*

$$M_T = \sum_{i\in T} m_i, \qquad \bar{z}_T = M_T^{-1} \sum_{i\in T} m_i z_i, \qquad \bar{w}_T = M_T^{-1} \sum_{i\in T} m_i w_i.$$

On the original normalized equations,

$$U_T = \sum_{i\in T} m_i (z_i - \bar{z}_T)(w_i - \bar{w}_T) - \sum_{i\in T} \sum_{j\notin T} m_i m_j \delta_{ij}^3 z_{ij} (w_i - \bar{w}_T). \tag{9}$$

In particular, suppose T is a zw triangle, so its two internal separation orders are 1. If an exterior body j has a w-only edge to $p \in T$ with z-order $s \in (1,2]$, then

pair	$\operatorname{ord} z$	$\operatorname{ord} w$	$\operatorname{ord}(\delta^3 z)$
jp	s	$4-3s$	$4s-6$
$ji,\ i \in T\setminus\{p\}$	1	$4-3s$	$(9s-13)/2$

and the contribution of this body to the exterior sum in (9) *is*

$$O(\varepsilon^{4s-5}) + O\Big(\varepsilon^{(9s-11)/2}\Big). \tag{10}$$

The internal inertia in (9) *is* $O(\varepsilon^2)$.

Proof. Multiply every original z_i row with $i \in T$ by $m_i(w_i - \bar{w}_T)$ and sum. The internal pair ij contributes $m_i m_j \delta_{ij}^3 z_{ij} w_{ij} = m_i m_j \delta_{ij}$. The remaining terms are exactly the exterior sum displayed in (9); its left side is the internal inertia because $\sum_{i\in T} m_i(w_i - \bar{w}_T) = 0$.

For any pair whose separation orders are x, y, the signed tie gives

$$\operatorname{ord}\delta = -\frac{x+y}{2}, \qquad \operatorname{ord}(\delta^3 z) = -\frac{x+3y}{2}, \qquad \operatorname{ord}(\delta^3 w) = -\frac{3x+y}{2}. \tag{11}$$

A w stroke has $\operatorname{ord}(\delta^3 w) = -2$, giving $y = 4 - 3s$ on jp. Since $s > 1$ and $4 - 3s < 1$, the identities through p give exact orders $1, 4 - 3s$ on every other pair ji. Finally $w_i - \bar{w}_T = O(\varepsilon)$, which proves (10). No assumption about the absolute cluster centre, or about the signs of the complex inverse distances, enters this argument. □

Lemma 4.8 (An unmarked finite pair)**.** *Let ij be a pair with $m_i + m_j > 0$, and write R_i, R_j for the two original z forces after removing their mutual interaction. On a germ, if $(w_i - w_j)(R_i - R_j) = o(1)$, then δ_{ij} tends to a finite nonzero complex value d and $d^3 = (m_i + m_j)^{-1}$.*

Proof. Subtract the original z_i, z_j rows and use the original signed tie $\delta_{ij}^2 z_{ij} w_{ij} = 1$. These give the exact on-solution identity

$$1 = (m_i + m_j)\delta_{ij}^3 + \delta_{ij}^2 (w_i - w_j)(R_i - R_j). \tag{12}$$

If $\delta_{ij} \to 0$, the right side tends to zero. If its modulus diverges, divide by δ_{ij}^3: the last term becomes $o(1)/\delta_{ij}$, giving $m_i + m_j = 0$, a contradiction. The remaining nonzero limit has the stated cube on every phase. □

4.6. **No. 105: incompatible centre and pole signs.** For illustration, No. 105 has just the signed inverse distance poles $25, 36$ of order -1. Its original centres give $YP/(XQ) = AD/(BC) > 0$ for four positive group masses A, B, C, D. Its source rows and signed ties give $YP = -m_2 m_5 D_{25}^4$ and $XQ = m_6^2 D_{36}^4$, while constancy of U forces $m_2 m_5 D_{25} + m_3 m_6 D_{36} = 0$. Eliminating the pole ratio makes $YP/(XQ) < 0$, a contradiction. That local argument does not substitute for the remaining 27 diagram proofs and their valuation-completeness checks.

4.7. **No. 20: second-order matching and energy.** In No. 20, only the outer vertices $1, 2, 3$ have w-circles; the inner zw triangle $T = \{4, 5, 6\}$ is uncircled. Write $p(1) = 6$, $p(2) = 5$, $p(3) = 4$ for the three matches. Every outer–inner w separation has order -2. On a marked match, the original w-force order and signed tie therefore give

$$\operatorname{ord}\delta_{ip(i)} = 0, \qquad \operatorname{ord} z_{ip(i)} = 2.$$

The inner z- and w-differences have exact order 1. Hence all six z coordinates differ by $O(\varepsilon)$, and the original mass centre gives $z_a = O(\varepsilon)$ for every a. On a nonmatched outer–inner edge,

$$z_{ia} = z_{p(i)a} + O(\varepsilon^2), \qquad \operatorname{ord}\delta_{ia} = \tfrac{1}{2}, \qquad \operatorname{ord}(\delta_{ia}^3 z_{ia}) = \tfrac{5}{2}.$$

Apply Lemma 4.6 to T, with any inner reference vertex. Its reference and relative-inertia terms are $O(\varepsilon^2)$. The matched exterior z forces are $O(\varepsilon^2)$, and the remaining exterior z forces are $O(\varepsilon^{5/2})$; multiplication by the inner relative w coordinates gives respectively $O(\varepsilon^3)$ and $O(\varepsilon^{7/2})$. Thus

$$U_{456} = O(\varepsilon^2).$$

This estimate uses the original exterior forces and precedes the second-order matching calculation.

Let $W_1, W_2, W_3 \neq 0$ be the outer w-leads. Since the inner vertices are uncircled, their original centre gives $\sum_{i=1}^{3} m_i W_i = 0$, so the three leads cannot all coincide. If they were pairwise distinct, every outer edge would have z-order 1 and w-order -2. Its inverse distance and z force would consequently have orders $1/2$ and $5/2$. The original outer z rows would then give $z_i = O(\varepsilon^2)$, and the matched ties would put all three inner z positions at $O(\varepsilon^2)$, contrary to their exact order-one differences. Hence exactly one outer pair has equal leads, say $W_i = W_j \neq W_k$.

The inverse distances on ik, jk tend to zero with order $1/2$. Moreover, $z_{ij} = z_{p(i)p(j)} + O(\varepsilon^2)$ has exact order 1. Subtracting the original z_i, z_j rows, whose forces other than their mutual interaction are $O(\varepsilon^2)$, gives

$$\big(1 - (m_i + m_j)\delta_{ij}^3\big) z_{ij} = O(\varepsilon^2).$$

Thus $\delta_{ij} \to e_{ij} \neq 0$ with $e_{ij}^3 = (m_i + m_j)^{-1}$. If $d_i = \lim \delta_{ip(i)}$, all fifteen pairs have now been accounted for: the inner total potential tends to zero, the six nonmatched outer–inner inverse distances and the two unequal-lead outer inverse distances tend to zero, and only the three matches and ij remain. Consequently

$$c_0 = \lim U = \sum_{i=1}^{3} m_i m_{p(i)} d_i + m_i m_j e_{ij}.$$

For the equal pair (i, j) and unmatched outer body k, all three choices are related by relabelling. If p, q, r are their matched inner vertices, the outer z rows and the inner centre give $(m_i, m_j) = \lambda(m_p, m_q)$, $\lambda > 0$. Put $(a, b, c) = (m_p, m_q, m_r)$, let $u = (z_p - z_r)^* \neq 0$ and $v = (w_p - w_r)^* \neq 0$ be the inner leading differences, and write $t_0 = D_{qr}/D_{pr}$ for the ratio of the nonzero signed inverse-distance leads. The leading inner z centre gives $(z_q - z_r)^* = -au/b$. The original leading z_r row is then $au(D_{pr}^3 - D_{qr}^3) = 0$, so $t_0^3 = 1$. The pr, qr ties give $(w_q - w_r)^*/v = -(b/a)t_0$; the third tie and the zero z_p force give

$$\left(1 + \frac{b}{a} t_0\right)^3 = \frac{b^3}{(a+b)c^2} > 0.$$

For $t_0 \in \{\omega, \omega^2\}$, $\omega^3 = 1 \neq \omega$, the cube on the left has nonzero imaginary part if $a \neq b$ and equals -1 if $a = b$. Hence every positive-mass leading branch has $t_0 = 1$, and the tie equation yields $(ab)^3 = c^2(a+b)^4$. Moreover, using the zero z_p force to eliminate D_{pq}^3, the original inner w_p row has leading internal force

$$F_{T,p}^w = \frac{c(a-b)}{a} D_{pr}^3 v = \frac{m_i}{m_p} W_i = \lambda W_i \neq 0.$$

The displayed sign follows the convention $F_{T,p}^w = \sum_{j \in T \setminus \{p\}} m_j D_{pj}^3 (w_p - w_j)^*$. Thus $a \neq b$ follows from an original row, including when the mass polynomial has a repeated shape root. We have proved the necessary condition

$$(ab)^3 = c^2(a+b)^4, \qquad a \neq b. \tag{13}$$

It is not itself an exclusion: its positive zero set is nonempty. For completeness, eliminating the two leading inner z forces from the three signed ties gives the shape polynomial

$$P(t) = c^2(bt^3 + a)(a + bt)^3 - (ab)^3 t^3.$$

At the root selected by the original rows, $P(1) = -(ab)^3 + c^2(a+b)^4 = 0$ and $P'(1) = 3c^2(a + b)^3(b - a) \neq 0$. The other algebraic roots of P are not silently identified with the actual ratio t_0: the original z_r row and positive-mass tie above have already forced $t_0 = 1$. The full inner z rows have dimensionless force residual $O(\varepsilon^3)$, since their internal forces have order -2 and their coordinate and matched exterior terms have order ε. Local elimination at this simple root gives

$$t - 1 = O(\varepsilon^3), \qquad \frac{z_q - z_r}{z_p - z_r} + \frac{a}{b} = O(\varepsilon^3), \qquad a(z_p - z_r) + b(z_q - z_r) = O(\varepsilon^4).$$

The original k-th outer z row has matched force coefficient $m_r d_k^3 = 1$ and every other force later than ε^2; hence $z_r = o(\varepsilon^2)$ and the inner mass-weighted z sum is $o(\varepsilon^2)$. Define the matched offsets by $z_\ell - z_{p(\ell)} = \varepsilon^2 A_\ell + o(\varepsilon^2)$ for $\ell = i, j, k$. Their original signed ties make every A_ℓ nonzero. Summing the original outer z rows with mass weights cancels the unmarked ij force; the unmatched outer-to-inner forces are $O(\varepsilon^{5/2})$. The exact global z centre and the inner estimate then give

$$m_i A_i + m_j A_j + m_k A_k = 0. \tag{14}$$

The preceding finite-edge reduction gives the complete constant $U = I = c_0 > 0$. Keeping all signed cube phases, the real positivity of c_0 and the offset identity force their phases to be 1. Put $A = a + b$, $B = c^{2/3}$, $H = a^{5/3} + b^{5/3}$, $d = \sqrt{AH/B}$ and $x = d/A \geq 19/12$. With $m_k = \lambda d$, the physical Hölder inequality $\sigma^3 \leq M c_0^3$ would imply

$$(\lambda x + 1 + x)^3 \leq \big(\lambda(1 + x) + 9/8\big)(27/2 + \lambda^2/16). \tag{15}$$

The difference of left and right increases with $x \geq 19/12$; at $x = 19/12$ its ascending λ coefficients are $(3547/1728, -1829/576, 22301/1152, 1645/432)$. The middle quadratic has discriminant test $4ac - b^2 = 148167571/995328 > 0$, making that difference strictly positive. Thus no positive-mass No. 20 first limit survives any of the three equal-pair branches. The simple-root and signed-phase checks are required certificate inputs.

4.8. **No. 57: two outer triangle regimes.** The outer w-triangle 123 has a common z-order $\alpha \in (1, 2]$, and its inner zw triangle is 456.
No. 57: the range $1 < \alpha < 2$. Put $A = \{1, 2, 3\}$, $B = \{4, 5, 6\}$ and $b = 4 - 3\alpha$. The internal orders of A are (α, b), and those of B are $(1, 1)$. The original w centre forces the two cluster leading values to be distinct: if they were equal, all six nonzero circled leading values would agree. Hence the bridge 36 is maximal, with orders $(2, -2)$ and $\delta_{36} \to d \neq 0$. All nine cross pairs have the following exact orders:

pair	ord z	ord w	ord δ	ord$(\delta^3 z)$
36	2	-2	0	2
16, 26	α	-2	$(2 - \alpha)/2$	$3 - \alpha/2$
$ij,\ i \in A,\ j = 4, 5$	1	-2	1/2	5/2

These entries follow by adding the internal differences to z_{36}, w_{36}; in each addition the two competing orders are distinct. Apply (9) first to B. Its relative w coordinates are $O(\varepsilon)$, so the table gives

$$U_B = O(\varepsilon^2) + O(\varepsilon^3) + O(\varepsilon^{4-\alpha/2}) + O(\varepsilon^{7/2}) = O(\varepsilon^2).$$

For A, the relative w coordinates are $O(\varepsilon^b)$ and its internal inertia is $O(\varepsilon^{\alpha+b})$. Thus

$$U_A = O(\varepsilon^{4-2\alpha}) + O(\varepsilon^{6-3\alpha}) + O(\varepsilon^{7-7\alpha/2}) + O(\varepsilon^{13/2-3\alpha}) \longrightarrow 0. \tag{16}$$

Every exponent is positive for $1 < \alpha < 2$. The table also proves that every cross inverse distance except 36 tends to zero, so $c = m_3 m_6 d$. In the original mass-weighted w rows of A, only 36 contributes at order -2: the other cross w-forces have orders $1 - 3\alpha/2 > -2$ and $-1/2$. If W_A, W_B are the cluster leading values, then

$$M_A W_A = m_3 m_6 d^3 (W_A - W_B), \qquad M_A W_A + M_B W_B = 0.$$

Since $W_A \neq W_B$, this gives $m_3 m_6 d^3 = M_A M_B / M$ and $M c^3 = M_A M_B (m_3 m_6)^2$. The strict inequality $\sigma > M_A M_B > m_3 m_6$ contradicts physical Hölder.

If $\alpha = 2$, the outer triangle is maximal. When $W_3 \neq W_{456}$, maximal 13 and 36 force the forbidden closing stroke 16. When $W_3 = W_{456}$, write $\gamma = \operatorname{ord} z_{36} \in (1, 2)$ on the nonmaximal w-only stroke 36. Its signed tie gives $\operatorname{ord} w_{36} = 4 - 3\gamma$ and $\operatorname{ord} \delta_{36} = \gamma - 2$. The original body-3 z-row has a nonzero 36 force at order $4\gamma - 6 < 2$. The nearest 34, 35 crossed forces have order at least $(9\gamma - 13)/2$, later by $(\gamma - 1)/2 > 0$; the coordinate term is later by $4(2 - \gamma) > 0$, and all

remaining terms are later. Thus both maximal branches are impossible even without a physical point.

4.9. **No. 51: the triangle and pendant at 16.** Let $\alpha \in (1,2]$ be the common z-order on its w-only triangle 236 and $\gamma \in (1,2]$ the z-order on pendant 16. The four endpoint choices are exhaustive. If both equal 2, maximal 16 and 26 require a missing 12 stroke. If both are less than 2, the six circled w-leads coincide, contrary to their positive-mass centre. For $\alpha = 2$, $\gamma < 2$, the original body-1 z-row has its nonzero 16 force uniquely earliest at $4\gamma - 6$. Finally, for $\alpha < 2$, $\gamma = 2$, the least seven-product order is $12 - 7\alpha$; a first limit needs $12/7 < \alpha < 2$.
Joint energy argument for No. 51 and No. 55. Put $T = \{2,3,6\}$, $Q = \{4,5,6\}$ and $b = 4 - 3\alpha$, where $12/7 < \alpha < 2$. The internal orders of T are (α, b), and those of Q are $(1,1)$. The pendant is maximal: it is 16 in No. 51 and 13 in No. 55. For $i = 2,3$ and $j = 4,5$, adding differences through body 6 gives

$$\operatorname{ord} z_{ij} = 1, \qquad \operatorname{ord} w_{ij} = b, \qquad \operatorname{ord} \delta_{ij} = \frac{3\alpha - 5}{2}, \qquad \operatorname{ord}(\delta_{ij}^3 z_{ij}) = \frac{9\alpha - 13}{2}. \tag{17}$$

In No. 51, the pairs $12, 13$ have orders $(\alpha, -2)$ and $14, 15$ have orders $(1, -2)$. The original body-1 z row therefore has its pendant force at order 2, the $12/13$ forces at order $3 - \alpha/2 > 2$, and the $14/15$ forces at order $5/2$. Consequently

$$z_1 = O(\varepsilon^2), \qquad z_6 = O(\varepsilon^2). \tag{18}$$

In No. 55, the pairs $12, 16$ have orders $(\alpha, -2)$ and $14, 15$ again have orders $(1, -2)$. The same original row gives

$$z_1, z_3 = O(\varepsilon^2), \qquad z_6 = O(\varepsilon^\alpha). \tag{19}$$

For U_Q, the exterior bodies $2, 3$ each attach to 6 with order α. Lemma 4.7 bounds their complete contributions by $O(\varepsilon^{4\alpha-5}) + O(\varepsilon^{(9\alpha-11)/2})$. In No. 51 the remaining exterior body 1 attaches maximally to 6 and contributes $O(\varepsilon^3) + O(\varepsilon^{7/2})$. In No. 55 its pair with 6 has orders $(\alpha, -2)$ and its pairs with $4, 5$ have orders $(1, -2)$; its contribution is therefore $O(\varepsilon^{4-\alpha/2}) + O(\varepsilon^{7/2})$. In both diagrams,

$$U_Q = O(\varepsilon^2) + O(\varepsilon^{4\alpha-5}) + O(\varepsilon^{(9\alpha-11)/2}) \longrightarrow 0. \tag{20}$$

For U_T, use the reference-body identity with reference 6, rather than assigning a separate subsystem equation to the shared body 6. Its relative inertia is $O(\varepsilon^{4-2\alpha})$. Equations (18) and (19) bound the reference term by $O(\varepsilon^{6-3\alpha})$ and $O(\varepsilon^{4-2\alpha})$, respectively. The four exterior interactions from $i = 2, 3$ to $j = 4, 5$ contribute

$$O\Big(\varepsilon^{(9\alpha-13)/2+b}\Big) = O\Big(\varepsilon^{(3\alpha-5)/2}\Big).$$

The interactions with body 1 contribute $O(\varepsilon^{7-7\alpha/2})$ in No. 51, and $O(\varepsilon^{6-3\alpha}) + O(\varepsilon^{7-7\alpha/2})$ in No. 55. Thus in both cases

$$U_T = O(\varepsilon^{4-2\alpha}) + O(\varepsilon^{6-3\alpha}) + O(\varepsilon^{(3\alpha-5)/2}) + O(\varepsilon^{7-7\alpha/2}) \longrightarrow 0. \tag{21}$$

All exponents are positive on $12/7 < \alpha < 2$. The pair sets inside T and Q are disjoint. Together with the four pairs in (17) and the five pairs incident to body 1, they exhaust the fifteen pairs. The former four inverse distances tend to zero; among the latter five, only the pendant has a nonzero finite limit d. Writing $r = 6$ for No. 51 and $r = 3$ for No. 55, we obtain

$$c = m_1 m_r d, \qquad m_r d^3 = \frac{M - m_1}{M}, \qquad Mc^3 = m_1^3 m_r^2 (M - m_1).$$

Here the cube relation follows from the original body-1 w row and the full w centre: its only order--2 force is the pendant force, while bodies $2, \dots, 6$ share one leading w value. Finally $\sigma > m_1(M - m_1)$ and $M - m_1 > m_r$ give $\sigma^3 > Mc^3$, contrary to physical Hölder.

4.10. **No. 40: two maximal strokes force the missing edge.** Body 1 has no w circle, whereas bodies 3 and 6 have w circles. Therefore the marked edges 13 and 16 both have exact w-order -2. Their marked w-force order and the signed ties give $\operatorname{ord} z_{13} = \operatorname{ord} z_{16} = 2$. Consequently $z_{36} = z_{31} + z_{16} = O(\varepsilon^2)$. Put $x = \operatorname{ord} z_{36} \geq 2$ and $y = \operatorname{ord} w_{36} \geq -2$, the latter inequality following from normalization. The exact signed tie implies

$$\operatorname{ord}(\delta_{36}^3 w_{36}) = -\frac{3x+y}{2} \leq -2.$$

Strict inequality violates normalization; equality requires a 36 w stroke. The diagram has no such stroke. This excludes No. 40 without any potential cancellation or any assumption on the order of 25.

4.11. **No. 55: the triangle and pendant at 13.** Here the w-only triangle is 236 but the pendant is 13, not 16 as in No. 51. Write $\alpha, \gamma \in (1, 2]$ for their z-orders. With $\alpha = \gamma = 2$, consecutive maximal 13 and 23 require a missing 12 stroke; with $\alpha, \gamma < 2$, the original w-centre forbids a common nonzero lead at all six bodies. When $\alpha = 2$, $\gamma < 2$, the original body-1 z-row has the 13 force uniquely first at $4\gamma - 6$, earlier than its coordinate and all exterior forces. In the remaining branch $\gamma = 2$, $\alpha < 2$, a first-limit pole requires $12/7 < \alpha < 2$ by the product order $12 - 7\alpha$. The joint argument in Section 4.9, in particular (19) and (20)–(21), proves that both complete internal potentials tend to zero. Only the finite pendant 13 remains, and

$$m_3 d^3 = (M - m_1)/M, \qquad c = m_1 m_3 d, \qquad M c^3 = m_1^3 m_3^2 (M - m_1). \tag{22}$$

The strict inequalities $\sigma > m_1(M - m_1)$ and $M - m_1 > m_3$ again contradict physical Hölder. All four branches are excluded with the correct pendant-specific reference estimate.

4.12. **No. 44: an unpaired star force.** The three w-only spokes $16, 26, 36$ meet the inner zw triangle 456. Positive mass centre excludes no maximal spoke; two maximal spokes would require a missing outer closing w-stroke. Thus exactly one is maximal, which we label 16. Put $a = \operatorname{ord} z_{26}$, $b = \operatorname{ord} z_{36}$, both in $(1, 2)$. If $a = b$, the missing 23 signed tie has w-force order at most -2, either drawing an absent maximal stroke or violating normalization. Otherwise swap 2 and 3 to take $a < b$. The original body-1 z-row and maximal 16 tie give $z_1, z_6 = O(\varepsilon^2)$. In the original body-2 z-row the nonzero 26 force has order $4a - 6$. Its coordinate, 21, 23 and 24/25 forces are later by

$$6 - 3a, \qquad \frac{9}{2}(2 - a), \qquad \frac{9}{2}(b - a), \qquad \frac{a-1}{2}, \tag{23}$$

all strictly positive. The unmarked 23 inverse distance was retained in this comparison; its possible pole cannot cancel the unique 26 term. The $b < a$ case is the body-3 image. Thus no positive-mass complex No. 44 germ exists.

4.13. **No. 49: all extremal sets of the w-path.** The w-only path $1 - 6 - 3 - 2$ has maximal edge set one of $\{16\}$, $\{23\}$, $\{36\}$ or $\{16, 23\}$. The empty set violates the original w-centre; consecutive maximal 16/36 or 23/36 would draw a missing closing edge. For only 16 or only 23, equality of the two remaining nonmaximal z-orders likewise forces the absent closing w-stroke. For unequal orders, whichever nonmaximal edge has smaller order contributes a unique nonzero $4s - 6$ term to its original outer z-row. For only 36, subtracting the original body-2/3 z-rows leaves the nonzero 23 term of order $4s - 6$ earlier than the position difference of order s and all exterior forces of order 2. When both 16 and 23 are maximal, retain the unmarked pair 12. Write $g = \operatorname{ord} z_{36} \in (1, 2)$, $\lambda = \operatorname{ord} w_{12}$ and $k = 4g - 6 < 2$. The maximal ties give $z_{16}, z_{23} = O(\varepsilon^2)$, so $z_{12} = -z_{36} + O(\varepsilon^2)$ has exact order g. The missing 12 stroke forces $\lambda < 4 - 3g$. Consequently its original z-force has order

$$\operatorname{ord}(\delta_{12}^3 z_{12}) = -\frac{g + 3\lambda}{2} = k + \frac{3}{2}(4 - 3g - \lambda) > k.$$

In the original body-2 z-row, the forces from 23, 24/25 and 26 have orders 2, 5/2 and $3-g/2$, respectively, all greater than k. Together with the preceding estimate for 21, this gives

$$z_2 = o(\varepsilon^k), \qquad z_3 = z_2 - z_{23} = o(\varepsilon^k).$$

In the original body-3 z-row, the 36 force is nonzero at exact order k. The 32 and 31 forces have orders 2 and $3-g/2$, while the 34/35 forces have order

$$\frac{9g-13}{2} = k + \frac{g-1}{2} > k.$$

Thus every other term in that row, including its left side, is $o(\varepsilon^k)$, a contradiction. All four maximality sets fail, including the potentially unbounded 12-distance branch.

4.14. **No. 45: the two possible potential-pole regimes.** The zw triangle is $Q = \{4,5,6\}$, the exterior w-only edges are $16, 26, 35$, and all six bodies have w circles. At least one exterior edge is maximal, since otherwise all six circled leading w values coincide, contrary to their mass centre. The edges 16 and 26 cannot both be maximal: their z differences would force $z_{12} = O(\varepsilon^2)$, and normalization would then require the missing 12 w stroke. Thus, up to interchanging bodies 1 and 2, the possible maximal sets are $\{16\}$, $\{35\}$ and $\{16, 35\}$.

Suppose first that only 16 is maximal. Put $a = \operatorname{ord} z_{26}$ and $b = \operatorname{ord} z_{35}$, both in $(1,2)$. The original body-1 z row gives $z_1, z_6 = O(\varepsilon^2)$: its 16 force has order 2, its 12 force has order $3 - a/2 > 2$, and its 13/14/15 forces have order 5/2. Hence z_2 has exact order a, while z_{23} has exact order 1. If $a \neq b$, then

$$\operatorname{ord} w_{23} = 4 - 3\max\{a,b\}, \qquad \operatorname{ord}(\delta_{23}^3 z_{23}) = \frac{9\max\{a,b\} - 13}{2} > 4a - 6.$$

The 24/25 forces have order $(9a-13)/2 > 4a-6$, and the 21 force has order $3 - a/2 > 4a - 6$. Since $a > 4a - 6$, the 26 force is the unique earliest term in the original body-2 z row, a contradiction.

It remains to consider $a = b = s \in (1,2)$. Write $\lambda = \operatorname{ord} w_{23}$. The separation identities give $\lambda \geq 4 - 3s$, while $\operatorname{ord} z_{23} = 1$. All terms in the original body-2 z row except its 26 and 23 forces are later than $4s - 6$. Their cancellation therefore requires

$$-\frac{1+3\lambda}{2} = 4s - 6, \qquad \lambda = \frac{11-8s}{3} > 4 - 3s, \qquad \operatorname{ord}\delta_{23} = \frac{4s-7}{3} > s - 2.$$

The strict inequality for λ shows that w_{26} and w_{35} have the same nonzero leading coefficient u. Write

$$\delta_{26} = A\varepsilon^{s-2}(1 + o(1)), \qquad \delta_{35} = D\varepsilon^{s-2}(1 + o(1)), \qquad ADu \neq 0.$$

Bodies $2, 3$ and Q have the same nonzero circled leading w value W. Their original w_2, w_3 rows give

$$W = m_6 A^3 u = m_5 D^3 u, \qquad (D/A)^3 = m_6/m_5 > 0.$$

Indeed the 23 w-force has order $(4s-10)/3 > -2$, each nonmatched force into Q has order $(3s-7)/2 > -2$, and the remaining exterior forces are also later than -2.

We now retain the complete internal potential. Lemma 4.7 gives

$$U_Q = O(\varepsilon^2) + O(\varepsilon^{4s-5}) + O(\varepsilon^{(9s-11)/2}) = o(\varepsilon^{s-2}). \tag{24}$$

The final equality uses $s > 1$; it does not require $U_Q \to 0$. The nonmatched outer-to-inner inverse distances have order $(3s-5)/2 > s-2$, and the 23 inverse distance has order $(4s-7)/3 > s-2$. The inverse distances on $12, 13$ tend to zero, while 16 has a finite limit. Consequently the complete potential satisfies

$$U = (m_2 m_6 A + m_3 m_5 D)\varepsilon^{s-2} + o(\varepsilon^{s-2}).$$

Its constancy forces $D/A = -m_2 m_6/(m_3 m_5) < 0$, contradicting the positive cube ratio above.

Suppose next that only 35 is maximal. Put $a = \operatorname{ord} z_{16}$ and $b = \operatorname{ord} z_{26}$. If $a = b$, then $\operatorname{ord} z_{12} \geq a$ and $\operatorname{ord} w_{12} \geq 4 - 3a$; the signed tie forces the missing 12 w stroke or violates

normalization. Thus, after interchanging $1,2$, take $a < b$. The original body-3 z row gives $z_3, z_5 = O(\varepsilon^2)$. The zw edge 56 then implies that z_6, z_1, z_2 have exact order 1. The 12 z-force has order $(9b-a-12)/2$, later than both the 16 force of order $4a-6$ and the 26 force of order $4b-6$. All remaining forces in the respective original body-1 and body-2 z rows are later than those marked forces: the nonmatched inner forces exceed them by $(a-1)/2$ and $(b-1)/2$, and the body-3 forces have order $5/2$. The two original rows therefore require $4a-6 = 4b-6 = 1$, contrary to $a < b$.

Finally suppose that 16 and 35 are maximal, while $s = \operatorname{ord} z_{26} \in (1,2)$. Multiply the original body-1 and body-3 z rows by m_1, m_3 and add. Their mutual 13 interaction cancels exactly; every remaining force is $O(\varepsilon^2)$. The maximal ties yield

$$m_1 z_6 + m_3 z_5 = O(\varepsilon^2), \qquad z_{56} \text{ has exact order 1.}$$

Since $m_1, m_3 > 0$, both z_5 and z_6 have exact order 1, as does $z_2 = z_6 + O(\varepsilon^s)$. In the original body-2 z row, the 26 force has order $4s-6$; its 24/25 forces have order $(9s-13)/2$, its 21 force has order $3-s/2$, and its 23 force has order $5/2$. Thus $4s-6 = 1$, giving $s = 7/4$. The unmarked pair 13 must still be retained. Its separation z_{13} has exact order 1, and subtraction of its two original rows gives

$$\big(1 - (m_1+m_3)\delta_{13}^3\big) z_{13} = O(\varepsilon^2).$$

Hence $\delta_{13}^3 = (m_1+m_3)^{-1} + O(\varepsilon)$, so this inverse distance has a finite nonzero limit. For the internal triangle, (10) now gives

$$U_Q = O(\varepsilon^2) + O(\varepsilon^{4s-5}) + O(\varepsilon^{(9s-11)/2}) = O(\varepsilon^2).$$

All exterior inverse distances other than 26 are finite or tend to zero: $16, 35, 13$ are finite, and every remaining pair follows from the distinct orders just listed. The nonzero pole $m_2 m_6 \delta_{26}$ of order $s-2 = -1/4$ is therefore the unique pole of the complete potential, contradicting its constancy. All three maximality sets are excluded.

4.15. **No. 31: a finite unmarked pair and the bridge gap.** The vertices $3,4,5,6$ have w-circles, whereas $1,2$ are uncircled and have no strokes. For $i \in \{1,2\}$ and $a \in \{3,4,5,6\}$, the w separation has order -2 and its unmarked w force is $o(\varepsilon^{-2})$. Thus $\delta_{ia} = o(1)$ and $\delta_{ia}^3 z_{ia} = \delta_{ia}/w_{ia} = o(\varepsilon^2)$. In particular, the remaining forces in the original z_1, z_2 rows satisfy $R_1 - R_2 = o(\varepsilon^2)$; since $w_{12} = o(\varepsilon^{-2})$, Lemma 4.8 gives $\delta_{12} \to e \neq 0$ with $e^3 = (m_1+m_2)^{-1}$.

A nonmaximal bridge 36 would make all four circled w-leads equal, contradicting their original mass centre. Hence 36 is maximal, $z_{36} = O(\varepsilon^2)$, and $\delta_{36} \to d \neq 0$. The nonmatched edges $34, 35$ have z-order 1, w-order -2, and inverse-distance order $1/2$. The original body-3 z row therefore gives $z_3 = O(\varepsilon^2)$, and then $z_6 = O(\varepsilon^2)$. Apply Lemma 4.6 to 456 with reference vertex 6. Its reference and relative terms have orders at least 3 and 2; its exterior terms from body 3 have order $7/2$, and those from bodies 1 and 2 are $o(\varepsilon^3)$. Thus $U_{456} = O(\varepsilon^2)$. All remaining inverse distances other than $12, 36$ tend to zero. The original body-3 w row and mass centre give $d^3 = (m_4+m_5+m_6)/[m_6(m_3+m_4+m_5+m_6)]$, so only 12 and 36 remain in $c = \lim U$.

Let $A = m_1+m_2$, $C = m_3+m_4+m_5+m_6$, $X = m_1 m_2$, and let Y be the inner pair-mass sum. Then

$$M^{1/3} c < X + Y + \frac{XC}{3A} + \frac{YA}{3C} \le \sigma - \frac{19}{24} AC < \sigma. \tag{25}$$

This contradicts Hölder at the physical point for every signed phase of the two finite edges; it does not exclude an unrelated complex-only curve.

4.16. **No. 50: five path branches and the resonance.** The possible maximal sets on w-only edges $16, 23, 35$ are $\{16\}$, $\{23\}$, $\{35\}$, $\{16, 23\}$ and $\{16, 35\}$; no maximum violates the w-centre, and maximal 23 and 35 together require a missing 25 edge. In the only-16 and only-35 cases, a missing outer tie or a unique earlier z-force excludes unequal-order branches; the only-35 case also follows by subtracting the original body-$2/3$ z-rows. For both 16 and 23, the original

weighted outer rows cancel the unmarked 12 force. Their remaining 35 edge would be the unique signed potential pole. For both 16 and 35, the 12 and 13 w-differences cannot simultaneously reach the order needed to cancel 23, since their difference is w_{23}; the endpoint $\operatorname{ord} z_{23} = 7/4$ again leaves a unique 23 potential pole.

It remains to treat the only-23 branch in which $\operatorname{ord} z_{16} = \operatorname{ord} z_{35} = s$, with $1 < s < 2$. The w-stroke equations and the signed ties give

$$a := \operatorname{ord} \delta_{16} = \operatorname{ord} \delta_{35} = s - 2, \qquad b := \operatorname{ord} w_{16} = \operatorname{ord} w_{35} = 4 - 3s.$$

The maximal edge 23 has separation orders $(2, -2)$ and $\operatorname{ord} \delta_{23} = 0$. Every internal edge of the zw triangle 456 has separation orders $(1, 1)$. Consequently

ij	$\operatorname{ord} z_{ij}$	$\operatorname{ord} w_{ij}$	$\operatorname{ord} \delta_{ij}$
$12, 24, 26$	1	-2	$1/2$
25	s	-2	$(2-s)/2$
$14, 15, 34, 36$	1	b	$k := (3s-5)/2.$

For example, $z_{12} = z_{65} + z_{16} - z_{23} - z_{35}$ has the nonzero order-one coefficient of z_{65}, while $w_{15} = w_{16} + w_{65}$ has the nonzero order-b coefficient of w_{16}. The other entries follow by the same difference identities. The five force orders in the original z_2 row are $5/2, 2, 5/2, 3 - s/2, 5/2$, in body order $1, 3, 4, 5, 6$. Thus

$$z_2, z_3 = O(\varepsilon^2), \qquad z_5 = O(\varepsilon^s), \qquad z_1 = \zeta\varepsilon + o(\varepsilon), \qquad \zeta \neq 0. \tag{26}$$

In particular $z_{13} = \zeta\varepsilon + o(\varepsilon)$, and all six z positions are $O(\varepsilon)$.

Suppose first that $1 < s < 7/4$, and put $h = 4s - 6 < 1$ and $t = (s-1)/3 > 0$. The 16 force in the original z_1 row has order h. The coordinate term has order 1, the 12 force has order $5/2$, and the $14, 15$ forces have order $(9s-13)/2 = h + (s-1)/2 > h$. Hence the 13 force must have order h as well. Its signed tie determines

$$\operatorname{ord} \delta_{13} = q := \frac{4s-7}{3} = a + t < 0, \qquad \operatorname{ord} w_{13} = \frac{11 - 8s}{3} = b + t. \tag{27}$$

The exact identity $w_{13} = w_{16} - w_{35} + w_{65}$, with $\operatorname{ord} w_{65} = 1 > b + t$, shows that w_{16} and w_{35} have the same nonzero leading coefficient u. Write

$$\begin{aligned} &\delta_{16} = \varepsilon^a A(\varepsilon), && A(0) = A_0 \neq 0, \\ &\delta_{35} = \varepsilon^a D(\varepsilon), && D(0) = D_0 \neq 0, \\ &w_{16} = u\varepsilon^b(1 + o(1)), && w_{13} = v\varepsilon^{b+t}(1 + o(1)), \\ &\delta_{13} = E_0\varepsilon^q(1 + o(1)). \end{aligned} \qquad uvE_0 \neq 0.$$

It follows for the full Puiseux functions that

$$\frac{w_{35}}{w_{16}} = 1 - \frac{v}{u}\varepsilon^t + o(\varepsilon^t). \tag{28}$$

In the sum of the original z_1, z_3 rows weighted by m_1, m_3, the 13 force cancels exactly. At order h, the remaining two terms and their signed ties give

$$L_0 := m_1 m_6 A_0 + m_3 m_5 D_0 = 0. \tag{29}$$

The original z_1 row itself, using $\delta_{ij}^3 z_{ij} = \delta_{ij}/w_{ij}$, gives

$$\frac{m_6 A_0}{u} + \frac{m_3 E_0}{v} = 0, \qquad m_1 m_3 E_0 = m_3 m_5 D_0 \frac{v}{u}. \tag{30}$$

The z equations above are used at order h. We determine the next potential coefficient from the original w rows through order $-2 + t$. Put $T = m_1 + m_4 + m_5 + m_6$. The exterior w forces

other than $16, 23, 35$ have the following orders:

ij	$\operatorname{ord}(\delta_{ij}^3 w_{ij})$
$12, 24, 26$	$-1/2$
13	$-2+4t$
$14, 15, 34, 36$	$-2+9t/2$
25	$1-3s/2.$

All these orders exceed $-2+t$; for the last entry the difference is $(20-11s)/6>0$. The original w_1 row and the sum of the original w_2, w_3 rows weighted by m_2, m_3 therefore yield

$$w_1 = m_6\delta_{16}^3 w_{16} + o(\varepsilon^{-2+t}),$$
$$m_2w_2 + m_3w_3 = m_3m_5\delta_{35}^3 w_{35} + o(\varepsilon^{-2+t}).$$

The internal 23 force cancels in the second equation. The full mass centre also gives

$$m_2w_2 + m_3w_3 = -Tw_1 + O(\varepsilon^b), \qquad b-(-2+t) = \frac{19-10s}{3} > 0.$$

Consequently

$$Tm_6\delta_{16}^3 w_{16} + m_3m_5\delta_{35}^3 w_{35} = o(\varepsilon^{-2+t}). \tag{31}$$

Since $3a+b=-2$, division by $\delta_{16}^3 w_{16}$ gives

$$\left(\frac{D}{A}\right)^3 \frac{w_{35}}{w_{16}} = -\frac{Tm_6}{m_3m_5} + o(\varepsilon^t), \qquad \left(\frac{D_0}{A_0}\right)^3 = -\frac{Tm_6}{m_3m_5}.$$

Taking the cube root selected by the actual Puiseux branch and using (28), we obtain

$$\frac{D/A}{D_0/A_0} = 1 + \frac{v}{3u}\varepsilon^t + o(\varepsilon^t). \tag{32}$$

Thus (29) and (30) imply

$$\begin{aligned} m_1m_6\delta_{16} + m_3m_5\delta_{35} &= \varepsilon^a A\left(m_1m_6 + m_3m_5\frac{D}{A}\right) \\ &= \frac{1}{3}m_1m_3E_0\varepsilon^q + o(\varepsilon^q). \end{aligned} \tag{33}$$

This calculation retains every earlier correction to the full functions A and D.

We next control the complete internal triangle potential. Multiply the original z_5, z_6 rows by m_5w_{54}, m_6w_{64} and sum. The signed ties give the exact identity

$$U_{456} = \sum_{i\in\{5,6\}} m_iz_iw_{i4} - \sum_{i\in\{5,6\}}\sum_{j\in\{1,2,3\}} m_im_j\delta_{ij}^3 z_{ij}w_{i4}. \tag{34}$$

The first sum is $O(\varepsilon^2)$. In the second sum, the $16, 35$ z forces have order h, the $15, 36$ forces have order $h+(s-1)/2$, the 25 force has order $3-s/2$, and the 26 force has order $5/2$. Since w_{54}, w_{64} have order 1,

$$U_{456} = O(\varepsilon^2) + O(\varepsilon^{4s-5}) = o(\varepsilon^q), \qquad (4s-5)-q = \frac{8(s-1)}{3} > 0.$$

All remaining exterior potentials are later as well: their orders are $k, 1/2, (2-s)/2, 0$, and $k-q=(s-1)/6>0$. The unmarked 13 potential must, however, be retained at order q. Combining it with (33) yields the complete expansion

$$U = \frac{4}{3}m_1m_3E_0\varepsilon^q + o(\varepsilon^q). \tag{35}$$

Its coefficient is nonzero and $q<0$, contrary to constancy of U. This excludes the whole range $1<s<7/4$, including the first-limit window $13/8<s<7/4$.

At $s = 7/4$, (26) still holds, while $a = -1/4$, $b = -5/4$ and $h = 1$. The original z_3 row has coordinate term $O(\varepsilon^2)$ and a nonzero order-one 35 force. Its only possible order-one counterterm is 13. Thus $\operatorname{ord}\delta_{13} = 0$ and $\operatorname{ord} w_{13} = -1 > b$, so w_{16}, w_{35} again share their leading coefficient u. Equation (34) gives $U_{456} = O(\varepsilon^2)$, and all other exterior potentials are bounded. Constancy of U forces its order-$-1/4$ coefficient $L_0 = m_1m_6A_0 + m_3m_5D_0$ to vanish. But the original weighted $z_1 + z_3$ rows at order 1 give $m_1\zeta = L_0/u$, contradicting $\zeta \neq 0$.

Finally, for $7/4 < s < 2$, the same weighted $z_1 + z_3$ rows cancel the 13 force exactly. Their left side has the nonzero order-one coefficient $m_1\zeta$; their $16, 35$ forces have order $h > 1$, their 23 force has order 2, and every other force is strictly later than order 1. This is again impossible.

4.17. **No. 32: the missing 12 tie contradicts body 2.** The original w-centre of its zw triangle 456 and w-only triangle 236 makes the three 236 w-strokes maximal, with nonzero signed internal inverse-distance limits. The original body-3 z-row and its three maximal ties then give $z_2, z_3, z_6 = O(\varepsilon^2)$, without using the unmarked 12 edge. With R_i the original body-i z-force excluding pair 12, the graph gives $R_1 = O(\varepsilon^2)$, $R_2 = O(\varepsilon^2)$ and $w_{12} = o(\varepsilon^{-2})$. The exact equation in Lemma 4.8 forces a finite nonzero $e = \lim \delta_{12}$ with $e^3 = 1/(m_1 + m_2)$. Only now apply the original body-2 z-row: its 12 force and $z_2, R_2 = O(\varepsilon^2)$ force $z_{12} = O(\varepsilon^2)$. But the original signed 12 tie has

$$\delta_{12}^2 z_{12} w_{12} = o(1)$$

instead of 1. This excludes even fixed-positive-mass complex germs of No. 32.

4.18. **No. 84: optional outer edges and a four-cycle.** Its marked pairs are $14, 23, 56$, each maximal. The original relative $w_1 - w_4$, $w_2 - w_3$ and $z_5 - z_6$ rows, with their nonzero marked differences, give separately

$$(m_1 + m_4)d_{14}^3 = (m_2 + m_3)d_{23}^3 = (m_5 + m_6)d_{56}^3 = 1.$$

The four unmarked outer edges are $12, 13, 24, 34$. An edge with distinct leading w values has fading inverse distance. Equal-lead outer edges form a matching because $W_1 \neq W_4$ and $W_2 \neq W_3$. For an optional edge ij, remove its mutual interaction from the two original z rows and call the residual forces R_i^{ij}, R_j^{ij}. Their difference and the exact signed ij tie give

$$(1 - (m_i + m_j)\delta_{ij}^3)z_{ij} = R_i^{ij} - R_j^{ij}, \qquad 1 = (m_i + m_j)\delta_{ij}^3 + \delta_{ij}^2 w_{ij}(R_i^{ij} - R_j^{ij}).$$

The marked-partner forces are $O(\varepsilon^2)$, the other missing forces are $o(\varepsilon^2)$, and the 5/6 crossings are $O(\varepsilon^4)$, so $R_i^{ij} - R_j^{ij} = O(\varepsilon^2)$ and $w_{ij} = o(\varepsilon^{-2})$. If δ_{ij} is bounded, the last term tends to zero; if it diverges, division by δ_{ij}^3 contradicts $m_i + m_j > 0$. Thus every optional finite limit satisfies $e_{ij}^3 = (m_i + m_j)^{-1}$.

The seven possible matchings, including the empty one, lie in one of the two four-cycles $\{14, 23, 12, 34\}$ and $\{14, 23, 13, 24\}$. Every outer-to-5/6 inverse distance fades. The product P_{14} has order -16, so this really would be a first-limit pole. Write the chosen cycle's masses in cyclic order (a, b, c', d) and put $Q = a + b + c' + d$, $D = ac' + bd$, $B = m_5 + m_6$, $\mu = m_5m_6$, and $M = Q + B$. Taking the modulus of each permitted signed cube-root phase gives

$$|c| \le V_4 + \frac{\mu}{B^{1/3}}, \qquad V_4 = \sum_{xy \text{ on cycle}} \frac{xy}{(x + y)^{1/3}}.$$

The exact rational cycle identity

$$2D - \sum_{xy \text{ on cycle}} \frac{xy(Q - x - y)}{x + y} = (ac' - bd)^2 \left(\frac{1}{(a + b)(c' + d)} + \frac{1}{(a + d)(b + c')} \right) \ge 0 \tag{36}$$

and $\sum_{xy \text{ on cycle}} xy/(x+y) \le Q/2$, together with $\mu/B \le B/4$, give by strict cube-root concavity $M^{1/3}c < \sigma - D/3 - 3QB/4 < \sigma$. Indeed, apply $(1 + t)^{1/3} < 1 + t/3$ to each cycle edge with $t = (M - x - y)/(x + y) > 0$ and to the 56 term with $t = Q/B > 0$; the two displayed

bounds contribute $QB/6$ and $QB/12$ to the upper estimate. The positive point instead requires $\sigma \le M^{1/3}c$ by Hölder. Thus no optional finite-edge pattern survives.

4.19. **No. 36: two matches and the hidden 23 edge.** Body 1 is uncircled and has no strokes. If neither 26 nor 35 is maximal, the five circled w-leads violate the original w-centre. If only one is maximal, the original row at its outer endpoint and the matched tie put its inner endpoint at $O(\varepsilon^2)$; the still-unmarked 16 or 15 tie would then tend to zero instead of 1.

Suppose both matches $26, 35$ are maximal. Their signed ties give $z_{26}, z_{35} = O(\varepsilon^2)$ and nonzero finite limits for δ_{26}, δ_{35}. Write $W_T \neq 0$ for the common inner w-lead. Body 1 is uncircled, while all other vertices have nonzero w-leads. Every w_{1j} therefore has order -2. Since none of these edges has a w stroke, its original w force is $o(\varepsilon^{-2})$, whence

$$\delta_{1j} = o(1), \qquad \delta_{1j}^3 z_{1j} = \frac{\delta_{1j}}{w_{1j}} = o(\varepsilon^2), \qquad z_1 = o(\varepsilon^2).$$

We first show that $W_2 = W_3$. Otherwise every nonmatched force in the original z_2, z_3 rows has a w separation of order -2 and no w stroke, so it is $o(\varepsilon^2)$ by its signed tie. The matched forces are $O(\varepsilon^2)$; hence $z_2, z_3 = O(\varepsilon^2)$. The matched ties would imply $z_6, z_5 = O(\varepsilon^2)$, contradicting the exact order-one difference z_{65}. Put $W_B = W_2 = W_3$. Maximality gives $W_B \neq W_T$, and the original centre gives

$$BW_B + TW_T = 0, \qquad B = m_2 + m_3, \qquad T = m_4 + m_5 + m_6.$$

Every nonmatched edge from $\{2,3\}$ to $\{4,5,6\}$ has z-order 1, w-order -2, inverse-distance order $1/2$ and z-force order $5/2$. All z differences among these five vertices are $O(\varepsilon)$. Their original mass centre, together with $z_1 = o(\varepsilon^2)$, gives $z_a = O(\varepsilon)$ for $a = 2, \dots, 6$.

Apply Lemma 4.6 to 456, with reference vertex 4. The reference and relative-inertia terms are $O(\varepsilon^2)$. The two matched exterior terms are $O(\varepsilon^3)$, the nonmatched exterior terms from bodies 2 and 3 are $O(\varepsilon^{7/2})$, and those from body 1 are $o(\varepsilon^3)$. Therefore $U_{456} = O(\varepsilon^2)$.

The difference $z_{23} = z_{65} + O(\varepsilon^2)$ has exact order 1. Subtracting the original z_2, z_3 rows gives

$$(1 - B\delta_{23}^3) z_{23} = O(\varepsilon^2),$$

so $\delta_{23} \to e \neq 0$ and $e^3 = B^{-1}$. At order -2, the original w_2, w_3 rows contain only their matched forces: the hidden 23 force is of strictly later order, the other outer–inner forces have order $-1/2$, and the forces from body 1 are $o(\varepsilon^{-2})$. Hence

$$W_B = m_6 d_{26}^3 (W_B - W_T), \qquad W_B = m_5 d_{35}^3 (W_B - W_T),$$

which yields

$$d_{26}^3 = \frac{T}{m_6(B+T)}, \qquad d_{35}^3 = \frac{T}{m_5(B+T)}.$$

All five inverse distances incident to body 1 and all four nonmatched outer–inner inverse distances tend to zero. Together with the internal total-potential estimate, this accounts for all fifteen pairs and proves

$$c = \lim U = m_2 m_3 e + m_2 m_6 d_{26} + m_3 m_5 d_{35}.$$

With $X = m_2 m_3$, $C = B + T$ and

$$\sigma_C = X + BT + m_4 m_5 + m_4 m_6 + m_5 m_6,$$

the relevant mass identity is

$$(B - X/B)^3 - (m_2^3 + m_3^3) = \frac{X^2}{B}\left(3 - \frac{X}{B^2}\right) > 0. \tag{37}$$

Here $X/B^2 \le 1/4$, so the strict positivity follows directly. Since

$$c = Xe + m_2 m_6 d_{26} + m_3 m_5 d_{35},$$

we take the modulus of each signed root and use their specified cubes to obtain

$$C^{1/3}|c| \le X(C/B)^{1/3} + T^{1/3}\big(m_2 m_6^{2/3} + m_3 m_5^{2/3}\big).$$

Hölder's inequality and $m_5 + m_6 < T$ give

$$m_2 m_6^{2/3} + m_3 m_5^{2/3} \le (m_2^3 + m_3^3)^{1/3}(m_5 + m_6)^{2/3} < (m_2^3 + m_3^3)^{1/3} T^{2/3}.$$

Combining this with (37) and $(1+t)^{1/3} < 1 + t/3$ for $t > 0$, we find

$$\begin{aligned} C^{1/3}|c| &< X(1 + T/B)^{1/3} + T(B - X/B) \\ &< X + BT - \frac{2XT}{3B} < X + BT < \sigma_C. \end{aligned}$$

Finally $M = C + m_1$, $\sigma = \sigma_C + m_1 C$ and $\sigma_C < C^2/2$. A second application of cube-root concavity yields

$$M^{1/3}|c| < \Big(1 + \frac{m_1}{3C}\Big)\sigma_C < \sigma_C + \frac{m_1 C}{6} < \sigma.$$

At the positive point $c > 0$, so this contradicts physical Hölder.

Lemma 4.9 (The two-edge wing energy gap)**.** *Let $S = m_2+m_3+m_4$, $Q = m_1+S$, $B = m_5+m_6$, $M = Q + B$, $\mu = m_5 m_6$ and $\tau = m_2 m_3 + m_2 m_4 + m_3 m_4$. If a limit has only finite edges $14, 56$, with $d_{14}^3 = S/(m_4 Q)$ and $d_{56}^3 = 1/B$, then for every signed phase*

$$M^{1/3}|c| < \sigma - \tau - \frac{5}{6}QB < \sigma, \qquad \sigma = m_1 S + \tau + QB + \mu. \tag{38}$$

Proof. Triangle inequality first gives $|c| \le v + \mu/B^{1/3}$ with $v = m_1 m_4^{2/3}(S/Q)^{1/3}$. Use $m_4 < S$, $m_1 S \le Q^2/4$, $\mu \le B^2/4$, and $(1+t)^{1/3} < 1 + t/3$ twice to obtain $M^{1/3} v < m_1 S + QB/12$ and $M^{1/3}\mu/B^{1/3} < \mu + QB/12$. Their sum leaves the stated strict deficit. □

4.20. **No. 86: the pure w-triangle wing.** The three pure-w edges of 234 have a common z-order $\alpha \in (1, 2]$; the attachment 14 has order $\gamma \in (1, 2]$. At $(2, 2)$ maximal 14 and 24 require a missing 12 stroke; with both orders below 2 all four outer w-leads coincide, contrary to their centre. For $\alpha = 2$, $\gamma < 2$, the original body-1 z-row has the 14 force uniquely earliest at $4\gamma - 6$. No. 86: the range $1 < \alpha < 2$, $\gamma = 2$. Put $T = \{2, 3, 4\}$ and $b = 4 - 3\alpha$. The internal separation orders are (α, b) and the pendant 14 is maximal, with orders $(2, -2)$. The original diagram has w circles only at $1, 2, 3, 4$ and z circles only at $5, 6$. Therefore every pair from $\{1, 2, 3, 4\}$ to $\{5, 6\}$ has orders $(-2, -2)$, inverse-distance order 2, and z-force order 4. The pairs $12, 13$ have orders $(\alpha, -2)$, so their z-force order is $3 - \alpha/2 > 2$. The original body-1 z row consequently gives

$$z_1 = O(\varepsilon^2), \qquad z_4 = O(\varepsilon^2).$$

Use the complete reference-body identity with reference 4 and sum the original rows only at $2, 3$. Its reference and relative terms have orders at least $6 - 3\alpha$ and $4 - 2\alpha$. Its exterior terms involving body 1 have order at least $3 - \alpha/2 + b = 7 - 7\alpha/2$; those involving bodies $5, 6$ have order at least $4 + b = 8 - 3\alpha$. Thus

$$U_T = O(\varepsilon^{6-3\alpha}) + O(\varepsilon^{4-2\alpha}) + O(\varepsilon^{7-7\alpha/2}) + O(\varepsilon^{8-3\alpha}) \longrightarrow 0. \tag{39}$$

Every exponent is strictly positive on $1 < \alpha < 2$. The $12, 13$ inverse distances tend to zero, and the eight cross-block inverse distances are $O(\varepsilon^2)$. The isolated z-circled pair 56 is maximal: its leading z centre makes the two circled values distinct, and its signed tie then gives $\operatorname{ord} w_{56} = 2$. Subtracting its original z rows shows $d_{56}^3 = 1/(m_5 + m_6)$, since all exterior z-forces have order 4. Similarly the original body-1 w row and the leading w centre give

$$d_{14}^3 = \frac{m_2 + m_3 + m_4}{m_4(m_1 + m_2 + m_3 + m_4)}.$$

The complete potential limit therefore contains exactly the two finite terms $c = m_1 m_4 d_{14} + m_5 m_6 d_{56}$. Lemma 4.9 now contradicts physical Hölder.

4.21. **No. 95: the two-colour triangle wing.** Unlike No. 86, the zw triangle 234 has internal z- and w-orders both equal to 1. A nonmaximal attachment 14 would make all four circled w-leads equal, contrary to their original mass centre. Thus 14 is maximal and $z_{14} = O(\varepsilon^2)$, while the unmarked edges $12, 13$ have z-order 1, w-order -2, and inverse-distance order $1/2$. Every crossing from 1234 to the isolated z-circled pair 56 has both separation orders -2, inverse-distance order 2, and z-force order 4. The original body-1 z row consequently gives $z_1 = O(\varepsilon^2)$ and then $z_4 = O(\varepsilon^2)$.

In the reference-4 identity of Lemma 4.6, the reference term has order at least 3, the relative term order 2, the exterior terms from body 1 order $7/2$, and those from bodies 5 and 6 order 5. Hence $U_{234} = O(\varepsilon^2)$. All exterior inverse distances tend to zero except those on $14, 56$. The original body-1 w row and the w centre give $d_{14}^3 = S/(m_4 Q)$; subtracting the original z_5, z_6 rows gives $d_{56}^3 = 1/B$. Therefore Lemma 4.9 applies and yields $M^{1/3}|c| < \sigma$, contrary to physical Hölder.

4.22. **No. 77: three matched pairs cannot feed two triangles.** The two z-only triangles 123,456 have common inner orders $\alpha \in (-2, 1)$ in z and $\beta = (4-\alpha)/3 > 1$ in w. Each matching w-pair $16, 25, 34$ is maximal and has a finite signed root $d_{i,p(i)}^3 = 1/(m_i + m_{p(i)})$ by its two original w-rows. Contract each triangle's original z-rows against its relative w-coordinates: the signed inner potential has positive lower orders $(4+2\alpha)/3, \beta+2$, and $\beta+3-\alpha/2$, and tends to zero. All six nonmatched crossings fade.

For pair masses $M_i = m_i + m_{p(i)}$ and products $\mu_i = m_i m_{p(i)}$, only these three bridges remain in $c = \sum_i \mu_i d_{i,p(i)}$. Strict concavity and $\mu_i \le M_i^2/4$ give

$$M^{1/3} c \le M^{1/3}|c| < \sum_i \mu_i + \frac{1}{12}\sum_i M_i(M - M_i) = \sigma - \frac{5}{6}\sum_{i<j} M_i M_j < \sigma. \tag{40}$$

At the positive point physical Hölder gives $\sigma \le M^{1/3} c$, a contradiction on every signed matching phase. The product P_{14} would have order $16(\alpha-1)/3 < 0$; thus the argument genuinely excludes a first-limit pole.

4.23. **No. 61: opposite signed cube ratios.** Its z-only strokes are 34,56, zw edges 35,46 and maximal w-only edges 16,25. The z-only differences 34 and 56 share one nonzero lead; their w differences have the same order $\beta = (4-\alpha)/3 > 1$, $-2 < \alpha < 1$. The identity

$$w_{35} - w_{46} = w_{34} - w_{56} = O(\varepsilon^\beta) = o(\varepsilon)$$

shows that the order-one differences w_{35} and w_{46} have the same nonzero leading coefficient v. The original leading z_3, z_4, w_3, w_4 rows and the two zw-edge ties, with signed roots $A = D_{35}$, $B = D_{46}$ and nonzero shape leads x, y, v, S, H, give

$$\begin{aligned} m_4 S + m_5 A^3 x = 0, \qquad & -m_3 S + m_6 B^3 y = 0, \\ H = m_5 A^3 v, \qquad & H = m_6 B^3 v, \\ A^2 x v = 1, \qquad & B^2 y v = 1. \end{aligned} \tag{41}$$

All unmarked 36,45 terms are strictly later than these source rows throughout $-2 < \alpha < 1$. Eliminating the nonzero shape leads gives $B/A = -m_3 m_5/(m_4 m_6) < 0$, whereas the two w-rows give $(B/A)^3 = m_5/m_6 > 0$. The equations are incompatible for positive masses, independently of the complex signed-root choices.

Lemma 4.10 (The crossed four-cycle deficit)**.** *Let a, b, c', d be positive masses around a four-cycle, $Q = a + b + c' + d$, $D = ac' + bd$ and $S = ab + bc' + c'd + da$. Let the outer pair have mass $A = m_1 + m_2$ and product $X = m_1 m_2$, and set $M = A + Q$, $\sigma = X + S + D + AQ$. If only the four cycle edges and the outer pair have finite inverse-distance limits with cubes equal to the inverse sums of endpoint masses, then for every signed phase $M^{1/3}|c| < \sigma$.*

Proof. The exact rational identity

$$2D - \sum_{xy \text{ on cycle}} \frac{xy(Q-x-y)}{x+y} = (ac' - bd)^2 \left(\frac{1}{(a+b)(c'+d)} + \frac{1}{(a+d)(b+c')} \right) \geq 0 \tag{42}$$

and $\sum xy/(x+y) \leq Q/2$, $X/A \leq A/4$ combine with strict cube-root concavity after the triangle inequality to give $M^{1/3}|c| < X + S + 2D/3 + AQ/4 = \sigma - D/3 - 3AQ/4 < \sigma$. □

4.24. **No. 100: the marked outer pair on a four-cycle.** The original inner four-body rows make z-only 34,56 and w-only 36,45 maximal. Their four relative source rows give finite signed roots $d_{ij}^3 = 1/(m_i + m_j)$ for each marked cycle edge; the unmarked inner diagonals have both coordinate differences maximal and their inverse distances tend to zero. The isolated circled w-only pair 12 is also maximal and its original relative row gives $d_{12}^3 = 1/(m_1 + m_2)$. Hölder after Lemma 4.10 gives a further condition $m_3 m_5 = m_4 m_6$, but it does not avoid Lemma 4.10: $M^{1/3}|c| < \sigma$ contradicts physical Hölder at the positive point. Here order $P_{12} = -8$, so a pole was possible before that energy contradiction.

4.25. **No. 99: an unmarked outer pair on the same cycle.** The same four marked inner edges have the same finite cube roots as No. 100. Pair 12 is not marked and cannot be discarded. With R_i the original z-forces at bodies 1,2 excluding 12, all eight outer-to-inner signed ties make $R_1 = O(\varepsilon^4)$, while $w_{12} = o(\varepsilon^{-2})$. The exact row/tie equation of Lemma 4.8 forces $d_{12}^3 = 1/(m_1 + m_2)$, and every complete phase contributes to the same five finite edges and no others, contributing to c. Lemma 4.10 again gives $M^{1/3}|c| < \sigma$, contradicting Hölder only at the positive point of the same curve. The missing 12 edge changes the proof of its finite limit, not the final strict mass gap.

Lemma 4.11 (Two original-row poles cannot cancel)**.** *Suppose the only signed inverse-distance poles are 25,36 with a common order and common nonzero leading z-difference t. If the original body-2/3 z-rows have no other leading force, their signed coefficient satisfies $m_5 D_{25}^3 t = m_6 D_{36}^3 t \neq 0$. On a fixed-mass curve finite constant U would require $m_2 m_5 D_{25} + m_3 m_6 D_{36} = 0$. At positive masses these cannot hold simultaneously.*

Proof. The source rows give $(D_{36}/D_{25})^3 = m_5/m_6 > 0$, whereas cancellation of the only two potential poles gives $D_{36}/D_{25} = -m_2 m_5/(m_3 m_6) < 0$. Cubing a negative real number contradicts the positive source ratio. The roots need not be real before the potential equation imposes that ratio. □

4.26. **No. 112: the zw-edge double pole.** The original circled-component rules force maximal z-edges 15,46 and maximal w-only edges 14,23,56. The two zw edges 25,36 have common order $(1,1)$ and their inverse distances both have order -1. The original ties make the unmarked edges 12,34,26,35 fade; all other exterior inverse distances are finite or fade. The original body-2/3 z-rows each have only their respective 25/36 force at order -2. Their z-differences have the same nonzero leading coefficient because the maximal 23 and 56 differences are $O(\varepsilon^2)$. Lemma 4.11 excludes this sole maximality branch, even on a fixed-mass complex solution curve.

4.27. **No. 109: both reflected double-pole branches.** The three isolated circled w-pairs 14,23,56 are maximal. Of the z-strokes, Rule 1 and the missing 12 and 34 closing edges allow only 15,46 or 25,36 maximal. In the first case the nonmaximal 25/36 z-order is common $\alpha \in (-2,1)$, while their inverse distance has pole order $-(\alpha+2)/3$. The original body-2/3 z-rows again have only these two earliest forces; all twelve other inverse distances are finite or fade. Lemma 4.11 forbids their potential-pole cancellation at positive masses. The graph involution (1 2)(3 4), with masses permuted, maps the second maximality branch to the first. Thus no branch remains; the complex-only signed roots need not be positive.

4.28. **No. 111: the missing 56 w-stroke.** The maximal original z-edges 15,46 and w-edges 14,23 force two z-leading groups X, Y and two w-leading groups P, Q. The original w-centre gives

$$\frac{P}{Q} = -\frac{m_3 + m_4 + m_6}{m_1 + m_2 + m_5} < 0. \tag{43}$$

The edge 56 is *not* w-stroked; put $\gamma = \operatorname{ord} z_{56} \in [1, 2)$. Its tied w-force is later than the original z_2, z_3, w_5, w_6 leading rows by $3(2-\gamma)/2 > 0$. Their only two inverse-distance poles are 25,36, each of order -1. Write

$$\begin{aligned} z_{25} &= x\varepsilon + o(\varepsilon), & z_{36} &= y\varepsilon + o(\varepsilon), \\ w_{25} &= v\varepsilon + o(\varepsilon), & w_{36} &= u\varepsilon + o(\varepsilon), \\ \delta_{25} &= D_{25}\varepsilon^{-1} + o(\varepsilon^{-1}), & \delta_{36} &= D_{36}\varepsilon^{-1} + o(\varepsilon^{-1}), \end{aligned}$$

with nonzero coefficients. The leading z-coefficients at bodies 2 and 3 are both Y, while the leading w-coefficients at bodies 5 and 6 are P and Q, respectively. Their four original coordinate rows and the two signed ties give

$$\begin{aligned} Y &= m_5 D_{25}^3 x, & Y &= m_6 D_{36}^3 y, \\ P &= -m_2 D_{25}^3 v, & Q &= -m_3 D_{36}^3 u, \\ D_{25}^2 xv &= 1, & D_{36}^2 yu &= 1. \end{aligned}$$

Consequently, for $a = m_2 m_5$ and $b = m_3 m_6$,

$$YP = -aD_{25}^4, \qquad YQ = -bD_{36}^4.$$

Constancy of U forces $aD_{25} + bD_{36} = 0$. Hence

$$\frac{P}{Q} = \frac{a}{b}\left(\frac{D_{25}}{D_{36}}\right)^4 = \frac{a}{b}\left(-\frac{b}{a}\right)^4 = \frac{b^3}{a^3} > 0,$$

contrary to the original centre equation. No equality between the two zw-edge z-coefficients x, y is used. The strict force-order gap also covers the endpoint $\gamma = 1$. The 33 still-relevant published relation labels are

$$\begin{aligned} D_{\mathrm{CC}} = \{&7, 9, 11, 12, 13, 14, 15, 16, 17, 18, 19, 27, 28, 29, 30, 43, \\ &46, 47, 48, 58, 62, 76, 101, 102, 103, 104, 106, 107, 108, 110, \\ &114, 115, 117\}. \end{aligned} \tag{44}$$

Table 2 identifies the Chang–Chen proof owner and label count for every graph in the roster.

Published section	Still-relevant graph numbers	Count
§6.2	101, 102, 103, 104, 106, 107, 108, 110, 114, 115, 117	11
§6.3	7, 9, 12, 15, 17, 19, 28, 30, 76	9
§6.4	43, 48, 58	3
§6.5	46, 62	2
§6.6	47	1
§6.7	13, 29	2
§6.8	27	1
§6.9	18	1
§6.10	11	1
§6.11	16	1
§6.12	14	1
Total		33

TABLE 2. Proof-owner lookup for all 33 surviving published relation diagrams.

For each graph r in Table 2, the published Algorithm III mass locus is proper and algebraic by Chang–Chen's Theorem 6.1 [6] and the corresponding section. Choose a nonzero polynomial Q_r vanishing on that locus. A first limit of graph r then implies $Q_r(m) = 0$. A body permutation replaces it by $Q_r(\pi m)$, still nonzero; the finite orbit product is therefore a proper necessary polynomial for every labelled occurrence of r.

This places the 33 published diagrams in E_{40}. The lookup audits graph-to-section scope, not the original elimination computations or an expanded formula for every Q_r; those remain attributed to the cited theorem rather than claimed as local proofs. Together with the 7 derived labels $\mathcal{D}_{\text{new}}$ defined in the Introduction, the roster (44) accounts for all of $\mathcal{D}_{40}$.

The 24 labels historically open in the Chang–Chen partition remain historically open there, even though 7 now have positive-tier necessary mass equations and No. 52 is excluded.

5. The 24 historically unclassified diagrams

These diagrams were *not* assigned a mass equation by Chang–Chen. Of the 24, 17 have no positive-point first limit, and the other 7 require proper mass relations. Among the 17 exclusions, nine need no mass equation because a seven-product cannot diverge at their first limit; the other eight are excluded at strictly positive masses by the mechanisms below. No exclusion on arbitrary complex masses is inferred unless the indicated original-row argument itself provides one.

No.	Reason it cannot be a positive first limit
5, 67, 69, 73, 80	connectivity lower bounds make each seven-product bounded
71	remaining P_{56} case needs original body-1/2 rows
65	its first-limit seven-products are bounded
74, 75	five-body zw-clique plus satellite order bound
6	all pair distances shrink, forcing $I \to 0$
52	both remaining valuation endpoints fail the original-row and Hölder tests
64	original K4 contraction and strict Hölder gap
79	bridge order/shape sector and all-wing Hurwitz certificate
81	three finite pair energies contradict physical Hölder
82	original body-1 torque has one earliest term
89	original wing torque forces an impossible pair-root sum
90	original upper torque has one earliest crossed term
	17 diagrams in 12 proof groups

Table 3. Proof-owner map for the historical-open graphs removed only from the positive first-limit obligation.

The 17 first-limit exclusions.

Lemma 5.1 (Connectivity bound for seven-products)**.** *For a pair ij let L_{ij} be 2 on a zw edge, 0 on a single-colour stroke, -1 on an unmarked pair joined by a stroke path in at least one colour, and -4 on an unmarked pair joined by no such path in either colour. In particular, the same conservative bound -1 is used when an unmarked pair is joined by paths in both colours. Then every germ obeying the diagram has*

$$\operatorname{ord}(z_{ij}w_{ij}) \geq L_{ij}, \qquad \operatorname{ord} P_{ij} \geq L_{ij} + \sum_{\{k,\ell\}\cap\{i,j\}=\varnothing} L_{k\ell}. \tag{45}$$

If all fifteen sums on the right are nonnegative, the diagram cannot be an unbounded-seven-product first limit.

Proof. A stroke path of one colour makes every consecutive separation in the opposite colour have order at least 1, and hence so does the endpoint difference. All position orders are at least -2 in either colour. The stated L_{ij} follow, using the signed tie on an edge. The product P_{ij} contains the pair ij and its six disjoint pairs, so its order is the sum of their squared-distance orders. □

5.1. **No. 5: two triangles and bounded products.** For the pinned prism graph, the fifteen sums in Lemma 5.1 have minimum 0, attained only at $P_{15}, P_{16}, P_{24}, P_{26}, P_{34}, P_{35}$.

Every other sum is strictly positive. No seven-product can diverge, so this graph cannot be a first limit; the bound does not exclude complex germs elsewhere.

5.2. **No. 6: all distances vanish.** Every pair is a zw edge, so all squared distances $z_{ij}w_{ij}$ tend to zero. The exact centred inertia identity gives

$$MI = \sum_{i<j} m_i m_j z_{ij} w_{ij} \longrightarrow 0.$$

This contradicts $I = c > 0$ on a curve through a positive configuration, independently of a choice of signed inverse-distance phase.

5.3. **No. 67: a bounded K4 first limit.** The fifteen sums of Lemma 5.1 are nonnegative for this zw-K4 diagram. Their minimum 0 occurs only at P_{36}, P_{46}, P_{56}; the other twelve products have strictly positive lower bounds. Hence no first-limit seven-product pole is compatible with the graph.

5.4. **No. 69: a bounded K4 first limit.** The additional w-circle pattern does not weaken any of the pairwise lower bounds used in Lemma 5.1. All fifteen product sums are nonnegative, with equality only at P_{36}, P_{46}, P_{56}. Thus No. 69 is not a first limit, without claiming nonexistence of complex-only configurations of that pattern.

5.5. **No. 73: a bounded K4 first limit.** The zw-K4 edge and stroke-path bounds likewise give nonnegative order on every seven-product, with the only zero lower bounds at P_{36}, P_{46}, P_{56}. The circles of No. 73 affect leading coordinates, not this valid distance lower bound; a pole at a first limit is impossible.

5.6. **No. 80: the prism with three matches.** The six zero lower bounds from Lemma 5.1 occur at $P_{14}, P_{15}, P_{24}, P_{26}, P_{35}, P_{36}$. The nine other products have positive lower bounds, so none of the fifteen can diverge on a germ with the No. 80 strokes and circles.

5.7. **No. 71: the one missing connectivity order.** The graph-only bound is nonnegative except for $\operatorname{ord} P_{56} \geq -1$. If this product had a pole, put $b = \operatorname{ord} w_{16}$ and $b' = \operatorname{ord} w_{25}$. The circle normalization gives $b, b' \geq -2$, while the z stroke path 1–6–5–2 gives $\operatorname{ord} z_{12} \geq 1$ and hence $\operatorname{ord} r_{12}^2 \geq -1$. Therefore the graph lower bound $4 + \operatorname{ord} r_{12}^2 + 2(1+b) + 2(1+b')$ can be negative only when $b + b' < -7/2$, which with $b, b' \geq -2$ forces both $b, b' < -3/2$. Define the original signed z force $Z_{12} = \delta_{12}^3 z_{12}$. Subtracting the exact original body-2 z row from the body-1 row gives

$$z_{12} = (m_1 + m_2) Z_{12} + \sum_{j=3}^{6} m_j \big(\delta_{1j}^3 z_{1j} - \delta_{2j}^3 z_{2j}\big).$$

Here $\operatorname{ord} z_{16} = (4-b)/3 > 11/6$ and likewise $\operatorname{ord} z_{25} > 11/6$. The 16 and 25 force orders are $-(2+4b)/3$ and $-(2+4b')/3$, both greater than $4/3$; all other satellite force orders are $-(1+3b)/2$ or $-(1+3b')/2$, both greater than $7/4$. Thus the sum is $o(\varepsilon^{4/3})$. But $z_{12} = z_{16} + z_{65} + z_{52}$ has order 1 because the 56 edge has a nonzero z lead and the other two summands have later order. Since $m_1 + m_2 > 0$, $\operatorname{ord} Z_{12} = 1$. The signed 12 tie gives $\operatorname{ord} Z_{12} = -(\operatorname{ord} z_{12} + 3 \operatorname{ord} w_{12})/2$, so $\operatorname{ord} w_{12} = -1$ and $\operatorname{ord}(z_{12} w_{12}) = 0$. This raises the sole negative graph bound for P_{56} to 0; therefore it cannot have a first-limit pole.

5.8. **No. 65: a circled pair outside a zw-K4.** The isolated circled w-pair 12 has maximal w-stroke and z_{12} order 2. All cross w-separations have order -2. If a cross z-separation has order $\alpha \le 2$, its contribution to the original outer-body z-force has order $3 - \alpha/2 \ge 2$. The original pair z-rows put both outer z-positions at order at least 2. Mass centre and order-one K4 z-separations put each K4 z-position at order at least 1, improving every cross squared-distance bound to -1. Using 2 on K4 edges and 0 on edge 12, the resulting fifteen seven-product lower bounds have minimum 0. Thus No. 65 cannot be an unbounded-product first limit; the coarse connectivity bound alone would have been negative.

Lemma 5.2 (An external body at a zw-K5)**.** *Suppose five bodies form a zw clique and p is the remaining body. The centre-of-mass and original p-row bounds give* $\operatorname{ord}(z_{pk}w_{pk}) \ge -1$ *for every clique body k. Consequently the seven-products on clique pairs have order at least 5, and those on satellite pairs at least 11.*

Proof. Internal clique differences have order 1 in both colours. Writing clique positions relative to their mass centre, each relative coordinate has order at least 1. If at least one colour of the satellite position has order at least 1, the other has order at least -2, yielding the required pair-square bound. If both satellite orders are below 1, all five crossed differences share the corresponding leading order. If their sum $A + B$ were negative, every signed inverse distance would have order $-(A + B)/2 > 0$, so every term of the original satellite z-row would have order $A - 3(A + B)/2 > A$. Its nonzero left-hand coordinate of order A could not be balanced, regardless of signed leading-force cancellation. Thus $A + B \ge 0$. For a product based on a clique pair, that pair contributes 2 and its complementary four bodies give three clique pairs and three satellite pairs, so the lower bound is $2 + 3 \cdot 2 - 3 = 5$. For a product based on a satellite pair, the complementary four bodies lie in the clique, giving the lower bound $-1 + 6 \cdot 2 = 11$. □

5.9. **No. 74: the isolated satellite.** Its ten inner edges form a zw-K_5 with body 1 as the satellite. Lemma 5.2 gives order at least 5 for every seven-product based inside the clique and at least 11 for every satellite product. No. 74 cannot be a first limit.

5.10. **No. 75: the extra w-stroke satellite.** The added w-stroke does not invalidate the same K_5 relative-centre and original satellite-row estimate: all satellite squared-distance orders remain at least -1. Applying Lemma 5.2 again gives a minimum product order of 5, excluding a first-limit pole on No. 75.

5.11. **No. 64: the circled pair and its energy gap.** The energy argument is different for No. 64: write $A = m_1 + m_2$, $B = \sum_{i=3}^{6} m_i$, $\mu = m_1 m_2$ and $M = A + B$. Its original pair row forces $\delta_{12}^3 \to 1/A$; contracting only the four K4 z-rows against their centred w-differences makes their signed inner potential tend to zero. The cross potentials also fade, leaving $c = \mu d$ with $d^3 = 1/A$. Strict cube-root concavity, $\mu \le A^2/4$, and positive internal pair weights imply

$$M^{1/3}c \le \mu(1 + B/A)^{1/3} < \mu + \frac{\mu B}{3A} \le \mu + \frac{AB}{12} < \sigma, \tag{46}$$

contrary to $\sigma \le M^{1/3}c$ at a positive point. This excludes only the positive-point tier; the complex signed K4 configuration itself is not claimed impossible.

5.12. **No. 81: the three finite-pair energy gap.** The original opposite-colour rows give finite inverse distances on pairs 12,34,56 with cubes $1/M_P$ for their respective total pair masses. The two isolated satellites' orders and the original pair torques remove all other finite inverse distances. Let μ_P be the product of the two masses in pair P, and let M be the six-body total mass. Then $|c| \le \sum_P \mu_P / M_P^{1/3}$. Since $\mu_P \le M_P^2/4$ and $(1 + x)^{1/3} < 1 + x/3$ for $x > 0$,

$$M^{1/3}|c| < \sum_P \mu_P + \frac{1}{12}\sum_P M_P(M - M_P) = \sigma - \frac{5}{6}\sum_{P<Q} M_P M_Q < \sigma. \tag{47}$$

At the positive point on the same curve Hölder instead gives $\sigma \le M^{1/3}c$. Hence No. 81 cannot occur on a positive-point curve, without claiming a complex-only diagram exclusion.

Lemma 5.3 (Signed-pair noncancellation)**.** *Let $m_5, m_6 > 0$, and let $Z_5, Z_6, W_j, D_{j5}, D_{j6}$ be nonzero leading coefficients satisfying the pair-centre equation and the two leading signed crossed ties*

$$m_5 Z_5 + m_6 Z_6 = 0, \qquad -D_{j5}^2 Z_5 W_j = 1, \qquad -D_{j6}^2 Z_6 W_j = 1.$$

Then

$$(m_5 D_{j5})^2 - (m_6 D_{j6})^2 = -\frac{m_5^3 + m_6^3}{m_5 Z_5 W_j} \neq 0. \tag{48}$$

Consequently $m_5 D_{j5} + m_6 D_{j6} \neq 0$ for every choice of complex square-root signs satisfying these ties.

Proof. The pair-centre equation gives $Z_6 = -m_5 Z_5/m_6$. Therefore

$$(m_5 D_{j5})^2 - (m_6 D_{j6})^2 = -\frac{m_5^2}{Z_5 W_j} + \frac{m_6^2}{Z_6 W_j} = -\frac{m_5^3 + m_6^3}{m_5 Z_5 W_j}.$$

Its numerator is strictly positive and its denominator is nonzero. The two terms cannot cancel. □

5.13. **No. 82: the earliest satellite torque.** The exact graph has w-edges $23, 24, 34$, w-circles at 3 and 4, a z-edge 56, and z-circles at 5 and 6. There are no other circles or edges. We prove that the coordinate rows force the small coordinates needed to prevent cancellation in the body-1 torque.

We use the diagram normalization in which coordinates and force terms have order at least -2. A circle or stroke means that its corresponding coordinate or force has order exactly -2; an absent circle or stroke means strictly larger order. For any pair, writing $\alpha = \operatorname{ord} z_{ij}$ and $\beta = \operatorname{ord} w_{ij}$, the signed tie gives

$$\operatorname{ord} \delta_{ij} = -\frac{\alpha+\beta}{2}, \qquad \operatorname{ord}(\delta_{ij}^3 z_{ij}) = -\frac{\alpha+3\beta}{2}, \qquad \operatorname{ord}(\delta_{ij}^3 w_{ij}) = -\frac{3\alpha+\beta}{2}. \tag{49}$$

Thus, when $\operatorname{ord} w_{ij} = -2$, an absent w-stroke forces $\operatorname{ord} z_{ij} < 2$; the colour-dual assertion also holds.

Proposition 5.4. *For positive masses, No. 82 cannot occur as a singular Puiseux germ of the normalized equations.*

Proof. Write W_3, W_4 for the nonzero order--2 coefficients of w_3, w_4, and Z_5, Z_6 for those of z_5, z_6. Summing the original mass-weighted w-rows over 234 cancels the internal forces. Every exterior w-force is later than order -2, and body 2 has no w-circle. Similarly, summing the original mass-weighted z-rows over 56 cancels their mutual force. Hence

$$m_3 W_3 + m_4 W_4 = 0, \qquad m_5 Z_5 + m_6 Z_6 = 0. \tag{50}$$

Positive masses give $W_3 \neq W_4$ and $Z_5 \neq Z_6$. Since body 2 is uncircled, the three w-differences on 234 have order -2, as does z_{56}. Their strokes and (49) imply

$$\begin{gathered} \operatorname{ord} z_{23} = \operatorname{ord} z_{24} = \operatorname{ord} z_{34} = 2, \qquad \operatorname{ord} w_{56} = 2, \\ \operatorname{ord} \delta_{23} = \operatorname{ord} \delta_{24} = \operatorname{ord} \delta_{34} = \operatorname{ord} \delta_{56} = 0. \end{gathered} \tag{51}$$

First we determine the small coordinates from the original rows. Because body 1 has no w-circle, w_{13}, w_{14} have order -2. The missing w-strokes force $\operatorname{ord} z_{13}, \operatorname{ord} z_{14} < 2$. Their difference has order 2, so both have a common order $a \in (-2, 2)$. In the original body-3 z-row, the two internal forces are $O(\varepsilon^2)$, the 13 force has order $3 - a/2 > 2$, and the 35 and 36 forces have order 4. Therefore $z_3 = O(\varepsilon^2)$. The internal differences in (51) now give

$$z_2, z_3, z_4 = O(\varepsilon^2), \qquad \operatorname{ord} z_1 = a \in (-2, 2). \tag{52}$$

For each $j = 1, 2$, both z_{j5} and z_{j6} have order -2. Their missing z-strokes force $\operatorname{ord} w_{j5}, \operatorname{ord} w_{j6} < 2$. Since their difference has order 2, these orders agree; denote the common value by $b_j \in (-2, 2)$. In the original body-5 w-row, the 56 force is $O(\varepsilon^2)$, the forces to $j = 1, 2$ have orders $3 - b_j/2 > 2$, and the forces to 3 and 4 have order 4. It follows that

$$w_5, w_6 = O(\varepsilon^2), \qquad \operatorname{ord} w_j = b_j \quad (j = 1, 2). \tag{53}$$

In particular, write

$$z_1 = Z_1 \varepsilon^a (1 + o(1)), \qquad w_j = W_j \varepsilon^{b_j} (1 + o(1)) \quad (j = 1, 2),$$

where Z_1, W_1, W_2 are nonzero.

Next we show that $b_1 = b_2$. The exact torque obtained from the original mass-weighted rows at 5 and 6 is

$$\sum_{k \in \{5,6\},\ j \in \{1,2,3,4\}} m_k m_j \delta_{kj}^3 (w_k z_j - z_k w_j) = 0. \tag{54}$$

For $j = 1, 2$ and $k = 5, 6$, the signed ties give

$$\delta_{kj} = D_{jk} \varepsilon^{1 - b_j/2} (1 + o(1)), \qquad -D_{jk}^2 Z_k W_j = 1.$$

By (52)–(53), $w_k z_j$ is strictly later than $z_k w_j$. Thus the total contribution of satellite j to (54) is

$$m_j (m_5 D_{j5} + m_6 D_{j6}) \varepsilon^{1 - b_j/2} + o(\varepsilon^{1 - b_j/2}). \tag{55}$$

Its coefficient is nonzero by (50) and Lemma 5.3. The remaining four terms, to bodies 3 and 4, have order 2. Since $0 < 1 - b_j/2 < 2$, unequal values of b_1, b_2 would leave exactly one earliest nonzero satellite contribution. Consequently

$$b_1 = b_2 =: b \in (-2, 2).$$

Finally, put $q = \operatorname{ord} w_{12} \geq b$. The signed tie ensures that q is finite. By (52), $\operatorname{ord} z_{12} = a$ and

$$\delta_{12} = D_{12} \varepsilon^{-(a+q)/2} (1 + o(1)), \qquad D_{12} \neq 0.$$

The exact body-1 torque is

$$\sum_{j=2}^{6} m_j \delta_{1j}^3 (z_1 w_j - w_1 z_j) = 0. \tag{56}$$

Crucially, $z_2 = O(\varepsilon^2)$ and $a < 2$ give

$$z_1 w_2 - w_1 z_2 = Z_1 W_2 \varepsilon^{a+b} (1 + o(1)).$$

The 12 term therefore has the nonzero leading coefficient $m_2 D_{12}^3 Z_1 W_2$ and order

$$e = b - \frac{a}{2} - \frac{3q}{2}.$$

This nonvanishing also holds when $q = b$: the potential cancellation is prevented by the already established smallness of z_2.

For $j = 3, 4$, the inverse distance has order $1 - a/2$, while the torque determinant has order $a - 2$; the corresponding torque term has order $1 - a/2$. For $j = 5, 6$, the analogous order is $1 - b/2$. Their differences from e are

$$1 - \frac{a}{2} - e = \frac{2 + b}{2} + \frac{3}{2}(q - b) > 0, \qquad 1 - \frac{b}{2} - e = \frac{2 + a}{2} + \frac{3}{2}(q - b) > 0.$$

Thus the 12 term is uniquely earliest in (56), a contradiction. □

5.14. **No. 89: a forced nonzero crossed root sum.** Let $a = \operatorname{ord} z_1$ and $b = \operatorname{ord} w_1$ lie in $(-2, 2)$ on its isolated satellite. The original z_2, z_3 rows first give $z_2, z_3 = O(\varepsilon^2)$: their internal terms have order 2, their opposite-wing terms order 4, and their body-1 terms order $3 - \operatorname{ord} z_{1j}/2 > 2$. The $24, 34$ w strokes make $z_4 = O(\varepsilon^2)$. The dual original w_5, w_6 rows similarly give $w_5, w_6 = O(\varepsilon^2)$. Thus z_{14}, w_{14} have respective orders a, b. The 14 tie gives $\operatorname{ord} \delta_{14} = -(a+b)/2$; its force in the original body-1 z row has order $a - 3(a+b)/2$. Every competing wing force is later by at least $3(2+b)/2$ or $(2+a)/2$; the body-1 w row has the symmetric positive gaps $(2+b)/2$ and $3(2+a)/2$. Since the 14 leading coefficient is nonzero by its signed tie, its unique force must balance the coordinate term of order a in the z row (and order b in the w row). Hence $a + b = 0$. In the original 456 wing torque, the satellite-to-56 contribution then has order $1 + a/2$; the other wing and crossed contributions have orders at least 2 and $2 + a$, strictly later. Hence the leading coefficient must satisfy $m_5 D_{15} + m_6 D_{16} = 0$, contrary to Lemma 5.3. This argument uses no reality of the crossed roots.

5.15. **No. 90: the unique 41 crossing.** The exact graph has w-edges $12, 13, 23$ and w-circles at 2 and 3, together with z-edges $45, 46, 56$ and z-circles at 5 and 6. There are no crossed strokes, and bodies 1 and 4 are uncircled.

Proposition 5.5. *For positive masses, No. 90 cannot occur as a singular Puiseux germ of the normalized equations.*

Proof. Let W_2, W_3 denote the nonzero order--2 coefficients of w_2, w_3, and Z_5, Z_6 those of z_5, z_6. Summing the original mass-weighted w-rows over 123 and the mass-weighted z-rows over 456 gives

$$m_2 W_2 + m_3 W_3 = 0, \qquad m_5 Z_5 + m_6 Z_6 = 0.$$

There are no crossed strokes, so every exterior force in these sums is later than order -2. Positivity makes the two differences of circled coefficients nonzero. As bodies 1 and 4 are uncircled, all three marked edges in each triangle are maximal. By (49),

$$\begin{aligned} &\operatorname{ord} w_{12} = \operatorname{ord} w_{13} = \operatorname{ord} w_{23} = -2, \qquad \operatorname{ord} z_{12} = \operatorname{ord} z_{13} = \operatorname{ord} z_{23} = 2, \\ &\operatorname{ord} z_{45} = \operatorname{ord} z_{46} = \operatorname{ord} z_{56} = -2, \qquad \operatorname{ord} w_{45} = \operatorname{ord} w_{46} = \operatorname{ord} w_{56} = 2, \end{aligned} \tag{57}$$

and all six internal inverse distances have order zero.

Since w_2, w_3 have order -2 and w_4 has larger order, the missing 24 and 34 w-strokes force $\operatorname{ord} z_{24}, \operatorname{ord} z_{34} < 2$. The order-two internal differences make their common order equal to

$$a := \operatorname{ord} z_{41} \in (-2, 2).$$

Likewise, the missing 15 and 16 z-strokes and the order-two internal w-differences give

$$b := \operatorname{ord} w_{41} \in (-2, 2).$$

In the original body-2 z-row, the internal forces are $O(\varepsilon^2)$, the 24 force has order $3 - a/2 > 2$, and the 25 and 26 forces have order 4. Hence $z_2 = O(\varepsilon^2)$, and the internal differences give

$$z_1, z_2, z_3 = O(\varepsilon^2), \qquad w_4, w_5, w_6 = O(\varepsilon^2). \tag{58}$$

For the second assertion, use the original body-5 w-row: its internal forces are $O(\varepsilon^2)$, the 51 force has order $3 - b/2 > 2$, and the 52 and 53 forces have order 4; then use the internal w-differences. In particular, $z_4 = z_{41}(1 + o(1))$ and $w_1 = -w_{41}(1 + o(1))$.

Summing the original mass-weighted torques over 456 cancels every internal interaction and yields the exact nine-term identity

$$\sum_{k \in \{4,5,6\},\ l \in \{1,2,3\}} m_k m_l \delta_{kl}^3 (w_k z_l - z_k w_l) = 0. \tag{59}$$

The identity

$$w_4 z_1 - z_4 w_1 = z_{41} w_{41} + z_1 w_{41} - z_{41} w_4$$

and (58) show that the latter two terms have orders at least $2+b$ and $a+2$, both strictly later than $a+b$. Therefore the 41 contribution is

$$m_1 m_4 \delta_{41}^3 (w_4 z_1 - z_4 w_1) = m_1 m_4 \delta_{41} (1 + o(1)), \tag{60}$$

with a nonzero coefficient at order $-(a+b)/2$.

The other eight crossings fall into exactly three classes:

(k,l)	count	$\operatorname{ord}\delta_{kl}$	$\operatorname{ord}(w_k z_l - z_k w_l)$	torque order
$(4,2),(4,3)$	2	$1-a/2$	$a-2$	$1-a/2$
$(5,1),(6,1)$	2	$1-b/2$	$b-2$	$1-b/2$
$k\in\{5,6\},\ l\in\{2,3\}$	4	2	-4	2

For instance, in the first class $w_4 z_l = O(\varepsilon^4)$, whereas $z_4 w_l$ has order $a-2<0$, so there is no cancellation in the determinant. The other classes follow in the same way from (58). Their order gaps above (60) are, respectively,

$$1+\frac{b}{2}>0, \qquad 1+\frac{a}{2}>0, \qquad 2+\frac{a+b}{2}>0.$$

Thus the 41 contribution is uniquely earliest in the complete torque identity, a contradiction. □

5.16. **No. 79: the bridge shape and Hurwitz sector.** Put $\alpha = m_2 m_6$ and $\beta = m_3 m_5$. Let Z_{26}, Z_{35} and W_{26}, W_{35} be the leading coefficients of the two bridge coordinate differences. Summing the original mass-weighted z rows over either triangle cancels all internal forces; at the first bridge order only 26 and 35 remain. The same holds for the w rows. Thus

$$\alpha d_{26}^3 Z_{26} + \beta d_{35}^3 Z_{35} = 0, \qquad \alpha d_{26}^3 W_{26} + \beta d_{35}^3 W_{35} = 0.$$

The original two-triangle row reduction identifies the common projective ratio $r = Z_{26}/Z_{35} = W_{26}/W_{35}$. Hence, in this signed-distance orientation,

$$r = -\frac{\beta d_{35}^3}{\alpha d_{26}^3}, \qquad c = \alpha d_{26} + \beta d_{35} > 0, \qquad K = \alpha d_{26}^3 + \beta d_{35}^3 = \frac{M_A M_B}{M} > 0.$$

For real $r<0$, the bridge roots have the same real sign; $K>0$ and $c>0$ make both positive. Weighted Hölder gives $c^3 \le (\alpha+\beta)^2 K$. Since the total pair sum satisfies $\sigma > M_A M_B > \alpha+\beta$, we obtain

$$M c^3 \le M(\alpha+\beta)^2 K = M_A M_B (\alpha+\beta)^2 < \sigma^3,$$

contrary to physical Hölder $\sigma^3 \le M c^3$. For $0<r<1$, the first triangle has masses (a,b,d) and shape sextic $F_{a,b,d}(r)=0$. Here $U=(a+b)r+d$, $V=br+a+d$, and $W=a(1-r)$ are positive with $V-U=W>0$. The identity

$$d^2 U - b^2 V = (d^2-b^2)U - b^2 W$$

shows that $d^2U - b^2V < 0$ if $d \le b$. Thus its signed shape numerator

$$\mathcal{Q} = a^2 W (d^2 U - b^2 V) - b^2 d^2 U V$$

is negative, making $F_{a,b,d}(r) = \mathcal{Q}^3 - 27 a^4 b^4 d^4 U^2 V^2 W^2 < 0$, a contradiction. Therefore $d>b$. With the second triangle masses $\lambda(A,d,b)$, this gives $p=\alpha=\lambda b^2 < n = \beta = \lambda d^2$. The case $r>1$ is reduced to $0<1/r<1$ by the bridge and wing swap. Now take the positive-real phase of c and write the signed bridge distances as $x>0$ and $-y<0$. With $K = px^3 - ny^3 > 0$ and $c = px - ny > 0$, put $q = ny/(px) \in (0,1)$. Then

$$p^2 K - c^3 = p^3 x^3 \left[3q(1-q) + (1 - p^2/n^2) q^3\right] > 0. \tag{61}$$

But $\sigma > M_A M_B > p$ makes $\sigma^3 > M p^2 K > M c^3$, contrary to physical Hölder. Thus a positive-point first limit requires a nonreal r.

For a triangle with positive masses (a,b,d), put

$$u=(a+b)r+d, \qquad v=br+a+d, \qquad w=a(1-r), \qquad \tau=\frac{w}{u}, \quad \nu=\frac{w}{v}, \quad R=-1+\frac{a^2}{b^2}\nu-\frac{a^2}{d^2}\tau.$$

The cleared signed shape polynomial is

$$\mathcal{Q} = d^2a^2uw - a^2b^2vw - b^2d^2uv = b^2d^2uvR, \qquad F_{a,b,d}(r) = \mathcal{Q}^3 - 27a^4b^4d^4u^2v^2w^2.$$

Thus, whenever $uv \neq 0$, its root equation is the exact identity

$$\frac{F_{a,b,d}(r)}{b^6d^6u^3v^3} = R^3 - \frac{27a^4}{b^2d^2}\tau\nu = 0. \tag{62}$$

This is the signed triangle equation obtained from its original force rows and ties.

For equal bridge weights $\alpha = \beta = h > 0$, conjugacy at a complex point at infinity does *not* follow from conjugacy on the physical arc. Instead, the original cluster w-momentum and constant potential give, on the same complex solution curve and hence at its Puiseux limit,

$$h(d_{26} + d_{35}) = c > 0, \qquad h(d_{26}^3 + d_{35}^3) = K = \frac{M_AM_B}{M} > 0.$$

With $s = c/h$ and $T = K/h$, the two signed leading roots are zeros of the real-coefficient polynomial

$$z^2 - sz + \frac{s^3 - T}{3s} = 0.$$

Its discriminant is $(4T - s^3)/(3s) \in \mathbb{R}$. The two bridge coefficients are nonzero because the corresponding maximal signed-distance strokes persist on an actual No. 79 first limit. Thus they are either both real, yielding a real $r = -d_{35}^3/d_{26}^3$, which was just excluded, or they are a nonreal conjugate pair, giving $|r| = 1$. No conjugation invariance of the complex branch is used.

Write the two triangle mass triples as (a, b, d) and $\lambda(A, d, b)$, with $\lambda > 0$, and set $H = a+b+d$, $J = A + b + d$, and $D = b^2 + d^2$. Then $M_A = H$, $M_B = \lambda J$, and the common bridge weight is $h = \lambda bd$. Let

$$I_A = ab + ad + bd, \qquad I_B = \lambda^2(Ad + Ab + bd), \qquad \Gamma = 1 + 2\sqrt{\frac{I_AI_B}{M_AM_B}} > 1.$$

The total pair-mass sum is $\sigma = M_AM_B + I_A + I_B$, so $\sigma \geq M_AM_B\Gamma$. Also $M_AM_B = \lambda HJ > 4\lambda bd = 4h$. Write

$$d_{26} = m(1 + ix), \qquad d_{35} = m(1 - ix),$$
$$m = \frac{c}{2h} > 0, \qquad f = 1 - 3x^2 = \frac{K}{2hm^3} > 0,$$
$$r = -\left(\frac{1 - ix}{1 + ix}\right)^3.$$

Physical Hölder, $\sigma^3 \leq Mc^3$, and $K = M_AM_B/M$ give

$$f = \frac{4h^2M_AM_B}{Mc^3} \leq \frac{4h^2M_AM_B}{\sigma^3} \leq \frac{4h^2}{(M_AM_B)^2\Gamma^3} < \frac{1}{4\Gamma^3} < \frac{1}{4}.$$

Hence $1/4 < x^2 < 1/3$. For $s = x^2$,

$$\operatorname{Re} r = -\frac{(1 - s)(1 - 14s + s^2)}{(1 + s)^3}, \qquad \frac{d}{ds}\operatorname{Re} r = \frac{6(s - 3)(3s - 1)}{(s + 1)^4} > 0 \quad (1/4 < s < 1/3).$$

At $s = 1/4$ this real part is $117/125$. Since $|r| = 1$, $\delta := |1 - r|$ therefore satisfies $\delta^2 = 2(1 - \operatorname{Re} r) < 16/125$. The exact denominator norms are

$$|u|^2 = H^2 - (a + b)d\delta^2, \qquad |v|^2 = H^2 - b(a + d)\delta^2.$$

Each subtracted mass product is at most $H^2/4$; consequently $|u|, |v| > q_0H$ for $q_0 = 24/25$. The same Hölder bound, retaining Γ, gives

$$f < \frac{4b^2d^2}{H^2J^2\Gamma^3}, \qquad \delta = \frac{2f}{(1 + x^2)^{3/2}} < \frac{64b^2d^2}{5\sqrt{5}\,H^2J^2\Gamma^3} < \frac{6b^2d^2}{H^2J^2\Gamma^3}.$$

Put $\kappa = D/(bd) \geq 2$ and $z = a^3\delta/(bdH)$. Equation (62) gives

$$|R| \geq 1 - \frac{\kappa z}{q_0}, \qquad \left|\frac{27a^4}{b^2d^2}\tau\nu\right| < \frac{27z^2}{q_0^2}.$$

If $z \leq q_0/(4\kappa)$, the first bound gives $|R| \geq 3/4$, whereas the second is at most $27/(16\kappa^2) \leq 27/64 = (3/4)^3$, contradicting the cubic equation (62). Every root must therefore have $z > q_0/(4\kappa)$. Combining this with the upper bound on δ yields

$$J^2\Gamma^3 < 25D\left(\frac{a}{H}\right)^3.$$

The wing-swap identity $F_{A,d,b}(r) = r^6 F_{A,b,d}(r^{-1})$ and $|r| = 1$ give the same argument for the other triangle, so $H^2\Gamma^3 < 25D(A/J)^3$. Multiplying and using $\Gamma > 1$ proves

$$\frac{H^5J^5}{a^3A^3} < 625D^2. \tag{63}$$

The nonnegative identity

$$108(1+t)^5 - 3125t^3 = (2t-3)^2(27t^3 + 216t^2 + 76t + 12) \tag{64}$$

applied with $t = a/(b+d)$ and $A/(b+d)$ gives $H^5/a^3, J^5/A^3 \geq (3125/108)(b+d)^2$. Since $(b+d)^2 > D$, their product is strictly greater than $625D^2$, contradicting (63). For unequal bridge weights put $P = \alpha+\beta$, $p = \alpha/P = (1-\theta)/2$, $q = \beta/P = (1+\theta)/2$, where $\theta = (\beta-\alpha)/P$; use the graph involution to take $\theta > 0$. Let $m = c/P > 0$ and parameterize the surviving nonreal bridge-root half-sheet by

$$d_{26} = m(1 + q\theta t + iq\sqrt{L}), \qquad d_{35} = m(1 - p\theta t - ip\sqrt{L}), \qquad L = 3\theta^2t^2 + 6t, \quad t > 0.$$

Here t and $\sqrt{L}$ parameterize the real and imaginary parts of the complex leading coefficients; they are not asserted to be real functions along the complex curve. The fixed real values of c and K make this parameterization valid at the selected Puiseux branch without conjugating it. Put

$$N_1 = |d_{26}/m|^2 = (1+q\theta t)^2 + q^2L, \qquad N_2 = |d_{35}/m|^2 = (1-p\theta t)^2 + p^2L,$$

and $f_0(t) = K/(Pm^3)$. Direct expansion of the bridge cubic moment and squared moduli gives

$$\begin{aligned} N_1 - N_2 &= 4\theta t(2 + \theta^2 t) > 0, \\ |r|^2 &= \frac{q^2N_2^3}{p^2N_1^3}, \\ q^2N_2^3 - p^2N_1^3 &= \frac{\theta f_0(t)}{4}\Big[4 + (1-\theta^2)\big(4\theta^2(1+\theta^2)t^3 \\ &\qquad + (15\theta^2+9)t^2 + 12t\big)\Big] > 0. \end{aligned}$$

Thus $|r| > 1$ on this half-sheet despite $N_1 > N_2$. The same cubic moment yields

$$f_0(t) = 1 - \frac{1-\theta^2}{2}t(3 + 2\theta^2 t)^2.$$

Let $\eta = \sqrt{1-\theta^2}$. If I_A, I_B are the two internal pair-mass sums, then M_AM_B contains the four products $\alpha, \beta, m_2m_5, m_3m_6$, while I_A+I_B contains $m_2m_3+m_5m_6$. The identities $m_2m_5\, m_3m_6 = \alpha\beta = m_2m_3\, m_5m_6$ and AM–GM give $M_AM_B > P(1+\eta)$ and $I_A + I_B \geq 2\sqrt{\alpha\beta} = P\eta$. Since $\sigma = M_AM_B + I_A + I_B$, the function $x/(x+P\eta)^3$ is decreasing for $x \geq P(1+\eta)$, and hence

$$f_0(t) < E := \frac{M_AM_BP^2}{\sigma^3} < \frac{1+\eta}{(1+2\eta)^3}.$$

The first strict inequality is physical Hölder inequality at a positive six-body configuration; equality would require all fifteen pair distances to be equal. The function f_0 decreases strictly in t. To verify $(1+\eta)/(1+2\eta)^3 < f_0(1/5)$ exactly, substitute

$y = \eta/(1-\eta)$ and clear positive denominators: the coefficient list in ascending powers of y is $(1250, 12711, 54243, 125127, 167471, 129401, 53453, 9385, 175)$, all positive. Thus $t > 1/5$ for every unequal positive-bridge candidate.

For $0 < \theta \le 9/20$, one has $\eta > 89/100$, $p \ge 11/40$, and $N_1 > 1$. Therefore $E < (1+\eta)/(1+2\eta)^3 < 11/120$ and, from the exact weighted bridge identity,

$$p(d_{26}/m)^3(1-r) = f_0(t), \qquad \delta = |1-r| = \frac{f_0(t)}{pN_1^{3/2}} < \frac{P^3}{\alpha H^2 J^2 N_1^{3/2}} < \frac{1}{3},$$

where H, J are the total masses of the two triangles, $M_A M_B = HJ$, and physical Hölder was used in the middle bound. Since $|r| > 1$, the denominator norms in (62) are

$$|u|^2 = H^2 + H(a+b)(|r|^2-1) - (a+b)d\delta^2, \qquad |v|^2 = H^2 + Hb(|r|^2-1) - b(a+d)\delta^2.$$

The added terms are nonnegative and each subtracted product is at most $H^2/4$; hence $|u|, |v| > q_0 H$. Applying the cubic argument above, a root on the first triangle requires

$$\frac{a^3\delta}{bdH} > \frac{q_0}{4\kappa_1}, \qquad \kappa_1 = \frac{D_1}{bd} \ge 2, \qquad D_1 = b^2 + d^2.$$

For the second triangle, the wing-swap identity at r^{-1} divides both $|1-r|$ and its two denominator norms by $|r|$, so these factors cancel and give

$$\frac{A^3\delta}{efJ} > \frac{q_0}{4\kappa_2}, \qquad \kappa_2 = \frac{D_2}{ef} \ge 2, \qquad D_2 = e^2 + f^2.$$

Multiplying, using $bd\,ef = \alpha\beta$, and then using the upper bound on δ, gives

$$\delta^2 > \frac{q_0^2(\alpha\beta)^2 HJ}{16a^3A^3D_1D_2}, \qquad \delta < \frac{P^3}{\alpha H^2 J^2 N_1^{3/2}}.$$

Identity (64) applied to both triangles gives $H^5/a^3 > C_0 D_1$ and $J^5/A^3 > C_0 D_2$, where $C_0 = 3125/108$. Combining these inequalities and substituting $\alpha = pP$, $\beta = qP$ proves

$$N_1(t,\theta)^3(1-\theta)^4(1+\theta)^2 < \frac{1024}{q_0^2 C_0^2}, \qquad q_0 = 24/25, \quad C_0 = 3125/108. \tag{65}$$

In the lower interval, substitution of this floor reverses the inequality up to a unique $\theta_* \in (3/8, 2/5)$. To state the exact Sturm input, set $\ell(\theta) = \theta/3 + 17/200$ and

$$D_h(\theta) = N_1(h(\theta),\theta)^3(1-\theta)^4(1+\theta)^2 - \frac{1024}{q_0^2 C_0^2}.$$

The primitive integer polynomials $125000 D_{1/5}$ and $4665600000000000000 D_\ell$ have degrees 18 and 24, respectively; their descending coefficients, including every sign and zero, are recorded in the static certificate. Signed Euclidean Sturm chains of P, P' give the following endpoint data:

$h(\theta)$	endpoints	$\operatorname{sgn} P$	variations V
$1/5$	$0, 3/8, 2/5$	$+,+,-$	$6, 6, 5$
ℓ	$3/8, 9/20$	$+,+$	$7, 7$

Thus $D_{1/5}$ has no zero on $[0, 3/8]$ and exactly one on $(3/8, 2/5)$; call it θ_*. On $3/8 \le \theta \le 9/20$, monotonicity of the bridge function strengthens the floor to $t > \ell(\theta)$. The corresponding D_ℓ is positive at both endpoints and has no zero between them. Neither root count follows merely from a few sampled values. Thus this moderate-asymmetry interval is excluded. On the opposite nonreal bridge half-sheet, $K \ge c^3/P^2$ with $P = \alpha + \beta$. Since $K = M_A M_B / M$, this gives $Mc^3 \le M_A M_B P^2$. Strict positivity gives $\sigma > M_A M_B > P$, and therefore $\sigma^3 > (M_A M_B)^3 > M_A M_B P^2 \ge Mc^3$, contradicting physical Hölder.

For a remaining unequal bridge candidate, the physical asymmetry satisfies $\theta > 9/20$. Set $\gamma = 1 + q\theta t > 0$. The bridge parameterization gives

$$\operatorname{Re}\big((d_{26}/m)^3\big) = \gamma\big(\gamma^2 - 3q^2 L\big) = -\gamma G(\theta, t), \qquad G(\theta,t) = 2(1+\theta)^2\theta^2 t^2 + \frac{(1+\theta)(9+7\theta)}{2}t - 1.$$

Both partial derivatives of G are positive for $\theta, t > 0$, and $G(9/20, 1/5) = 1591621/2000000 > 0$. Since $t > 1/5$, we have $\mathrm{Re}(d_{26}^3) < 0$. The mass-weighted shape ratio above and $K > 0$ now give

$$r = 1 - \frac{K}{\alpha d_{26}^3}, \qquad \mathrm{Re}\, r = 1 - \frac{K}{\alpha}\frac{\mathrm{Re}(d_{26}^3)}{|d_{26}^3|^2} > 1.$$

Put $a = m_1/m_3$, $A = m_4/m_5$, $x = m_2/m_3$ and $y = m_6/m_5$. The two closed triangles, after scaling their masses, are $(a, x, 1)$ and $(A, y, 1)$ and have the same shape r, while

$$xy = \frac{1-\theta}{1+\theta} < \frac{11}{29} < \left(\frac{5}{8}\right)^2. \tag{66}$$

At least one wing ratio is $\le 5/8$. For any positive triple $(a, x, 1)$ in that range, write its necessary shape sextic as $P_{a,x}(z) = -F_{a,x,1}(1+z, 1)$. For the triangle relative coordinates $(r, 1)$ put

$$U = (a+x)r + 1, \quad V = xr + a + 1, \quad W = V - U = a(1-r), \quad Q = a^2UW - a^2x^2VW - x^2UV,$$

$$F_{a,x,1}(r,1) = Q^3 + 27a^6x^4UV^2W^3 - 27a^6x^2U^2VW^3 + 27a^4x^2QUVW^2.$$

This is homogeneous of degree six before setting the second shape coordinate to 1. Positive masses and the original three distance ties require $UVW \neq 0$; the formula clears only those nonzero denominators. The exact rational Hurwitz certificate has positive leading coefficient, positive constant term $x^6(a+x+1)^6$, and all six leading Hurwitz minors strictly positive on $a > 0$, $0 < x \le 5/8$. The same degree-preserving substitution used below takes c_6 to a polynomial of v-degree 12 with 91 nonzero coefficients, all positive.

For reproducibility take canonical symbols (A, X) for (a, x), let $c_6, \dots, c_0$ be the coefficients of $P_{A,X}(z)$ in descending order, and define $\Delta_k = \det \mathcal{H}_{[k],[k]}$, where

$$\mathcal{H} = \begin{pmatrix} c_5 & c_3 & c_1 & 0 & 0 & 0 \\ c_6 & c_4 & c_2 & c_0 & 0 & 0 \\ 0 & c_5 & c_3 & c_1 & 0 & 0 \\ 0 & c_6 & c_4 & c_2 & c_0 & 0 \\ 0 & 0 & c_5 & c_3 & c_1 & 0 \\ 0 & 0 & c_6 & c_4 & c_2 & c_0 \end{pmatrix}.$$

The table uses $\Delta_k = u_k \prod_j G_{kj}^{e_{kj}}$. For each factor the entry (e, d_A, d_X, s, n) records its multiplicity, degrees in A, X, common sign s, and number n of nonzero coefficients after substituting $X = 5v/[8(1+v)]$ and multiplying by $(1+v)^{d_X}$. Each decomposition is checked by exact division by the displayed positive X and $A + X + 1$ factors, with zero remainder; the last G_{kj} is the resulting signed quotient, not an assumed irreducible polynomial.

k	$\mathrm{sgn}\, u_k$	(e, d_A, d_X, s, n) for each factor
1	−	(1,1,1,+,4),(1,11,11,-,78)
2	−	(3,1,1,+,4),(1,21,21,-,253)
3	+	(6,1,1,+,4),(1,30,30,+,496)
4	+	(2,0,1,+,1),(10,1,1,+,4),(1,36,36,+,703)
5	−	(6,0,1,+,1),(15,1,1,+,4),(1,39,39,-,820)

For $k = 5$ the last displayed factor may be defined directly as $G_5 = -\Delta_5/[X^6(A+X+1)^{15}]$: first compute the 5×5 Hurwitz determinant in the independent indeterminates $c_0, \dots, c_6$ (it has 13 terms), and only then substitute the sextic coefficients. Exact polynomial division must have zero remainder, and $(1+v)^{39}G_5(A, 5v/[8(1+v)])$ has 820 nonzero coefficients, all negative. Its irreducibility is not required for the sign conclusion.

Every $\Delta_1, \dots, \Delta_5$ is positive; the constant term $c_0 = X^6(A+X+1)^6 > 0$ gives $\Delta_6 = c_0\Delta_5 > 0$. All transformed factors retain their full v-degree, so the leading v coefficients of the same sign include $x = 5/8$. Routh–Hurwitz puts all sextic roots in $\mathrm{Re}\, z < 0$, so the required shape has $\mathrm{Re}\, r < 1$, a contradiction. The fixed/adaptive Sturm signs, opposite-sheet energy estimate and all Hurwitz minors are exact certificate obligations; the final sector argument alone does not exclude No. 79.

5.17. **No. 52: exclusion of both valuation endpoints.**

Proposition 5.6 (No. 52 valuation sieve)**.** *Let $B \neq 0$ be the common leading w coefficient on the zw-triangle $T = 456$ of No. 52. Every first-limit germ through a positive configuration has $W_2 = W_3 = B$ and $W_1/B = -(m_2 + m_3 + m_4 + m_5 + m_6)/m_1$. Its w-only triangle $Q = 235$ has common z-order $\alpha = 5/3$ or $\alpha = 7/4$.*

Proof. We first exhaust the outer leading w coefficients W_1, W_2, W_3. On a w-only edge, unequal leading w coefficients make the edge maximal and its z-separation $O(\varepsilon^2)$; equal coefficients make the edge nonmaximal, with z-order less than 2. If exactly two of W_2, W_3, B agree, the other two edges of the w-only triangle 235 are maximal. Their two-edge path forces the supposedly nonmaximal edge's z-separation to be $O(\varepsilon^2)$, a contradiction.

If W_2, W_3, B are pairwise distinct, all three triangle edges are maximal. When W_1 differs from all three, the original outer z-rows force $z_5, z_6 = O(\varepsilon^2)$, contrary to the zw-edge 56. When $W_1 = W_2$ or $W_1 = W_3$, the mass-weighted sum of the original z_1 row and corresponding outer z row cancels the unmarked mutual force. Its remaining forces have order at least 2, but its position has nonzero order-ε coefficient $m_1 z_6$, again impossible. Finally, if $W_1 = B$, the original z_1 row forces $\operatorname{ord} z_{16} = 7/4$ and $\operatorname{ord} \delta_{16} = -1/4$. The original T-row contraction makes its signed internal potential $O(\varepsilon^2)$, while all other exterior inverse distances are bounded. Thus U has the uncancelled pole $m_1 m_6 D_{16} \varepsilon^{-1/4}$, contradicting $U = c$.

Only $W_2 = W_3 = B$ remains. The mass-weighted w-centre then gives the asserted W_1/B, which differs from 1. The maximal edge 16 and its original z_1 row give $z_1, z_6 = O(\varepsilon^2)$. Since 56 has both colours, $z_5 = \varepsilon Z_5 + o(\varepsilon)$ with $Z_5 \neq 0$; the three Q edges have z-order $\alpha > 1$, so the same order-one coefficient occurs in z_2, z_3. On each of their original z-rows the internal Q force has order $4\alpha - 6$, while the missing-edge forces have strictly greater order. An internal force cannot balance $z_5 \varepsilon$ when $\alpha > 7/4$.

The 15 pair-square orders are 2 on T, $4 - 2\alpha$ on Q, 0 on 16, -1 on the other four edges from body 1, and $5 - 3\alpha$ on 24,26,34,36. The least of the 15 seven-product orders is $13 - 8\alpha$, attained at products 26,36,56. Thus $\alpha \leq 13/8$ has no unbounded seven-product. For $13/8 < \alpha < 5/3$ the inverse distances on the four missing edges can grow, so discarding them would be wrong. Instead the exact tie $\delta_{ij}^2 z_{ij} w_{ij} = 1$ gives, for $i \in \{2, 3\}$ and $j \in \{4, 6\}$,

$$\delta_{ij} - \delta_{ij}^3 z_{ij}(w_i - w_5) = \delta_{ij}^3 z_{ij}(w_5 - w_j). \tag{67}$$

Contracting only the original z_2, z_3 rows against $m_i(w_i - w_5)$ cancels these four crossed potential terms by (67). The remaining source, relative and crossed orders are respectively $5 - 3\alpha$, $4 - 2\alpha$, and $(9\alpha - 11)/2$, all positive; the T relative potential also tends to zero. Only $\delta_{16} \to d \neq 0$ remains finite, hence $c = \lim U = m_1 m_6 d$. Put $C = \sum_{i=2}^{6} m_i$, $M = m_1 + C$ and $\sigma = \sum_{i<j} m_i m_j$. The original w_1 row and mass centre give $d^3 = C/(M m_6)$, whence $M c^3 = m_1^3 m_6^2 C$. At a positive configuration, $\sum_{i<j} m_i m_j r_{ij}^2 = MI = Mc$ and $U = \sum_{i<j} m_i m_j / r_{ij} = c$. Hölder therefore gives $\sigma^3 \leq M c^3$. But $\sigma > m_1 C$ and $C > m_6$ imply $\sigma^3 > (m_1 C)^3 > m_1^3 m_6^2 C = M c^3$.

For $5/3 < \alpha < 7/4$ all missing-edge inverse distances fade. The original z_2, z_3 rows force zero individual Q internal z-forces; their signed ties and the original w_2, w_3 rows imply $F_0(m_2, m_3, m_5) = 0$ (defined in (69)). The same positive-mass signed-tie reduction as in Section 4.7 fixes the actual leading triangle ratio at 1; when $m_2 = m_3$, both internal w forces vanish, contrary to the common nonzero original w_2, w_3 coefficient B. Hence $m_2 \neq m_3$. With $w_2 - w_5 = p\varepsilon^{4-3\alpha} + o(\varepsilon^{4-3\alpha})$ and $p \neq 0$, the same original-row contraction yields

$$U_Q = z_5 p \frac{m_2^2 - m_3^2}{m_2} \varepsilon^{5-3\alpha}(1 + o(1)). \tag{68}$$

This is a nonzero pole. The T relative potential tends to zero, 16 stays finite and every other exterior inverse distance fades, so $U = c$ is impossible. Only the two stated endpoints remain. □

Before excluding the two endpoints, the original rows impose four intermediate necessary mass factors. With $a = m_2, b = m_3, d = m_5 > 0$,

$$\begin{aligned} F_0 &= (ab)^3 - d^2(a+b)^4, \\ F_{--} &= d(a+b)(a^2-b^2) - a^2b^2, \\ F_{-+} &= d(a+b)(a^2+b^2) - a^2b^2, \\ F_{+-} &= d(a+b)(a^2-b^2) + a^2b^2. \end{aligned} \tag{69}$$

Proposition 5.7 (No. 52 necessary mass locus)**.** *By Proposition 5.6, every positive-point No. 52 first limit satisfies $F_0F_{--}F_{-+}F_{+-} = 0$. This product is not the zero polynomial.*

Proof. When $\alpha = 5/3$, the two individual internal Q z-forces vanish; their three distance ties and the nonzero original w_2, w_3 forces give $F_0 = 0$. As above, their actual positive-mass triangle ratio is 1; if $a = b$ its internal w forces would vanish, contradicting $B \neq 0$. At $\alpha = 7/4$ write p, q for the leading coefficients of $w_2 - w_5, w_3 - w_5$, and u, v for those of $z_2 - z_5, z_3 - z_5$. The potentially divergent Q potential has leading coefficient $z_5(ap+bq)$. Since $U = c$ it must vanish. The internal pairwise antisymmetry of the original force rows yields

$$\sum_{i=2,3} m_i\big((z_i - z_5)F^w_{Q,i} - (w_i - w_5)F^z_{Q,i}\big) = 0. \tag{70}$$

Here $F^z_{Q,2} = F^z_{Q,3} = Z_5 \neq 0$ and $F^w_{Q,2} = F^w_{Q,3} = B \neq 0$, so $au + bv = 0$ as well. In particular $v/u = q/p = -a/b$. The three signed Q ties leave two independent signs, with

$$\frac{D_{23}}{D_{25}} = s_{23}\frac{b}{a+b}, \qquad \frac{D_{35}}{D_{25}} = s_{35}\frac{b}{a}, \qquad s_{23}, s_{35} \in \{\pm 1\}. \tag{71}$$

The difference of the original w_2, w_3 force requirements is, up to a nonzero factor,

$$d(a+b)(a^2 + s_{35}b^2) + s_{23}a^2b^2 = 0. \tag{72}$$

For $(s_{23}, s_{35}) = (+,+)$ this is strictly positive. The sign pairs $(-,-), (-,+), (+,-)$ give respectively $F_{--} = 0, F_{-+} = 0, F_{+-} = 0$. Thus every surviving endpoint lies on the asserted union; none is claimed to lift to a full solution germ. At the positive mass sample $(a, b, d) = (1, 2, 1)$ the four factors are $(-73, -13, 11, -5)$, with product $-52195 \neq 0$. □

Proposition 5.8 (No. 52 excluded as a positive-point first limit)**.** *No strictly positive mass vector admits a No. 52 first-limit germ through a positive configuration. The assertion does not exclude arbitrary complex-only curves with this diagram.*

Proof. Write $a = m_2$, $b = m_3$, $d = m_5$, $X = m_4$ and $Y = m_6$. By Proposition 5.6, only $\alpha = 5/3$ or $7/4$ can have a pole in a seven-product. At both endpoints the original body-4 and body-5 w-rows require, for the inner triangle 456, $F^w_{T,4} = B \neq 0$ and $F^w_{T,5}/F^w_{T,4} = R = 1 + (a+b)/d > 1$. Its three leading z-forces vanish, as does its signed leading potential. Normalize $m_5 = D_{46} = Z_{46} = W_{46} = 1$ and put $A = X/d$, $C = Y/d$ and $t = D_{56}$. Original signed ties and the zero z-forces give

$$C^2(A+t)^3(t^3+A) - A^3t^3 = 0, \qquad F^w_{T,5} = t^3F^w_{T,4}, \qquad F^w_{T,4} = \frac{C(A^2-t^4)}{A(t^3+A)}. \tag{73}$$

There is no lost root at $t = 0, -A$ or $t^3 = -A$. The nonreal cube phases of $t^3 = R > 1$ would require $(A+t)^3 = A^3R/[C^2(R+A)] > 0$; their imaginary parts vanish only when $|t| = A$, when $(A+t)^3 = -A^3 < 0$. Thus $t > 1$ is real and $Y < d$.

At $\alpha = 5/3$, the first two Q z-forces vanish and (69) gives $d = (ab)^{3/2}/(a+b)^2$ and $a \neq b$. Writing $a = x^2$, $b = y^2$, the seven original Q force/tie rows have a left kernel; in the order $(z_2, z_3, w_2, w_3, 23, 25, 35)$ and with second weight one it is

$$\left(-\frac{x^4}{y^4}, 1, \frac{x^4}{3y^4}, -\frac{x^4}{3y^4}, -\frac{x^5}{y(x^2+y^2)}, \frac{x^7}{y(x^2+y^2)^2}, \frac{x^5y}{(x^2+y^2)^2}\right). \tag{74}$$

Projecting the first original six-body crossed z_2, z_3 sources and all four signed crossed ties onto this vector makes the finite Q reference term equal the potential of the four finite crossed edges. The original reference-5 identity subtracts those same edges from the complete Q potential, so its finite coefficient vanishes. The w-source projection cancels. All other terms fade except the finite edge 16, so $c = U = m_1 Y D_{16} + U_\times$, with $D_{16}^3 = (a+b+X+d+Y)/(MY)$. The z-centre and crossed ties give, with $P = a+b$ and $v^2 = X/(X+P+d)$,

$$\frac{D_{24}^2}{D_{26}^2} = \frac{D_{34}^2}{D_{36}^2} = v^2, \qquad \frac{D_{34}^2}{D_{24}^2} = \frac{D_{36}^2}{D_{26}^2} = -\frac{a}{b}. \tag{75}$$

Each independently signed crossed pair has complex coefficient $a \pm i\sqrt{ab}$, of squared modulus aP. The left-kernel identity and triangle inequality imply

$$|U_\times|^3 \le |a-b| P^2 (Y + Xv)^2 < P^3 (Y + Xv)^2. \tag{76}$$

Normalize $P = 1$, write $\lambda = \sqrt{ab}/P \in (0, 1/2)$, $d = \lambda^3$, and $L = 1 + \lambda^3$. Since $v^2 = X/(X+L)$, the substitution $v = u/(1+u)$ gives

$$X = \frac{Lu^2}{1+2u}, \qquad v = \frac{u}{1+u}, \qquad Y = \lambda^3 q, \qquad u > 0, \quad 0 < q < 1.$$

Put $C_5 = L + X + Y$, $H_5 = \lambda^2 + \lambda^3 + L(X+Y) + XY$, and $S = Y + Xv$. We prove directly that

$$H_5^3 > C_5 S^2. \tag{77}$$

Set $J = X + \lambda^2$. Since $L > 1$ and $Y < \lambda^3$, we have

$$C_5 \le (L+Y)(X+1), \qquad S \le \lambda^3 + \frac{X^{3/2}}{\sqrt{X+1}}.$$

Cauchy–Schwarz gives

$$\left(X^{3/2} + \lambda^3 \sqrt{X+1}\right)^2 \le J\left[X^2 + \lambda^4 (X+1)\right] < J^3,$$

where the strict inequality follows from

$$J^2 - \left[X^2 + \lambda^4 (X+1)\right] = \lambda^2 (2 - \lambda^2) X > 0.$$

Moreover,

$$H_5 - (L+Y) J = (1 - \lambda^2)(\lambda^3 + Y) + \lambda^3 Y > 0.$$

Consequently,

$$C_5 S^2 < (L+Y) J^3 < (L+Y)^3 J^3 < H_5^3,$$

which proves (77) throughout the required parameter range.

For the extension to $m_1 > 0$, define for $s \ge 0$

$$G(s) = H_5 + s C_5 - s Y^{2/3} C_5^{1/3} - (s + C_5)^{1/3} S^{2/3}.$$

The inequality gives $G(0) > 0$. Moreover, $C_5 > 1$, $Y < \lambda^3 < 1/8 < C_5/8$, and $S < C_5$, so

$$G'(s) = C_5 - Y^{2/3} C_5^{1/3} - \frac{S^{2/3}}{3(s + C_5)^{2/3}} > \frac{3}{4} - \frac{1}{3} = \frac{5}{12}.$$

By (76) and $D_{16}^3 = C_5/(MY)$, the energy bound gives

$$M^{1/3} c \le m_1 Y^{2/3} C_5^{1/3} + M^{1/3} S^{2/3}, \qquad M = m_1 + C_5.$$

Since $\sigma = H_5 + m_1 C_5$, the positivity of $G(m_1)$ gives $\sigma > M^{1/3} c$, contrary to physical Hölder.

At $\alpha = 7/4$, the mass relation alone does not exclude $a = b$: $F_{-+}(a, a, a/4) = 0$. For $a = b$ the other factors are nonzero, so $(-,+)$ is the only equal-mass signed candidate. Put $p = w_{25}^* \ne 0$. Its signed ties give $w_{35}^* = -p$, $D_{23} = -D_{25}/2$ and $D_{35} = D_{25}$. Substitution into both original internal Q w-force rows gives $F_{Q,2}^w = F_{Q,3}^w = 0$, whereas the full original w_2, w_3 rows require the same nonzero coefficient B at this order. Thus $a = b$ is impossible. Only $a \ne b$ remains for the three positive signed Q branches in (69). Normalize $(z_{25}, w_{25}, D_{25}) = (1, 1, 1)$ and order

the seven Q variables as in the preceding kernel. For signs $(-,+),(-,-),(+,-)$, the original force/tie Jacobian determinants, after substituting the corresponding mass factor, are

$$-\frac{32b^8(a-b)^4(a^2+ab+b^2)^4}{a(a+b)^7(a^2+b^2)^4},\quad \frac{32b^8(a^2-ab+b^2)^4}{a(a-b)^4(a+b)^7},\quad -\frac{32b^8(a^2-ab+b^2)^4}{a(a-b)^4(a+b)^7}.$$

All are nonzero. Moreover the weighted w-shape row $(0,0,a,b,0,0,0)$ annihilates the adjugate against both common source columns $(1,1,0,0,0,0,0)^T$ and $(0,0,1,1,0,0,0)^T$, for every sign branch. The two source scalings of the homogeneous Q shape have determinant -2, so the formal implicit function theorem applies even at fractional Puiseux suborders. Direct expansion of all four original six-body Q rows with $\varepsilon = h^8$ shows no external coefficients at $h^9, h^{10}, h^{-15}, h^{-14}$; the first crossed z-force has relative order $3/8$, and all other noncommon sources are later. A finite Q reference-potential contribution would require relative order $1/4$ after its leading $\varepsilon^{-1/4}$ coefficient vanishes. It is therefore zero. The T potential and all other crossed contributions fade, leaving $c = m_1 Y D_{16}$. The original w_1 row gives $D_{16}^3 = (M-m_1)/(MY)$, while the positive mass-pair sum obeys $\sigma^3 > m_1^3 Y^2 (M-m_1) = Mc^3$, again contradicting physical Hölder. Both valuation endpoints are therefore impossible. □

The 7 derived relation obligations. The 7 arguments below explain separately how an unbounded first limit reaches a proper necessary mass locus. Each has a local positive-point first-limit proof. Table 4 records the proof status and the remaining independent audits of the stated certificates. Finiteness on the positive exceptional zero sets is not settled here.

No.	Necessary condition / method	Proof status / certificate audit
8	Two global-minimum triangle supports; six μ_6 factors, four resultant families and a direct order proof	the two no-match supports are covered by Lemma 5.9
63	Fully balanced K4; 32 signed affine-collinear branches and nine non-collinear Cayley–Menger cases	proved locally; independent replay audit of the 32 characteristic-zero resultants remains
66	Zero/nonzero K4 w-drift; mass differentials and a finite signed cover of one parameter	direct analytic proof; no diagram-specific computational certificate or coverage hypothesis
88	Pair/K4 split first crossed torque and nonuniform signed K4 root norm	proved locally; separately audit the degree-90 fiber, nine phase sheets and zero-star replay
93	Full intermediate order window; lower three-body Euler/Lagrange root and sign norm	proved locally; independently audit graph-order and crossed-tie guards
94	w-only K_4; finite distinct-spoke fibers, $H_3 = 0$ projective boundary census and first six-body torque norm	proved locally; independently audit the signed norm and boundary guards
98	Two closed three-body boundaries; signed first crossed torque and colour-block involution	proved locally; independently audit the cited graph rules and interval replay

TABLE 4. The 7 derived cases, their proofs and the independent certificate audits.

5.18. **No. 8: global minima and the finite mass-factor cover.** Set $Q = 236$ and $T = 456$. For positive a, b, c write

$$\begin{aligned}\mu_6(a,b,c) &= (ab)^3 - c^2(a+b)^4,\\ H_{a,b,c}(x) &= a^2(c+bx)^3(c+bx^3) - b^3c^3x^3.\end{aligned} \tag{78}$$

For each $S \in \{Q,T\}$ and $k \in S$ take the factor $\mu_6(m_i,m_j,m_k)$, where $\{i,j\} = S \setminus \{k\}$; these are six body-labelled factors, not just one symmetric triangle condition. Four more factor families

arise from eliminating centred shape ratios and the original mass-centre or force row. One representative from each family is

(79)
$$R_{2\times2} = \operatorname{Res}_r\big(H_{m_2,m_3,m_6}(r), (m_1+m_4+m_5+m_6)m_3r^3 + m_6(m_2+m_1+m_4+m_5+m_6)\big),$$

(80)
$$R_{3\times2} = \operatorname{Res}_s\big(H_{m_4,m_5,m_6}(s), m_5(m_1+m_4+m_5)s^3 + m_6(m_1+m_5)\big),$$

(81)
$$R_{\text{late}} = \operatorname{Res}_s\big(H_{m_4,m_5,m_6}(s), (m_2+m_3+m_6)m_5s^3 + m_6(m_2+m_3+m_6+m_4)\big).$$

For the fourth family set $M = \sum_{i=1}^6 m_i$, $u = m_3r^3/m_6$, $v = m_5s^3/m_6$, $A = m_2+m_3+m_6$ and $B = m_4+m_5+m_6$. At $W_3 - W_2 = Z_5 - Z_4 = 1$ the uncentred leading vectors are

$$z^0 = (0, -v, -v, 0, 1, -v), \qquad w^0 = (0, 0, 1, -u, -u, -u).$$

The exact cleared *full* mass-centre inertia is

(82)
$$\begin{aligned} N(r,s) &= m_6^2\left(M\sum_i m_i z_i^0 w_i^0 - \Big(\sum_i m_i z_i^0\Big)\Big(\sum_i m_i w_i^0\Big)\right) \\ &= m_6^2\left[M(-m_3v - m_5u + m_6uv) - (m_5 - Av)(m_3 - Bu)\right]. \end{aligned}$$

In particular the factor is the ordered polynomial resultant

(83)
$$R_{\text{double}} = \operatorname{Res}_r\big(H_{m_2,m_3,m_6}(r), \operatorname{Res}_s(H_{m_4,m_5,m_6}(s), N(r,s))\big).$$

Only the positive mass m_6 is cleared in N; the centring formula divides by $M > 0$. No factor depending on r or s is canceled from the resultant, so possible extraneous shape roots do not invalidate its necessity. The support partition and the nonzero leading-shape guards used in the resultants are checked below.

Lemma 5.9 (The two remaining No. 8 supports)**.** *Let a positive-point No. 8 first-limit germ have global minimum supports*

$$S_z = \{1,4,5\}, \qquad S_w = \{1,2,3,4,5,6\}.$$

Write $A = \min_i \operatorname{ord} z_i$, $B = \min_i \operatorname{ord} w_i$, and let Z_i, W_i denote the coefficients at orders A, B, respectively, with a zero coefficient for a later coordinate. If

$$Z_1 \neq Z_4, \qquad Z_1 \neq Z_5,$$

then every seven-product P_{ij} has nonnegative order. The same conclusion holds after the simultaneous involution $(2\ 4)(3\ 5)$ and $z \leftrightarrow w$.

Proof. Put $\rho = A + B$. If $\rho \geq 0$, every squared distance has order at least ρ, so the assertion is immediate. We therefore assume $\rho < 0$ and show that all fifteen seven-products have strictly positive order.

We first determine the difference orders from the two pure-colour triangles. For an edge put

$$\alpha_{ij} = \operatorname{ord} z_{ij}, \qquad \beta_{ij} = \operatorname{ord} w_{ij}.$$

The original distance tie gives

(84)
$$\operatorname{ord}(\delta_{ij}^3 z_{ij}) = -\frac{\alpha_{ij} + 3\beta_{ij}}{2}, \qquad \operatorname{ord}(\delta_{ij}^3 w_{ij}) = -\frac{3\alpha_{ij} + \beta_{ij}}{2}.$$

On the z-only triangle $T = 456$, the first expression is -2 and the second is greater than -2. Hence

$$\alpha_{ij} + 3\beta_{ij} = 4, \qquad \alpha_{ij} < 1 \qquad (ij \subset T).$$

The three α_{ij} are equal. Indeed, the minimum among the three orders of differences on a triangle occurs at least twice; if the orders were not all equal, their maximum would occur once. The

affine decreasing relation $\beta_{ij} = (4 - \alpha_{ij})/3$ would then give a unique minimum among the three β_{ij}, which is impossible. Since z_6 is later than z_4, z_5, the common z-difference order is A. Thus

$$A < 1, \qquad \alpha_{45} = \alpha_{46} = \alpha_{56} = A, \qquad \beta_{45} = \beta_{46} = \beta_{56} = h := \frac{4 - A}{3} > 1.$$

The colour-dual argument on the w-only triangle $Q = 236$ gives a common w-difference order $b < 1$, with common z-difference order $(4 - b)/3$. In particular $B \le b < 1$.

Because $h > 1 > B$, the coefficients W_4, W_5, W_6 are equal. The coefficient of the constant inertia I at the negative order ρ vanishes. Together with the z-centre it gives

$$0 = m_1 Z_1 W_1 + (m_4 Z_4 + m_5 Z_5) W_6 = m_1 Z_1 (W_1 - W_6),$$

so $W_1 = W_6$. If $b > B$, the triangle Q would also have $W_2 = W_3 = W_6$. All six W_i would then equal the same nonzero number, contrary to $\sum_i m_i W_i = 0$ and $\sum_i m_i > 0$. Consequently

$$b = B, \qquad \beta_{23} = \beta_{26} = \beta_{36} = B, \qquad \alpha_{23} = \alpha_{26} = \alpha_{36} = \frac{4 - B}{3}. \tag{85}$$

In particular W_2, W_3, W_6 are pairwise distinct. The support assumptions and the two no-match conditions now show that every edge outside Q has z-difference order A. Also,

$$\beta_{12} = \beta_{13} = \beta_{24} = \beta_{25} = \beta_{34} = \beta_{35} = B.$$

It remains to control the three differences from body 1 to T. Put

$$k_j = \operatorname{ord} w_{1j} \quad (j = 4, 5, 6), \qquad K = \max\{k_4, k_5, k_6\}.$$

Each k_j is finite by the original distance tie, and $k_j > B$ because $W_1 = W_4 = W_5 = W_6$. In the original w_1 row the terms from bodies 2 and 3 have order

$$-\frac{3A + B}{2} = B - \frac{3}{2}\rho > B.$$

Since $\operatorname{ord} w_1 = B$, at least one term from T must have order at most B. Formula (84) therefore implies

$$K \ge -3A - 2B. \tag{86}$$

The maximum K cannot occur at just one j. Otherwise, in the original z_1 row that edge would have the unique smallest force order, and its order would satisfy

$$-\frac{A + 3K}{2} \le 4A + 3B = A + 3\rho < A.$$

The coordinate z_1 has order A, while the terms from bodies 2 and 3 have order $A - 3\rho/2 > A$. The remaining terms from T have strictly larger order than the term corresponding to the unique maximum K. This would leave a unique nonzero lowest-order term in the equation, a contradiction.

Choose distinct $j, \ell \in T$ with $k_j = k_\ell = K$. Since every internal w-difference in T has order h, the triangle identity gives $h \ge K$. If the third k_r were less than K, the three orders k_r, K, h on the triangle $1rj$ would have a unique minimum. Hence

$$k_4 = k_5 = k_6 = k \le h.$$

Combining this with (86) yields

$$-3A - 2B \le \frac{4 - A}{3}, \qquad 4A + 3B = A + 3\rho \ge -2. \tag{87}$$

Since $A < 1$, this gives $\rho > -1$. Because also $A, B < 1$, we obtain $A, B > -2$.

For clarity, write the four squared-distance orders as

$$\eta_Q = \frac{4 + 2B}{3} > 0, \qquad \eta_T = \frac{4 + 2A}{3} > 0, \qquad \eta_1 = A + k \ge -2\rho, \qquad \rho < 0.$$

Here η_Q occurs on $23, 26, 36$, η_T on $45, 46, 56$, η_1 on $14, 15, 16$, and ρ on $12, 13, 24, 25, 34, 35$. The definition of the seven-products gives all fifteen orders in five groups:

ij	$\operatorname{ord} P_{ij}$	lower bound
$12, 13$	$3\rho + \eta_Q + 3\eta_T$	$(8-\rho)/3$
$14, 15, 45$	$2\rho + 3\eta_Q + \eta_T + \eta_1$	$3\eta_Q + \eta_T$
$16, 46, 56$	$4\rho + \eta_Q + \eta_T + \eta_1$	$8(1+\rho)/3$
23	$\eta_Q + 3\eta_T + 3\eta_1$	$\eta_Q + 3\eta_T - 6\rho$
$24, 25, 26, 34, 35, 36$	$3\rho + \eta_Q + \eta_T + 2\eta_1$	$\eta_Q + \eta_T - \rho$

For the first row, (87) gives $A \geq -2 - 3\rho$, and therefore

$$3\rho + \eta_Q + 3\eta_T = \frac{16 + 4A + 11\rho}{3} \geq \frac{8-\rho}{3}.$$

The remaining bounds follow directly from $\eta_1 \geq -2\rho$ and $\eta_Q + \eta_T = (8+2\rho)/3$. Every displayed lower bound is strictly positive because $-1 < \rho < 0$ and $\eta_Q, \eta_T > 0$. Thus no seven-product has a pole.

The proof permits equalities in the order comparisons and does not impose an upper bound on the orders of individual coordinates. In particular, a later coordinate among z_2, z_3, z_6 may vanish identically: only its zero coefficient at order A is used. All difference orders remain finite because $\delta_{ij}^2 z_{ij} w_{ij} = 1$. The simultaneous involution (2 4)(3 5) and $z \leftrightarrow w$ exchanges Q and T and proves the colour-dual case. □

Proposition 5.10 (No. 8 first-limit obstruction)**.** *Every positive-point No. 8 first limit lies on the zero set of one of the six μ_6 factors in* (78) *or an orbit member of R_{double} or the three resultants* (79)–(81)*. All these factor families are nonzero polynomials.*

Proof. Let $A = \min_i \operatorname{ord} z_i$ and $B = \min_i \operatorname{ord} w_i$. Every difference has orders at least A, B in its two colours; thus if a seven-product has negative order, $A + B < 0$. The z-only edges of T have a common separation order $a < 1$, while the w-only edges of Q have z-separation order $(4-b)/3 > 1$, where $b < 1$ is their common w-separation order. Some vertex of T has z-position order at most a, so $A < 1$. If no vertex of T attained A, the Q separations would prevent bodies 2,3 from attaining A as well: their z-coordinates differ from body 6 only above order 1. Body 1 would then be the sole z-minimum, contrary to the positive-mass z-centre. If exactly one vertex of T attained A, its two incident edge differences would have order A and the opposite edge would have strictly greater order, contradicting the common order a. Hence at least two vertices of T attain A. Interchanging z, w and T, Q proves that at least two vertices of Q attain B.

Write S_z, S_w for these global-minimum supports. The opposite-colour gaps on Q and T exceed $1 > A, B$, respectively. Thus $Q \cap S_z$ is empty or all of Q, and $T \cap S_w$ is empty or all of T. Each support has eight possibilities, giving precisely the following 64 necessary pairs. In the one-colour rows with 6 in both, "its triangle" is T for a z-minimal body 1 and Q for a w-minimal body 1.

Shared 6	Body 1 and triangle support	Pairs	Necessary outcome
Both	1 late in both colours	9	inertia
Both	1 in one colour; its triangle has two minima	12	μ_6
Both	1 in one colour; its triangle has three, the other two	4	μ_6 or R_{late}
Both	1 in one colour; both triangles have three minima	2	μ_6
Both	1 in both colours	9	R_{double}
Neither	1 in both colours	1	inertia
Neither	1 absent from at least one colour	3	μ_6
One colour	1 absent from the other colour	12	μ_6
One colour	1 only in the other colour	6	inertia
One colour	1 in both; the 6-triangle has two minima	4	$R_{2\times2}$
One colour	1 in both; the 6-triangle has three minima	2	$R_{3\times2}$ or analytic boundedness

At orders A, B let Z_i, W_i be the nonzero leading coefficients at a minimum, and set the coefficient to zero when that body is later. The original global centres and $I = c > 0$, with $A + B < 0$, give

$$\sum_{i=1}^{6} m_i Z_i = \sum_{i=1}^{6} m_i W_i = \sum_{i=1}^{6} m_i Z_i W_i = 0. \tag{88}$$

Moreover, the unmarked forces are strictly later than order -2 in either colour, while each marked triangle force has order -2 and every uncircled position is later. The two original wing rows of either triangle therefore have only internal stroke terms at order -2. We record the consequence when their two nonzero relative leading positions are mass-centred against the third. Label this triangle locally by $1, 2, 3$, with masses $(a, b, c) > 0$, and write its relative leading positions as $u, v, 0$, so that $au + bv = 0$ and $v = -au/b$. In the colour carrying the strokes, its two zero-force equations give

$$D_{12}^3 = -\frac{c}{a+b} D_{13}^3, \qquad D_{23}^3 = D_{13}^3.$$

Set $t = D_{23}/D_{13}$, so that $t^3 = 1$. The three signed edge ties give

$$\frac{D_{12}^2}{D_{13}^2} = \frac{ab}{(a+b)(a+bt)}.$$

Cubing this identity and squaring the first force identity yields

$$c^2(a+b)(a+bt)^3 = (ab)^3.$$

For $t = e^{2\pi i/3}$ the imaginary part of $(a + bt)^3$ is $3\sqrt{3}\, ab(a - b)/2$; the other primitive cube root gives its negative. When $a = b$, the cube is $-a^3 < 0$. Hence neither primitive cube root is possible, and $t = 1$ gives $\mu_6(a, b, c) = 0$. All divisions use positive masses and nonzero stroke leads; the edge ties also ensure $a + bt \neq 0$. If $6 \in S_z \cap S_w$, the Q z-lead is $Z_6 \neq 0$ and the T w-lead is $W_6 \neq 0$. Put $M_* = \sum_{i=2}^{6} m_i$, $M = m_1 + M_*$, $\Delta_z = m_4(Z_4 - Z_6) + m_5(Z_5 - Z_6)$ and $\Delta_w = m_2(W_2 - W_6) + m_3(W_3 - W_6)$. Substitution of the two centres into the original inertia

gives

$$\Delta_z = -m_1 Z_1 - M_* Z_6, \qquad \Delta_w = -m_1 W_1 - M_* W_6,$$
$$\sum_i m_i Z_i W_i = m_1 (Z_1 - Z_6)(W_1 - W_6) - M Z_6 W_6. \tag{89}$$

If body 1 is late twice, this is $-M_* Z_6 W_6 \neq 0$. If it joins just z, inertia instead gives $Z_1 = -M_* Z_6 / m_1$ and $\Delta_z = 0$; the w-only join gives the colour dual. When the corresponding T or Q edge-separation order equals A or B, this is a nontrivially centred relative triangle and gives μ_6. This includes all 12 cases with exactly two minimal vertices in that triangle.

For a three-minimum triangle with its edge order strictly later than the global minimum, all its leads equal the shared lead. If the opposite triangle has two minima, its original wing force/ties and full centre give R_{late} (or its exact dual), whether or not the first triangle's leading shape was degenerate. If the opposite triangle has three minima, all five inner bodies attain its global minimum. The leading body-1 coefficient in the joined colour is distinct from their common coefficient by (89); body 1 is late in the other colour. Every exterior inverse distance then has order $-(A+B)/2 > 0$. The original body-1 row cannot have its nonzero leading coordinate canceled, excluding this co-leading case. The surviving nondegenerate triangle gives μ_6.

If body 1 joins both colours, (89) instead requires

$$m_1 (Z_1 - Z_6)(W_1 - W_6) = M Z_6 W_6 \neq 0.$$

Uniform triangle-edge orders permit at most one Q wing to match W_1 and one T wing to match Z_1. Both are required by the two original body-1 rows: with only the Q match its z-row requires a match gap $P = -(A+B) > 0$ and its w-row requires $3P$, while the T-only equations give $3P$ and P in the opposite order. More explicitly, for a Q w-gap $q > 0$ and a T z-gap $p > 0$, the respective z-force orders minus A are $3(P-q)/2, (3P-p)/2$, while the w-force orders minus B are $(3P-q)/2, 3(P-p)/2$. The minimum in each row, including its coordinate term of relative order zero, must be attained at least twice. The nine possible pairs of first-two-term ties leave only $p = q = P$; the matched inverse distances consequently have order zero. Their contributions to the marked triangle wing rows are later than order -2. Both original internal shape equations $H = 0$ therefore hold, and the full-centre inertia numerator (82) gives R_{double}. The two independent wing swaps give all four match labels. If 6 is in neither support, the minima on the triangles are disjoint pairs 45 and 23. A body-1 minimum in both colours leaves the unique nonzero inertia lead $m_1 Z_1 W_1$. Otherwise it is absent from at least one colour; that colour's pair is mass-centred by (88) and gives its μ_6 factor.

Suppose $6 \in S_w \setminus S_z$; the other case is its exact colour-and-label dual. The only z-minimal T vertices are 4,5 and $W_4 = W_5 = W_6 \neq 0$. If body 1 is not z-minimal, the original z-centre isolates the 45 pair and gives μ_6, regardless of whether body 1 is w-minimal. If it is z-minimal but not w-minimal, (88) gives the nonzero inertia coefficient $-m_1 Z_1 W_6$. If it joins both, the same equations give $m_1 (Z_1 - Z_6)(W_1 - W_6) = 0$, so $W_1 = W_6$. With only two Q vertices w-minimal, its original w-rows and full w-centre give the $R_{2\times 2}$ effective-mass cubic in (79); the alternative late wing is obtained by swapping 2,3.

The two remaining support pairs have all of Q minimal in w and only 45 minimal in z, with body 1 in both minima, or their exact colour-and-label dual. In the canonical case their full supports are

$$S_z = \{1,4,5\}, \qquad S_w = \{1,2,3,4,5,6\}.$$

If Z_1 matches one of the T z-leads, the full original z-centre and the T wing force/ties give $R_{3\times 2}$, including the $4,5$ swap. If neither match occurs, Lemma 5.9 shows directly that every seven-product has positive order when $A + B < 0$. Thus neither no-match support can give a first limit. The simultaneous involution $(2\ 4)(3\ 5)$ and $z \leftrightarrow w$ covers the other support. This

completes the disjoint 64-case partition without an order cap or a terminal-cut certificate for these two cases.

For properness, the six μ_6 factors have product 15^6 at all-unit masses. At that same point the nested double-triangle resultant has value

$$R_{\mathrm{double}}(1,1,1,1,1,1) = 160352438096115455429 6875,$$

and $R_{2\times2}, R_{3\times2}, R_{\mathrm{late}}$ have values $20061, 1315, 6085$, respectively. The mass and total-mass denominators cleared

in these constructions are nonzero on the positive

cone; nonzero stroke leads and signed edge-tie roots

guard the other normalizations. The finite product over all body-label and colour orbits is therefore nonzero. Every positive-point No. 8 first limit therefore forces one of these factors to vanish.

□

5.19. **No. 63: the isolated fully balanced K_4.** The four bodies $3, 4, 5, 6$ form the isolated zw-clique. Its complete first-order balance follows from the pinned diagram and the original rows, without enumerating Type-2 order submatrices.

Lemma 5.11 (Direct No. 63 K_4 balance)**.** *On every germ with the No. 63 strokes and circles, all four internal z rows and all four internal w rows of the K_4 close at their first order.*

Proof. Every edge inside $C = \{3, 4, 5, 6\}$ is a stroke in both colours. No vertex in C is circled in either colour, and no edge from c to $\{1, 2\}$ is a stroke. By the diagram order rules each internal force term in an original z_i or u_i row, $i \in C$, has order -2. The position term and every external force term in that row have order strictly greater than -2. Taking the ε^{-2} coefficient of each of the eight original rows therefore yields precisely its sum of internal force leads equal to zero. This argument applies separately to each row and every valuation

of the pinned graph; no Type-2 enumeration is used. □

Proposition 5.12 (No. 63 mass confinement)**.** *Every No. 63 positive-point first limit satisfies a nonzero necessary polynomial in m_3, m_4, m_5, m_6.*

Proof. If the affine matrix $X = (\mathbf{1}, z, w)$ of the four-body core has rank 3, its symmetric mass-weighted stress matrix Ω annihilates X. The core has nonzero tied edges, so $\operatorname{rank}\Omega = 1$. Every off-diagonal entry is $-m_i m_j q_{ij}^3 \neq 0$. In the rank-one factorization $\Omega = \lambda\Delta\Delta^{\mathsf{T}}$, a zero cofactor Δ_i would zero all off-diagonal entries in row i, which is impossible. Thus every $\Delta_i \neq 0$; the signed ties likewise give $q_{ij} \neq 0$ for all six edges.

Its opposite-edge minors imply, for the nonzero signed inverse distance leads q_{ij},

$$q_{34}^3 q_{56}^3 = q_{35}^3 q_{46}^3 = q_{36}^3 q_{45}^3. \tag{90}$$

After $q_{34} = 1$, the two independent cube-root choices give nine shape sheets in three nonzero parameters. On each sheet the Cayley–Menger polynomial of the four squared distances is nonzero. Indeed, set $q_{34} = q_{35} = q_{36} = q_{56} = 1$ and take $(q_{46}, q_{45}) = (\xi, \eta)$ with $\xi, \eta \in \{1, 473, 31555\} \subset \mathbb{F}_{32029}$. Use the fixed order $(1, 31555, 473)$ for both variables; these are the three distinct cube roots of unity modulo 32029, and the nine evaluations in lexicographic order under that ordering of (ξ, η) are

$$(4, 28239, 3786, 28239, 27289, 32027, 3786, 32027, 4730), \tag{91}$$

All nine values are nonzero. Reduction at this prime sends each characteristic-zero cube-root sheet to one of these values, so no sheet has identically vanishing determinant. The tied shape locus therefore has dimension at most two. The rank-one factorization of Ω and its complementary-area cofactors $\Delta_i \neq 0$ give

$$\left(\frac{m_i}{m_j}\right)^2 = \frac{\Delta_i^2}{\Delta_j^2}\frac{q_{jk}^6}{q_{ik}^6} \qquad (k \neq i, j). \tag{92}$$

Each Δ_i^2 is a polynomial in the three squared distances of the complementary triangle, hence a rational function of the nonzero q_{ij} on its sheet. At fixed shape the displayed equations determine only finitely many mass ratios. The normalized mass image ($m_6 = 1$) therefore has dimension at most two in a three-dimensional mass chart; its algebraic closure has a nonzero defining polynomial. The only denominators in (92) are the already nonzero Δ_j and q_{ik}, so this dimension argument loses no zero-cofactor or zero-distance stratum.

If X has rank at most two, the four points are affine-collinear. There are five independent signed edge choices $e = (e_{35}, e_{36}, e_{45}, e_{46}, e_{56})$ with $e \in \{\pm 1\}^5$, hence 32 branches. Normalize $q_{34} = m_6 = 1$ and put $x = q_{35}$, $y = q_{36}$, $p = x - e_{35}$, $q = y - e_{36}$ and $d = e_{35}y - e_{36}x$. On the nonzero shape torus the three original triangle ties recover

$$q_{45} = -\frac{e_{45}x}{p}, \qquad q_{46} = -\frac{e_{46}y}{q}, \qquad q_{56} = -\frac{e_{56}xy}{d}.$$

The guard $xypqd \neq 0$ is automatic on a genuine branch: if p, q, d respectively vanished, the corresponding unreduced triangle tie would leave the nonzero term $e_{45}x$, $e_{46}y$ or $e_{56}xy$. The original body-3 and body-4 force rows give

$$\begin{aligned} F_4 &= m_4 + e_{36}y^2 + m_5 e_{35}x^2 = 0, \\ F_3 &= m_3 p^2 q^2 - e_{46}y^2 p^2 - m_5 e_{45}x^2 q^2 = 0. \end{aligned} \tag{93}$$

After these substitutions the original body-5 force row is $x^2y^2H_e/(p^2q^2d^2)$, where

$$H_e = e_{56}p^2q^2 - e_{35}e_{46}p^2d^2 + e_{36}e_{45}q^2d^2. \tag{94}$$

Consequently every such branch satisfies the mass equation

$$P_e = \mathrm{Res}_y(\mathrm{Res}_x(H_e, F_4), \mathrm{Res}_x(F_3, F_4)) = 0. \tag{95}$$

There is no cancellation of shape-dependent factors. The exact 32-branch certificate specified in Appendix A computes both inner resultants over the generic mass ring. In each branch they have y-degree eight, and specialization to $(m_3, m_4, m_5) = (2, 3, 5)$ commutes with both inner resultants without changing those degrees. The 32 outer resultants there are nonzero integers of 122–169 bits, so no P_e is the zero mass polynomial. Clear powers of m_6 to homogenize each equation from the chart $m_6 = 1$, multiply by the non-collinear mass equation, and complete the finite label orbit. The resulting polynomial is nonzero because each chart factor is nonzero. □

The local tests replay Lemma 5.11, the nine non-collinear values and the 32 signed-collinear characteristic-zero resultants. Neither a Type-2 cover nor a zero-cofactor exception is needed for this diagram. Independent review of the cited diagram rules and the replay computations remains a publication-level audit.

5.20. **No. 66: direct algebraic mass confinement.** The No. 66 diagram has a z-only complete graph on $C = \{3,4,5,6\}$, the three w-only edges $12, 16, 26$, and w-circles only at bodies 1 and 2. There are no other strokes or circles. We derive its core equations directly from the original six-body rows.

For a pure z-edge, writing $a_{ij} = \operatorname{ord} z_{ij}$ and $b_{ij} = \operatorname{ord} w_{ij}$, the force orders give $a_{ij} + 3b_{ij} = 4$ and $a_{ij} < 1$. On each triangle, the minimum of its three difference orders must occur at least twice. If its a_{ij} were unequal, their maximum would be unique, and the decreasing relation $b_{ij} = (4 - a_{ij})/3$ would give a unique minimum in the other colour. Thus all three orders agree. Applying this to the overlapping core triangles gives

$$\operatorname{ord} z_{ij} = \alpha \in (-2, 1), \qquad \operatorname{ord} w_{ij} = \beta := \frac{4-\alpha}{3} > 1 \qquad (i, j \in C,\ i \neq j).$$

Here the lower bound follows from the absence of z-circles.

Let b_1, b_2 be the nonzero order--2 coefficients of w_1, w_2. The global w-centre gives $m_1 b_1 + m_2 b_2 = 0$, so positivity implies $b_1 \neq b_2$. As body 6 is uncircled, all three outer w-differences have order -2. Their w-stroke force equations then give

$$\operatorname{ord} z_{12} = \operatorname{ord} z_{16} = \operatorname{ord} z_{26} = 2, \qquad \operatorname{ord} \delta_{12} = \operatorname{ord} \delta_{16} = \operatorname{ord} \delta_{26} = 0.$$

For $i \in \{1,2\}$ and $j \in \{3,4,5\}$, the triangle identity through body 6 gives $\operatorname{ord} z_{ij} = \alpha$ and $\operatorname{ord} w_{ij} = -2$. Hence these crossed z-forces have order $3 - \alpha/2 > 2$. The original z_1, z_2 rows, whose other forces have order 2, imply

$$z_1, z_2, z_6 = O(\varepsilon^2).$$

We may therefore define the core shape coefficients by

$$z_i = Z_i \varepsilon^{\alpha} + o(\varepsilon^{\alpha}), \qquad w_i - w_6 = W_i \varepsilon^{\beta} + o(\varepsilon^{\beta}) \qquad (i \in C),$$

with $Z_6 = W_6 = 0$. All core differences $Z_i - Z_j$ and $W_i - W_j$ are nonzero, and the global z-centre gives $\sum_{i=3}^{5} m_i Z_i = 0$. Write

$$\delta_{ij} = D_{ij} \varepsilon^{-(\alpha+2)/3} + o\big(\varepsilon^{-(\alpha+2)/3}\big), \qquad D_{ij}^2 (Z_i - Z_j)(W_i - W_j) = 1,$$

and set

$$F_i^z = \sum_{j \in C \setminus \{i\}} m_j D_{ij}^3 (Z_i - Z_j), \qquad F_i^w = \sum_{j \in C \setminus \{i\}} m_j D_{ij}^3 (W_i - W_j).$$

Every internal z-force has order -2. The position terms are later, as are the exterior forces: those at bodies 3,4,5 have order $3 - \alpha/2 > 2$, and those at body 6 have order 2. Thus $F_i^z = 0$ for every $i \in C$.

The internal w-force order and the exterior w-force order at bodies 3,4,5 are, respectively,

$$f = -\frac{4\alpha + 2}{3}, \qquad e = 1 - \frac{3\alpha}{2}, \qquad e - f = \frac{10 - \alpha}{6} > 0, \qquad \beta - f = \alpha + 2 > 0.$$

The three original rows give $w_3, w_4, w_5 = O(\varepsilon^f)$. Since all core w-differences have the later order β, all four core coordinates share a coefficient V at order f, possibly zero. This common coefficient is distinct from the relative shape coefficients W_i at order β. Taking the coefficient at order f in the three rows gives precisely

$$\begin{aligned} &F_i^z = 0 \quad (i \in C), \quad F_3^w = F_4^w = F_5^w = V, \\ &\sum_{i=3}^{5} m_i Z_i = 0, \qquad\qquad Z_6 = 0. \end{aligned} \tag{96}$$

The body-6 exterior w-forces have earlier order -2, so no assertion $F_6^w = V$ is made. The proof below treats both zero and nonzero drift directly from (96).

Finally, the squared-distance orders are $(2\alpha+4)/3$ on the six core edges, $\alpha-2$ on $13,14,15,23,24,25$, and zero on $12,16,26$. Consequently

$$\operatorname{ord} P_{36} = \operatorname{ord}\bigl(r_{36}^2 r_{12}^2 r_{14}^2 r_{15}^2 r_{24}^2 r_{25}^2 r_{45}^2\bigr) = 2\frac{2\alpha+4}{3} + 4(\alpha-2) = \frac{16}{3}(\alpha-1) < 0.$$

Thus P_{36} has a pole on this germ; the same is true of P_{46} and P_{56}.

Proposition 5.13 (No. 66 mass confinement)**.** *The signed core conditions* (96)*, together with the six nonzero distance ties, confine* (m_3,m_4,m_5,m_6) *to a proper algebraic subset of the four-dimensional mass space. This holds for both* $V=0$ *and* $V\neq 0$*. Consequently every No. 66 first limit satisfies a nonzero necessary mass polynomial, without an additional geometric coverage hypothesis or a computational certificate for this diagram.*

Proof. We work over $\mathbb{C}$ and retain all four mass variables; no mass is normalized to one. First translate W so that $W_6=0$. The original 36 tie gives $D_{36}^2 Z_3 W_3 = 1$, so the substitutions

$$Z_i \longmapsto Z_i/Z_3, \qquad W_i \longmapsto W_i/W_3, \qquad D_{ij} \longmapsto D_{ij}/D_{36}$$

preserve all the ties, the zero z forces and the equality of the three w forces, with V rescaled. They also preserve the z centre. Thus we may impose

$$(Z_3,W_3,D_{36}) = (1,1,1), \qquad (Z_6,W_6) = (0,0). \tag{97}$$

This uses only nonzero factors already guaranteed by the original ties and leaves the mass projection unchanged.

Let $\mathcal{S}$ be the normalized incidence defined by these conditions, with the four masses nonzero. It is a finite-type quasi-affine algebraic set. Suppose an irreducible component of $\mathcal{S}$ dominates the four-dimensional mass space, and let L be its function field. The four masses are then algebraically independent over $\mathbb{C}$. In characteristic zero they extend to a separating transcendence basis of L, so their differentials are linearly independent over L.

Put

$$U_C = \sum_{\substack{i<j\\ i,j\in C}} m_i m_j D_{ij}, \qquad Q_i = \sum_{\substack{j\in C\\ j\neq i}} m_j D_{ij}.$$

Pairing the internal forces and using the ties gives $U_C = \sum_{i\in C} m_i W_i F_i^z = 0$. Differentiating the same ties gives

$$\mathrm{d}D_{ij} = -\frac{D_{ij}^3}{2}\bigl((W_i-W_j)\,\mathrm{d}(Z_i-Z_j) + (Z_i-Z_j)\,\mathrm{d}(W_i-W_j)\bigr).$$

It follows exactly that

$$\mathrm{d}U_C = \sum_{i\in C} Q_i\,\mathrm{d}m_i - \frac{1}{2}\sum_{i\in C} m_i\bigl(F_i^w\,\mathrm{d}Z_i + F_i^z\,\mathrm{d}W_i\bigr).$$

Here $F_i^z=0$, $\mathrm{d}Z_6=0$ and $F_3^w=F_4^w=F_5^w=V$. Differentiating $\sum_{i=3}^5 m_i Z_i = 0$ therefore yields

$$0 = \mathrm{d}U_C = \sum_{i\in C}\left(Q_i + \frac{V}{2}Z_i\right)\mathrm{d}m_i.$$

Independence of the four mass differentials forces $Q_i = -(V/2)Z_i$. In particular, $Z_6=0$ gives

$$m_3 D_{36} + m_4 D_{46} + m_5 D_{56} = 0. \tag{98}$$

This identity has been deduced only on the hypothetical mass-dominating component; it is not imposed on every individual solution of the original core equations. No division by V has occurred.

Write

$$(a,b,c,d) = (m_3,m_4,m_5,m_6), \qquad (x,y) = (Z_4,Z_5),$$
$$(u,v) = (D_{46},D_{56}), \qquad (p,q,r) = (D_{34},D_{35},D_{45}).$$

The spoke ties and (98) give

$$W_4 = \frac{1}{u^2 x}, \qquad W_5 = \frac{1}{v^2 y}, \qquad v = -\frac{a+bu}{c}. \tag{99}$$

The z centre and the original body-6 z row are

$$a + bx + cy = 0, \qquad a + bu^3 x + cv^3 y = 0. \tag{100}$$

If $u^3 = v^3$, these two equations force $u^3 = v^3 = 1$. Equation (98) then gives $a + b\eta + c\zeta = 0$ for cube roots of unity η, ζ. This contradicts algebraic independence of a, b, c; equivalently, the product of these nine nonzero linear forms is a nonzero mass polynomial. Hence $u^3 - v^3 \neq 0$ at the generic point under consideration, and (100) gives

$$x = \frac{a(v^3-1)}{b(u^3-v^3)}, \qquad y = \frac{a(1-u^3)}{c(u^3-v^3)}. \tag{101}$$

Let $K = \mathbb{C}(a, b, c)$. All of v, x, y, W_4, W_5 are now rational functions of u over K. The three remaining signed ties are

$$\begin{aligned} p^2(1-x)(1-W_4) &= 1, \\ q^2(1-y)(1-W_5) &= 1, \\ r^2(x-y)(W_4-W_5) &= 1. \end{aligned} \tag{102}$$

The original body-3 and body-4 z rows respectively give

$$d = -bp^3(1-x) - cq^3(1-y), \tag{103}$$

$$0 = ap^3(x-1) + cr^3(x-y) + du^3 x. \tag{104}$$

If u is algebraic over K, all quantities on the right of (103) are algebraic over K, so d is algebraic over K, contradicting mass dominance. We may therefore assume that u is transcendental over K. Then $E = K(u, p, q, r) \subset L$ is a finite extension of $K(u)$, of degree at most eight, and (104) is an identity in E.

Extend the valuation at $u = \infty$ from $K(u)$ to E and normalize it so that $\nu(u) = -1$ and $\nu(K^*) = 0$. Such an extension exists for every finite field extension; its value group can have fractional elements. Put $T = b^3 + c^3$. Equations (99)–(101) give

$$x = x_0 + O(u^{-1}), \quad y = y_0 + O(u^{-1}), \qquad x_0 = -\frac{ab^2}{T}, \quad y_0 = -\frac{ac^2}{T}.$$

Algebraic independence of a, b, c makes $x_0, y_0, 1-x_0, 1-y_0, x_0-y_0$ all nonzero in K. Moreover the two w coordinates have the same leading term:

$$W_4 = -\frac{T}{ab^2}u^{-2} + O(u^{-3}), \qquad W_5 = -\frac{T}{ab^2}u^{-2} + O(u^{-3}).$$

Their difference cannot be identically zero, by the third tie in (102). Thus $\nu(W_4 - W_5) \geq 3$, and that tie implies $\nu(r) \leq -3/2$. The first two ties give $\nu(p) = \nu(q) = 0$, and (103) gives $\nu(d) \geq 0$. In (104) the three terms have valuations

$$\nu\big(ap^3(x-1)\big) = 0, \qquad \nu\big(cr^3(x-y)\big) \leq -\frac{9}{2}, \qquad \nu(du^3 x) \geq -3.$$

The middle term is uniquely of least valuation and cannot cancel, a contradiction. This comparison applies to every signed field component of the finite cover. The place over infinity is used only to test an algebraic identity; it need not be a physical first-limit branch.

No irreducible component of $\mathcal{S}$ can therefore dominate mass space. Since there are finitely many components, the union of the Zariski closures of their mass projections is proper. The incidence is defined over $\mathbb{Q}$, so elimination gives a nonzero polynomial in $\mathbb{Q}[m_3, m_4, m_5, m_6]$ vanishing on its entire mass image. The mass image is a cone, since $(m, V) \mapsto (tm, tV)$ preserves the equations; a nonzero homogeneous polynomial may consequently be chosen. Viewing it as a

six-mass polynomial and taking its finite label-orbit product gives P_{66}. Special masses at which a generic denominator or a displayed limit coefficient vanishes have not been discarded: the argument rules out dominance of the full original incidence, and its proper closed mass image contains those specializations as well. □

5.21. **No. 88: crossed torque and the four-body root norm.** The original pair/K_4 chart has bodies $1,2,3,4$ in the K_4 core and $5,6$ in the separated pair. Its first crossed torque and eight signed ties require

$$T=\sum_{i=1}^{4}\sum_{a=5}^{6} m_i m_a C_{ia}=0, \qquad C_{ia}^2=-\frac{1}{Z_a W_i}. \tag{105}$$

Write $u_i=m_i^2/W_i$ and let e_j be the elementary symmetric functions of $u_1,\dots,u_4$. The product over all 16 signs of a four-root sum is the square of

$$H=(e_1^2-4e_2)^2-64e_4. \tag{106}$$

All denominators here are nonzero on the circled core. If $H\neq 0$, no signed four-root sum vanishes. For a fixed K_4 shape, (105) then implies $m_5^3A_\eta^2+m_6^3A_\xi^2=0$ with both signed shape sums $A_\eta, A_\xi\neq 0$; this is a nontrivial equation in the two pair masses. It remains to confine K_4 shapes with $H=0$ to a proper core-mass locus.

Proposition 5.14 (No. 88 positive-mass six-mass locus)**.** *Every No. 88 positive-point first limit lies on a nonzero (possibly unexpanded) polynomial in all six masses.*

Proof. For affine-collinear K_4 shapes, the two original force-slope rows have rank two for generic shape. To replay every sign, normalize the four positions to $\xi=(0,1,u,v)$, set $L=\prod_{i<j}(\xi_i-\xi_j)^2$, and choose $\varepsilon_{ij}\in\{\pm1\}$ with $\varepsilon_{12}=1$. Define

$$F_i=\sum_{j>i}\frac{m_j\varepsilon_{ij}}{(\xi_i-\xi_j)^2}-\sum_{j<i}\frac{m_j\varepsilon_{ji}}{(\xi_j-\xi_i)^2} \qquad (1\le i\le 4).$$

The 2×4 mass-coefficient matrix of $L(F_3+(u-1)F_1-uF_2,\ F_4+(v-1)F_1-vF_2)$ has six maximal minors. For each of the $2^5=32$ independent sign classes their gcd in $\mathbb{Z}[u,v]$ is a nonzero constant, so any rank drop has zero-dimensional shape support. At $(u,v)=(2,3)$ the matrix has rank two in all 32 cases. In the standard increasing-free-column RREF kernel basis (v_0,v_1), the single choice $m=v_0-10v_1$ has nonzero total mass, nonzero centred coordinates, $H\neq 0$, and $F_2-F_1\neq 0$ for every sign class. The global sign reversal supplies the other 32 choices. At the finitely many rank-drop shapes these two rows cannot both have rank zero: that would equate the three signed cubes at every vertex and force all six K_4 cubes equal, contradicting their Gram ties. Thus even the rank-drop mass fibers have proper image in normalized four-mass space.

For non-collinear shapes, let $M_Q=\sum_{i=1}^4 m_i=1$ and $q_{ij}=1-D_{ij}^3$. The mass-weighted K_4 stress annihilates $(1,Z,W)$ and has rank one; rank zero would again imply six equal cubes. When all q_{ij} are nonzero, the complementary-area cofactors Δ_i fix the normalized mass ratios at each shape by

$$\frac{m_i}{m_j}=\frac{\Delta_i q_{jk}}{\Delta_j q_{ik}} \qquad (k\neq i,j). \tag{107}$$

Writing the six signed internal roots as $(x,y,z,t,v,w)=(D_{12},D_{13},D_{14},D_{23},D_{24},D_{34})$, two independent opposite-edge conditions are

$$(1-y^3)(v^3-1)+(1-z^3)(1-t^3)=0, \tag{108}$$

$$(1-x^3)(w^3-1)+(1-z^3)(1-t^3)=0. \tag{109}$$

Off $(1-x^3)(1-y^3)=0$, these monic cubic recoveries form a nine-sheet finite cover of (x,y,z,t). The four-point Cayley–Menger numerator has total degree ten in (t,v,w), so its finite norm has t-degree at most 90. At $(x,y,z)=(52,26,44)$ modulo 101, the recorded fiber has irreducible degree

90 with nonzero leading coefficient. Since both recovery cubics are monic, the normalized finite norm is an integral resultant after inverting $(1-x^3)(1-y^3)$. Removing its base-variable content, the full-degree irreducible specialization rules out a nontrivial factorization over $\mathbb{Q}(x,y,z)[t]$ by Gauss's lemma: both factors would retain positive t-degree modulo 101. This rational generic-fiber result does not by itself establish geometric irreducibility after extending constants, nor exclude extra components on the denominators' zero sets.

On each of the nine cube-phase sheets a component vertical over a divisor of (x,y,z) would divide both the Cayley–Menger value at $t=1$ and its derivative along that sheet. Their exact three-variable norms have constant GCD 4 on each phase orbit. The five phase-orbit representatives and their conjugates exhaust all nine sheets, so no vertical divisor survives where the cubic leading coefficients are units. If $1-x^3=0$ or $1-y^3=0$, one of q_{12}, q_{13} vanishes; the rank-one stress forces a complete zero star, treated separately below. The nine nonzero signed roots are units by their original ties. At the smooth full-tie point $(x,y,z,t,v,w) = (2,3,4,75,37\omega,18\omega)$ over $\mathbb{F}_{101^2}$, $\omega^2+\omega+1=0$, both orientations have nonzero H (coordinates $(93,99)$ and $(99,57)$ in the basis $(1,\omega)$). Its Jacobian in (t,v,w) is $(97,63)\neq 0$, so the point lifts in characteristic zero with all distance, stress and circle guards still units. On the saturated torus the degree-90 irreducible norm gives one component over $\mathbb{Q}(x,y,z)$; its geometric components are Galois conjugate. Since the equations and H have rational coefficients, vanishing of H on one such component would imply vanishing on all. The two orientation values above rule this out for either lift of the affine shape. The same smooth point has the three distinct spoke cubes and two distinct orientations, so neither the cube-coincidence equations nor the orientation discriminant can vanish identically on a geometric component: such a component would have all its Galois conjugates inside that locus, contrary to this point. Their intersections with the main three-dimensional shape cover have dimension at most two. The monic recovery cubics rule out components escaping to infinity in (v,w) while their leading coefficients are units; $t=0$ or a zero signed root is forbidden by the original distance ties. Thus every non-collinear first-limit shape lies either in the generic cover, a proper boundary divisor of that cover, or the zero-star stratum below. On every nonstar locus the mass ratios in (107) remain finite and uniquely fixed by the shape. Consequently $H=0$ cuts the generic shape dimension from three to two, while the already lower-dimensional boundary images are proper without a further cut.

If an edge has $q_{ij}=0$, rank-one factorization forces an entire zero star; two zero stars would make two tied vertices coincide. Fixing its three cube-one edges leaves finitely many phases of the three other roots. For any free nonstar edge jk, the Cayley–Menger determinant is quadratic in its squared distance r_{jk}^2, with leading coefficient $-2r_{i\ell}^2\neq 0$ on the opposite spoke $i\ell$. Thus it is not an identity on any zero-star phase. Each star shape has dimension at most two; its three nonstar mass ratios are fixed, with at most one remaining mass scale s. Taking the star's $W_i=0$, the nonzero noncircle coefficient K gives $\lim_{s\to 0} s^4 H = K^{-4}\neq 0$. Thus $H=0$ also removes that remaining free scale.

We have a finite union of proper $H=0$ images in the four core masses. Outside it, Yu–Zhu's finite complex four-body fibers at positive core masses [7] allow the nonzero pair-mass equation after (106) to be normed over the finite shapes and signed roots. Clearing their nonzero denominators gives a nonzero necessary six-mass polynomial. It says nothing about finiteness on its positive zero set. □

Lemma 5.15 (Certified root and signed-sum bounds)**.** *For a degree-five polynomial p, suppose five disjoint rational disks $|t-c|<r$ each satisfy*

$$|p(c)|_1 + \sum_{k=2}^{5} \frac{|p^{(k)}(c)|_1}{k!} r^k < r|p'(c)|_\infty,$$

where $|x+iy|_1 = |x|+|y|$ *and* $|x+iy|_\infty = \max(|x|,|y|)$. *Then the disks contain all five roots, one per disk. If outward rational rectangles enclose the resulting signed root sums, a strictly positive or strictly negative real or imaginary interval excludes zero.*

Proof. On each disk boundary the left side bounds the remainder after $p'(c)(t-c)$, while the right side is a lower bound for the linear term. Rouché's theorem gives one root per disk. Their disjointness and the degree give exhaustion. Outward interval operations contain every algebraic root and product used in a signed sum; separation of either coordinate from zero therefore excludes the sum from zero. □

5.22. **No. 93: an all-order three-body root-sum norm.** Here the isolated fully w-circled triangle is 123, while body 4 of the z-triangle 456 is not circled. The internal inverse distances of 123 have finite nonzero limits; its w positions have order -2 and its z differences order 2.

Proposition 5.16 (No. 93 lower-mass locus)**.** *The original graph valuation cover and crossed-tie remainder bounds are established below. The exact shape enumeration and certified root intervals are replayed in Appendix A. Every No. 93 positive-point first limit then lies on the zero set of a nonzero polynomial in* m_1, m_2, m_3, *independent of the other masses.*

Proof. Put $\alpha = \operatorname{ord}_\varepsilon z_4$. Non-circling gives $\alpha > -2$. The missing 4-to-123 w strokes imply $D_{4j} = o(1)$ for $j \le 3$. Summing the original lower z rows with mass weights cancels their internal forces. By the original crossed ties, $D_{4j}^3(z_j - z_4) = D_{4j}/(w_j - w_4) = o(\varepsilon^2)$: the lower w_j have order -2, while the uncircled w_4 is strictly later. This estimate does not assume $\alpha < 2$. The 5/6-to-123 sources have order at least 4. The lower mass centre and its order-two internal differences thus give $z_j = O(\varepsilon^2)$ for $j = 1,2,3$. If $\alpha \ge 2$, a 4-to-j tie would force D_{4j} to stay bounded away from zero, a forbidden w-stroke. Hence $-2 < \alpha < 2$.

Write $s = 1 - \alpha/2 \in (0,2)$. The 4-to-$j$ inverse distances have order s, whereas the 5/6-to-j distances have order 2. For $z_4 = \zeta\varepsilon^\alpha + o(\varepsilon^\alpha)$ and $w_j = W_j\varepsilon^{-2} + o(\varepsilon^{-2})$, the original signed crossed ties, inserted in the upper-cluster torque, give its strictly earliest term

$$\sum_{j=1}^{3} m_j K_j = 0, \qquad K_j^2 = -\frac{1}{\zeta W_j}, \qquad K_j \neq 0. \tag{110}$$

The corrections from the original ties have strictly later relative orders $2s, 4, 2s+4$; in particular this derivation includes $\alpha = 0$. The lower original z, w rows close to a centred multiplier-one three-body boundary because the 4-to-j z source has order $3s + \alpha = 3 - \alpha/2 > 2$, and the lower w corrections occur after order -2.

Independent centred z, w shapes are Lagrange type: their three internal cubes admit 27 phase choices, each with at most two tied shapes. Proportional shapes are Euler type, governed by eight signed quintics with at most five roots each. Thus there are at most $54 + 40$ closed lower shapes and eight crossed square-root signs for each. Take the product of the sums in (110) over all signs and then the finite algebraic norm over the signed shape roots. Square-root flips only permute the eight factors; after clearing nonzero circle and leading-coefficient denominators the result is a polynomial in m_1, m_2, m_3. If a leading coefficient or a cleared denominator vanishes on a special mass slice, include its nonzero mass polynomial (or its finite shape norm) as an additional factor; such a slice is not discarded.

At $(m_1, m_2, m_3) = (1, 2, 4)$, disjoint rational root enclosures for all 40 Euler and 54 Lagrange shape labels, with outward complex-rational interval evaluation, exclude zero from all $(40+54)\cdot 8 = 752$ signed sums. The independent root count uses Lemma 5.15 for the Euler quintics and exact quadratic discriminants for the Lagrange phases. For each signed sum, the interval replay gives a certified lower bound for its ℓ^∞ distance from zero. The least such bound is exactly 23830488781026267/781250000000000.

Its floor at scale 10^{24} is 30503025639713621760000000. The integer-only replay in Appendix A checks this margin and every root label. The exact root/label coverage therefore certifies nonidentity of the polynomial; it does not assert a germ on its positive zero set. □

5.23. **No. 94: exceptional-mass first limits.** No. 94 has the w-only K_4 on 1234, with w-circles at $1, 2, 3$, and a z-only upper triangle 456 circled at $5, 6$. The body 4 belongs to both components and is not circled. Original rows and ties exclude precisely two equal spoke inverse-distance cubes. All-equal spokes require a nonzero wing-mass condition. If the four-body z-centre $G = \sum_{i=1}^{4} m_i Z_i$ vanishes, the original fourth z row closes and the fully balanced K_4 condition applies.

For pairwise-distinct spokes with $G \neq 0$, the original wing w rows and the first independent z-row difference express all three wing-edge cubes in terms of the three signed spoke roots. The three remaining signed wing ties are quadratics in the common spoke scale. Form all three raw pairwise resultants of these quadratics and clear their reduced rational denominators. At the positive mass $(1, 2, 4, 1)$ modulo 17, the bivariate numerator multidegrees in (x, y) are $(102, 102)$, $(96, 102)$ and $(102, 96)$, and their common GCD in $\mathbb{F}_{17}[x, y]$ is 1. The full bidegrees are retained. This degree-preserving finite-field specialization rules out a common positive-dimensional factor in the characteristic-zero generic shape chart. In particular, the first pair at $x = 6$ has full y-degree 102 and GCD 1, while the first resultant at $y = 2, 6$ has full x-degree 102 and GCD 1; these slice checks are also replayed. Thus the generic signed K_4 shape fiber is finite. The inverted circle, collision, spoke and determinant guards are not imposed after specialization; a mass-dependent degree drop is confined to a separate proper lower-mass locus. The zero weighted-root-sum branch is defined by $u_j = -m_j^2/W_j$ for $j = 1, 2, 3$ and

$$H_3 = (u_1 + u_2 + u_3)^2 - 4(u_1u_2 + u_1u_3 + u_2u_3). \tag{111}$$

Lemma 5.17 (Wing-dominant No. 94 zero-sum tier)**.** *Suppose in a mass specialization toward an $H_3 = 0$ No. 94 boundary all three wing W-differences remain units, all three wing-edge inverse-distance cubes have the same nonzero leading order strictly before the spoke and coordinate terms, and their signed ties retain nonzero leading coefficients. Then the limiting positive wing masses lie on a proper algebraic locus.*

Proof. Write $(a, b, c) = (m_1, m_2, m_3)$ and normalize the relative wing w-leads to $(1, t, 0)$, with $t(1-t) \neq 0$. The original three w rows at their common earliest order force, after setting $D_{13}^3 = 1$ by a common nonzero scale,

$$D_{12}^3 = -\frac{c}{b(1-t)}, \qquad D_{23}^3 = -\frac{a}{bt}.$$

Let $U = 1/(D_{12}^2(1-t))$ and $V = 1/(D_{23}^2 t)$. The three original signed ties and additivity of the z-differences give $U + V = 1$, while $U^3 = b^2/[c^2(1-t)]$ and $V^3 = b^2/(a^2 t)$. Put

$$D = a^2c^2t(1-t), \quad N = D - a^2b^2t - b^2c^2(1-t), \quad F = N^3 - 27b^4D^2.$$

The exact two-cubic resultant of $U^3 - p$ and $(1-U)^3 - q$ is $(1-p-q)^3 - 27pq$; hence $F = 0$. The original w centre makes the three circled coefficients proportional to

$$(L_1, L_2, L_3) = (b(1-t) + c,\ a(t-1) + ct,\ -a - bt),$$

with every $L_i \neq 0$. Clearing their squares in $H_3 = 0$ yields the quartic

$$K = \sum_{i=1}^{3} m_i^4 \prod_{j \neq i} L_j^2 - 2 \sum_{1 \leq i < j \leq 3} m_i^2 m_j^2 L_i L_j L_k^2, \qquad \{k\} = \{1, 2, 3\} \setminus \{i, j\}.$$

At $(a, b, c) = (1, 3, 5)$ modulo 17, F and K retain degrees six and four; their signed coefficient lists are $(15, 15, 12, 5, 11, 8, 4)$ and $(10, 16, 10, 13, 14)$, and their resultant is $12 \neq 0$. Thus $\mathrm{Res}_t(F, K)$ is a nonzero necessary polynomial for this tier. It does not cover spoke-comparable or spoke-dominant limits. □

Lemma 5.18 (Arithmetic projective-fiber criterion)**.** *Let B be an integral open mass chart over $\mathbb{Z}_{(17)}$, with all inverted mass denominators units at a point m_0 of its special fiber. Let $X \to B$ be the projective closure, formed before reduction modulo 17, of an incidence cut out by the cleared original rows and signed ties on its guarded affine chart. If the entire geometric projective fiber $X_{m_0}(\overline{\mathbb{F}}_{17})$ is empty, then the characteristic-zero mass image of that incidence has proper algebraic closure in $B_{\mathbb{Q}}$. In particular it admits a nonzero necessary mass polynomial after clearing mass denominators.*

Proof. The morphism $X \to B$ is proper, so its image is closed. Since it misses m_0, this closed image cannot contain the dense generic fiber $B_{\mathbb{Q}}$; otherwise its closure would be all of the integral base B. Its intersection with $B_{\mathbb{Q}}$ is therefore a proper closed subset of the affine mass chart. A nonzero polynomial vanishes on it, and restriction to the guarded incidence preserves the necessary condition. □

The hypothesis concerns the closure of the guarded incidence, not the closure obtained by saturating away guard-zero points only after specialization: the latter can discard genuine specializations of generic solutions. All finite and infinite spoke, circle, collision, geometric-determinant and free-cube boundary charts must be accounted for when applying Lemma 5.18.

Lemma 5.19 (No. 94 zero-sum lower-mass image)**.** *The positive-mass No. 94 first limits on which $H_3 = 0$ and the three spoke cubes are pairwise distinct lie on a proper algebraic subset of the four lower masses.*

Proof. The signed spoke-gap ratio r satisfies a quartic $Q(r)$: for $q_i = D_{i4}^3$, put $h = q_3 - q_2$ and $q_2 = q_1 + rh$, $q_3 = q_1 + (r+1)h$. Clearing the denominators in $H_3 = 0$ gives, with $a = m_1^3$, $b = m_2^3$, $c = m_3^3$,

$$Q(r) = (ar - c)^2(r+1)^2 + b^2r^2 + 2br(r+1)(ar+c) = 0.$$

Its leading and constant coefficients are m_1^6 and m_3^6. It retains degree four and is irreducible at the positive mass $(1, 3, 5, 1)$ modulo 17, so the generic ratio cover is finite; its $r = 0, -1, \infty$ boundaries are not roots. The original wing w rows give

$$(W_1, W_2, W_3) = h\left(-\frac{1}{m_1}, \frac{1+r}{m_2}, -\frac{r}{m_3}\right).$$

If $h = 0$, all three spoke cubes are equal and belong to the separate all-equal branch. If $h \neq 0$, a wing circle or a wing collision forces respectively $r = -1, 0, -(m_1+m_2)/m_1, m_3/m_1$, or $-m_3/(m_2+m_3)$; $W_1 = 0$ would itself force $h = 0$. At $(m_1, m_2, m_3) = (1, 3, 5)$ modulo 17, the exact values of Q at those five ratios are $(15, 2, 5, 7, 12)$, all nonzero. Thus no circle or wing-collision guard can vanish on the surviving quartic ratio fiber, even along a mass specialization. The three original signed wing ties have explicitly guarded quadratic coefficients in their common free cube. At $(m_1, m_2, m_3, m_4) = (1, 3, 5, 1)$ modulo 17, the raw pairwise resultant multidegrees are $(102, 102)$, $(96, 102)$ and $(102, 96)$, and the H_3 numerator has multidegree $(12, 12)$. Eliminating y between each resultant and H_3 gives three degree-1608 polynomials in x. Their common factor has degree 1120 and factors exactly as

$$(x^4 + 12x^3 + 2x + 6)^{10}(x^2 + x + 1)^{360}(x - 1)^{360}.$$

Over the exact finite extension for each irreducible factor, the common y-fiber degrees are respectively $12, 12, 1$ for the factors of degrees $1, 2, 4$. The degree-four fiber meets only the geometric-determinant guard; the degree-one and degree-two fibers lie on the listed circle, collision, spoke or determinant guards. Saturating each fiber by all 13 explicit guards leaves degree zero. These computations establish the empty guarded affine $H_3 = 0$ fiber at the stated witness; the remaining projective boundary charts are checked separately below.

It remains to check the compactification rather than infer properness from an affine empty fiber. Suppose $D_{14} \to 0$ while D_{24}, D_{34} remain units. The five nonzero Q values above keep all

W_i and $W_i - W_j$ units. The spoke ties make $Z_2 - Z_4, Z_3 - Z_4$ units but $Z_1 - Z_4$ a pole, even if every Z_i itself moves. Thus the original signed wing ties give

$$\frac{D_{1j}^2}{D_{14}^2} \longrightarrow \frac{W_1}{W_1 - W_j} \neq 0 \qquad (j = 2, 3).$$

Both adjacent wing cubes and the spoke cube tend to zero; the original body-1 w row then gives $W_1 = 0$, a contradiction. Permuting the three wing bodies handles any single zero spoke. Two vanishing spokes with the third a unit instead force respectively $r = 0, -1$, or ∞ on the ratio cover, all forbidden by Q at the test fiber. This argument permits mass-varying coordinate poles.

If all three spokes vanish together, normalize the wing W_i by the common z, w rotation. Since Q excludes the five circle/collision ratios, all W_i and their pairwise differences remain units. The triangle inequality on the three $Z_i - Z_j$ orders, together with the original wing w rows, rules out a vanishing wing root: at least two wing roots would vanish and their common vertex would have nonzero W_i but no surviving w force. If a wing root diverges, each endpoint must have a second wing force at the same earliest order; hence all three wing roots diverge with the same order. The mass-weighted three original wing z rows and the spoke ties give

$$\sum_{i=1}^{3} m_i Z_i = m_4 \sum_{i=1}^{3} \frac{m_i D_{i4}}{W_i} \longrightarrow 0.$$

Since $m_1 + m_2 + m_3 = 9$ is a unit modulo 17 and $Z_i - Z_j \to 0$, every $Z_i \to 0$. The dominant wing z, w forces therefore both vanish. They are exactly the wing-dominant zero-force tier of Lemma 5.17, whose F, K resultant is $12 \neq 0$ at the test fiber.

If instead all wing roots remain units, their signed ties and the same mass-weighted identity keep every Z_i finite. The limiting wing is a noncolliding three-body central configuration with $H_3 = 0$. For a collinear wing set $\xi = (0, 1, 1 + t)$, $p_i = \xi_i - (m_2 + m_3(1 + t))/M_3$ with $M_3 = m_1 + m_2 + m_3$, and

$$A_1 = -m_2\varepsilon_{12} - \frac{m_3\varepsilon_{13}}{(1+t)^2}, \qquad A_2 = m_1\varepsilon_{12} - \frac{m_3\varepsilon_{23}}{t^2}.$$

Its eight signed Euler quintics are $M_3 t^2(1 + t)^2(p_2A_1 - p_1A_2)$, for $(\varepsilon_{12}, \varepsilon_{13}, \varepsilon_{23}) \in \{\pm 1\}^3$. The primitive H_3 numerator on the centred wing shape $(0, 1, 1 + t)$ is

$$8225t^4 + 24290t^3 + 15339t^2 - 3838t + 209;$$

each quintic retains degree five and has gcd 1 with this quartic modulo 17. For the 27 non-collinear cube phases let $\omega^2 + \omega + 1 = 0$ over $\mathbb{F}_{17^2}$. Normalize the three squared distances to $(1, \omega^j, \omega^k)$, with $j, k \in \{0, 1, 2\}$; the common phase has no effect. If the z shape is $(0, 1, t)$, its two orientations satisfy

$$t^2 + (\omega^k - 1 - \omega^j)t + \omega^j = 0.$$

Set $(s_{12}, s_{13}, s_{23}) = (1, \omega^j, \omega^k)$, center $W^0 = (0, s_{12}, s_{13}/t)$ by the three wing masses, and put $L_i = M_3 W_i$. The cleared norm is

$$K_3 = \sum_i m_i^4 \prod_{j \neq i} L_j^2 - 2 \sum_{i<j} m_i^2 m_j^2 L_i L_j L_k^2, \qquad \{k\} = \{1, 2, 3\} \setminus \{i, j\}.$$

The resultants of the displayed quadratic and K_3 have, for rows $j = 0, 1, 2$ and columns $k = 0, 1, 2$, residues

$$\begin{pmatrix} 4 & 9 & 9 \\ 11 + 14\omega & 4 + 12\omega & 8 + 15\omega \\ 14 + 3\omega & 10 + 2\omega & 9 + 5\omega \end{pmatrix} \pmod{17}.$$

All nine are nonzero. A common phase shift leaves each resultant unchanged, so the 27 phases and both orientations are exhausted.

If one spoke is asymptotically smaller than the other two, Q keeps the latter comparable and excludes every other mixed ratio pattern. Suppose it is D_{14}; by its original signed ties the adjacent wing roots D_{12}, D_{13} have its order. If $D_{14} \to 0$, the original body-1 w row again contradicts the unit W_1. Otherwise its original z row and the spoke tie bound Z_1, Z_4 by $O(D_{14})$. The w_2, w_3 rows force D_{23} to have the order of D_{24}, D_{34}; a still earlier 23 force could not be cancelled. Those three roots dominate all edges incident to body 1, so the leading three-body 234 forces vanish in both colours. Their signed ties make the two shapes proportional. With masses $(m_2, m_3, m_4) = (3, 5, 1)$ the normalized shape t satisfies one of $5t^2 = \pm 1$, $3(1-t)^2 = \pm 1$. The four integer resultants in sign order $(--, -+, +-, ++)$ are $(469, 229, 349, -11)$, with mod-17 residues $(10, 8, 9, 6)$. If instead the relatively smaller spoke is at body 2 or 3, the corresponding triples of masses and mod-17 resultants are $(1, 5, 1) : (16, 4, 16, 15)$ and $(1, 3, 1) : (3, 13, 3, 6)$. All twelve are nonzero; the same argument covers zero, unit and pole orders of the smaller spoke without setting its scale to one.

All-equal spoke cubes are a separate branch; at the test fiber, each of its nine signed phase choices has neither a common finite wing-tie root nor a root at infinite free cube. On the distinct spoke torus, a free-cube root at infinity requires all three quadratic leading coefficients to vanish. The third leading coefficient contains the square of the original source slope, a unit when the geometric determinant and circle guards are units. After the zero-, equal- and infinite-spoke strata above, the only remaining guarded projective boundary on this ratio chart is the geometric determinant's zero set.

At the test fiber the residual geometric factor is the irreducible fourth-power-free polynomial $x^4 + 12x^3 + 2x + 6$ over $\mathbb{F}_{17}$. For one root of its four conjugates, the signed spoke-gap cubic meets the geometric determinant at exactly one guarded point; Frobenius covers the other three. There the three normalized relative spoke z values satisfy

$$P_1 = P_2 = P_3 \neq 0, \qquad P_i = \frac{1}{d_{i4}^2 W_i},$$

with every $W_i, W_i - W_j$ and d_{i4} a unit. For any mass-parameter Puiseux arc specializing to this point, write $D_{i4} = s d_{i4}$; the original signed ties give, for each wing edge,

$$\frac{D_{ij}^2}{D_{i4}^2} = \frac{P_i W_i}{(P_i - P_j)(W_i - W_j)}.$$

Both the common spoke scale s and a common wing W scale cancel in this ratio, even if either has nonzero order. We normalize the common wing W scale so that all W_i and $W_i - W_j$ are units. Since $P_i - P_j \to 0$, every wing inverse distance strictly dominates its incident spokes. We now distinguish the order of s. If $s \to 0$, all three spokes vanish. This is the all-spokes-vanishing case treated above: vanishing wing roots contradict an original w row, finite nonzero wing roots give one of the excluded three-body central configurations, and diverging wing roots give the excluded wing-dominant tier.

Suppose instead that s is a unit or has a pole. Every wing inverse distance then has a pole. In an endpoint's original w row, its wing forces strictly dominate both the spoke force and the unit coordinate term. An earliest wing force must therefore cancel another incident wing force. Applying this at the three vertices shows that all three wing inverse distances have the same order. Their signed ties give $Z_i - Z_j \to 0$, while the mass-weighted original wing z rows give

$$\sum_{i=1}^{3} m_i Z_i = m_4 \sum_{i=1}^{3} \frac{m_i D_{i4}}{W_i} = O(s).$$

Since $m_1 + m_2 + m_3 = 9$ is a unit at the test fiber, each $Z_i = O(s)$. The spoke z forces are also $O(s)$, whereas a wing z force has the order of D_{ij}, by $D_{ij}^3 (Z_i - Z_j) = D_{ij}/(W_i - W_j)$. Thus the wing forces strictly dominate the spoke and coordinate terms in both colours. Lemma 5.17 applies. Its two necessary polynomials $F(t)$ and $K(t)$ retain degrees six and four at $(1, 3, 5)$

modulo 17, with resultant $12 \neq 0$. Together with the all-spokes-vanishing case, this excludes the geometric boundary for every order of the common spoke scale.

Indeed, if the scheme-theoretic projective closure of the original guarded incidence over $\mathbb{Z}_{(17)}$ had a point over this mass, curve selection after a finite extension would give a valuative arc whose generic point satisfies the original rows and signed ties. A common z, w rotation makes one wing W_i a unit; the quartic Q and its five guard values keep all wing circles and collisions units. The spoke valuations now fall into the finite distinct-root chart, the equal-root chart, the vanishing-spoke charts, or a chart with one spoke relatively smaller than the other two. These have respectively been checked by the guarded affine norm and Δ analysis, the nine phase ties, the zero-spoke/three-body checks, and the twelve signed zero-force resultants above. The only free-cube infinity chart away from Δ was excluded by its unit leading coefficient. These checks concern arcs of the pre-specialization closure; no guard is discarded only after reduction modulo 17. Thus its entire special fiber over the positive-mass witness is empty, as required by Lemma 5.18. Consequently its mass image cannot be dense; its algebraic closure has a nonzero defining polynomial. The same assertion holds on the original positive mass cone after clearing mass denominators and completing the finite label orbit. □

Proposition 5.20 (No. 94 positive-mass first-limit locus)**.** *Every No. 94 positive-point first limit satisfies a nonzero necessary six-mass polynomial, after completing all signed branches and body-label orbits.*

Proof. Exactly two equal spoke cubes contradict the original body-4 w row and the nonzero wing circles. The all-equal-spoke case has a proper necessary wing-mass factor from its original wing force rows and the signed Gram phases. If $G = 0$, the fourth lower z row closes and Proposition 5.12's balanced-K_4 mass-image argument applies. For pairwise-distinct spokes with $G \neq 0$, the original rows give the generically finite guarded shape cover above; every degree-drop lower-mass fiber belongs to a proper discriminant locus. Lemma 5.19 supplies a proper lower-mass factor on the $H_3 = 0$ branch. On every remaining finite shape, $H_3 \neq 0$. With C_{ja} the signed leading inverse distance on a crossed edge, the necessary first six-body crossed torque, obtained without closing the fourth lower z row, is

$$T_{94} = \sum_{j=1}^{3} m_j(m_5 C_{j5} + m_6 C_{j6}) = 0, \qquad C_{ja}^2 = -\frac{1}{Z_a W_j} \quad (j \leq 3,\ a = 5, 6). \tag{112}$$

Equation (112) has a signed upper-mass-ratio norm of degree 12 with leading coefficient $H_3^6 \neq 0$ by (111). The product of its field norms across the finitely many signed K_4 shape components is therefore a nonzero six-mass polynomial. Multiplying it by the equal-spoke, balanced-K_4, bad-fiber and $H_3 = 0$ lower-mass conditions, and completing the finite body-label orbit, gives a proper necessary mass locus for every positive-point No. 94 first limit.

This is a necessary condition: certain partial K_4 shapes still unfold across open positive mass sets but do not construct six-body germs. The positive zero set of the new mass equation, the shared-body higher jets and finiteness on exceptional masses remain open. The raw affine resultants, the H_3 elimination and its 13-guard saturation are replayed at the displayed finite-field witnesses in Appendix A. The zero-, equal- and infinite-spoke and geometric-determinant boundary cases are handled by the explicit Puiseux chart arguments above; the accompanying scripts replay the listed algebraic boundary certificates but do not enumerate every formal arc. Thus the proposition has no remaining global-coverage or saturation hypothesis, while the analytic boundary case split remains part of the proof. □

5.24. **No. 98: two finite boundaries and a crossed norm.** The graph has an isolated upper z-triangle 456 and lower w-triangle 123, with every vertex circled in its own colour. The diagram rules used in the cited Chang–Chen classification [5] make all upper internal (z, w, D) orders $(-2, 2, 0)$, all lower orders $(2, -2, 0)$, and all nine crossed orders $(-2, -2, 2)$. Indeed Rule 1e

forces a maximal stroke in each isolated circled triangle, Rule 2e gives all three strokes the same order, and Estimate 2 gives the displayed internal orders. Every upper z-circle and lower w-circle has position order -2, while vertices in the other block are not circled in that colour. Both crossed separations thus have order -2 and the signed ties force D_{ab} to have order 2.

The original upper w and lower z rows then bound their opposite-colour positions by $O(\varepsilon^2)$: internal forces have order 2 and crossed forces have order 4. The original 12 coordinate rows and 15 signed ties consequently yield two closed, mass-centred three-body boundaries for every valuation of this graph, not just one uniform-rate example.

Proposition 5.21 (No. 98 six-mass locus)**.** *The diagram chart and original-row transition reductions are established above. The root isolations and exact outward-interval torque bounds are replayed in Appendix A. Every No. 98 positive-point first limit then satisfies a nonzero necessary polynomial in all six masses; the Lagrange×Lagrange branch is excluded for all positive masses.*

Proof. For a three-body block C, independent centred Z, W vectors force its mass-weighted internal Laplacian to be the identity on the full centred two-plane. Hence its three signed inverse-distance cubes all equal $1/M_C$, where $M_C = \sum_{i\in C} m_i$. The three cube phases have 27 choices and the ties yield at most two Lagrange shapes per choice. If Z, W are proportional, the original two force rows and three ties give eight signed Euler quintics, with at most five roots each. These two cases exhaust the finite shapes when all circled coefficients and internal separations are nonzero.

Let $A = 123$ and $B = 456$, with masses M_A, M_B and internal pair sums σ_A, σ_B. If both blocks are Lagrange, the signed internal inverse distances have modulus $M_C^{-1/3}$, so $|U_C| \le \sigma_C/M_C^{1/3}$. The inequalities $\sigma_C \le M_C^2/3$ and $(1+x)^{1/3} < 1 + x/3$ for $x > 0$ give

$$M^{1/3}|c| \le M^{1/3}(|U_A| + |U_B|) < \sigma_A + \sigma_B + \frac{2M_AM_B}{9} < \sigma_A + \sigma_B + M_AM_B = \sigma. \tag{113}$$

All crossed inverse distances tend to zero. At a positive point on the same curve Hölder gives $\sigma \le M^{1/3}c$, a contradiction since $c > 0$.

For an Euler-containing branch, the original upper torque modulo its nine signed crossed ties has earliest coefficient

$$T = \sum_{a=4}^{6}\sum_{b=1}^{3} m_a m_b C_{ab} = 0, \qquad C_{ab}^2 = -\frac{1}{Z_aW_b}. \tag{114}$$

No uniformity of the individual first crossed forces is assumed. Choose $\alpha_a^2 = -1/Z_a$ and $\beta_b^2 = 1/W_b$; all crossed roots are $C_{ab} = \eta_{ab}\alpha_a\beta_b$ with $\eta \in \{\pm 1\}^9$. For upper Euler, norm T over all 512 signs and every upper Euler and lower Euler/Lagrange algebraic shape root. Flipping any square root permutes the sign product; after clearing nonzero circle and leading-coefficient denominators, the finite norm is a six-mass polynomial P_{up}. The vanishing loci of any cleared coefficients or denominators are included by their nonzero mass norms, so degree drops do not omit positive-mass branches.

At the positive vector $(1, 2, 4, 30, 50, 330)$ the upper and lower Euler quintics have 40 isolated roots each; the lower Lagrange cases give 54 signed-tie roots. Exact rational complex intervals exclude zero from all $40(40 + 54)2^9 = 1{,}925{,}120$ first-torque labels. Lemma 5.15 and the exact Lagrange discriminants certify root exhaustion; outward rectangle products enclose each of the nine crossed terms before their signed sum is tested. For each label, the interval replay gives a certified lower bound for its ℓ^∞ distance from zero. The least such bound is exactly 411152670810834199/390625000000000.

Its floor at scale 10^{24} is 1052550837275735549440000000. The independent replay in Appendix A checks root completeness, outward error bounds and this strictly positive integer margin; thus $P_{\mathrm{up}} \not\equiv 0$. The exact involution exchanging z, w while sending $(1, 2, 3) \leftrightarrow (4, 5, 6)$ takes a lower-Euler branch to upper Euler and preserves every original row, tie, mass centre, U and

I. Multiply P_{up} by its block-permuted counterpart; the dual and upper-Euler cases are both covered. Together with the double-Lagrange exclusion this gives a proper necessary six-mass locus, not a finiteness theorem on its positive zero set. □

Table 3 maps each historical exclusion to its specific branch-completeness and original-row proofs. Only after those inputs are certified can one subtract all 17 from the historical 24. The 7 graphs in $\mathcal{D}_{\mathrm{new}}$ are *not* removed: they still can be first limits on their necessary proper mass zero sets. The four factors in (69) do not define an additional exceptional obligation because both No. 52 endpoints fail the positive-point first-limit test.

Proof of Theorem 1.1. Suppose that $\mathcal{A}_m$ has infinitely many positive points. Lemmas 2.2 and 2.1 give a curve through positive configurations on which $U = I = c > 0$. By Lemma 2.3, a branch at infinity has an unbounded seven-product. The cited Chang–Chen classification assigns it to one of the 117 classes and removes the 31 Algorithm II cases.

The remaining exclusions are as follows. Lemma 4.5 and Section 4.7 exclude No. 4 and 20; Lemma 4.2 excludes No. 2 and 22; Lemmas 4.3 and 4.4 exclude No. 3 and 78. Sections 4.10, 4.12 and 4.14 exclude No. 40, 44 and 45; Sections 4.20 and 4.21 exclude No. 86 and 95; Sections 4.25 and 4.24 exclude No. 99 and 100; Sections 4.27, 4.26 and 4.28 exclude No. 109, 112 and 111; Sections 4.9 and 4.11 exclude No. 51 and 55; Sections 4.6 and 4.22 exclude No. 105 and 77; Sections 4.15, 4.17 and 4.19 exclude No. 31, 32 and 36; Sections 4.13 and 4.16 exclude No. 49 and 50; Sections 4.8, 4.23 and 4.18 exclude No. 57, 61 and 84. Lemma 5.1 and its five diagram evaluations exclude No. 5, 67, 69, 73 and 80, while Sections 5.2 and 5.7 exclude No. 6 and 71. Lemma 5.2 excludes No. 74 and 75; Sections 5.13, 5.14 and 5.15 exclude No. 82, 89 and 90; Section 5.12 excludes No. 81; Sections 5.11 and 5.8 exclude No. 64 and 65; Section 5.16 excludes No. 79; and Proposition 5.8 excludes No. 52. Chang–Chen [6, Table 6.1] removes No. 116 for positive masses.

The published relation theorem supplies a proper mass locus for each of the remaining 33 classes. Propositions 5.10, 5.12, 5.13, 5.14, 5.16, 5.20 and 5.21 supply one for each of the seven derived classes, using the specified certificate inputs where indicated. Proposition 5.13 supplies its necessary polynomial by a direct analytic proof, including zero drift. In particular, Lemma 5.9 covers the two remaining No. 8 supports, and the complete potential and torque arguments above establish the additional local exclusions. Hence every possible first limit lies in one of the 40 relation classes and forces its orbit-completed polynomial $P_r(m)$ to vanish. By (4), $m \in E_{40}$. □

Proof of Corollary 1.2. Apply Theorem 1.1 to a mass outside E_{40}. □

6. Part II boundary

Establishing the necessary mass conditions for the 33 published and seven derived diagrams is an input to Part I. Part II concerns finiteness on their positive zero sets, including intersections and possible singular curves. A vanishing necessary polynomial does not prove a curve exists; conversely a pointwise test excluding one formal boundary label does not close an entire exceptional-mass stratum. No Part II theorem is asserted in this draft.

Appendix A. Verification functions

The following independent scripts are in `verification/independent_review/`. Their generated certificates and resource-limited replay logs are stored beside them; the scope of each replay is recorded in `appendix_replay_manifest.json`. A replay checks only the exact computations it lists and does not replace the analytic arguments in the body.

- Graph census:
  `../verify_n6_structure.py`
  `../generate_117_diagrams.py`. The second script redraws the 117 cached graphs in the authors' worksheet; it does not rerun Algorithm I.

- No. 20 (Section 4.7):
  `verify_no20_energy.py`. This exact replay checks the mass relation and simple shape root, the original-row offset and signed-phase algebra, the stated lower bound $x \geq 19/12$, and the final strict Hölder gap. The valuation estimates giving the $O(\varepsilon^3)$ first correction and second-order offsets remain analytic proof steps.
- No. 8 (Proposition 5.10):
  `generate_no8_revised_order_certificate.py`. The final two no-match supports are proved analytically in Lemma 5.9; no terminal-cut manifest is used for them.
- No. 52:
  `generate_no52_certificate.py`
  `verify_no52_valuation_partition.py`.
- No. 63 (Proposition 5.12):
  `generate_no63_certificate.py`.
- No. 66 (Proposition 5.13):
  `verify_no66_function_field_cover.py`.
- No. 79:
  `generate_no79_bridge_certificate.py`
  `verify_no79_shape_energy.py`
  `generate_no79_sturm_certificate.py`
  `verify_no79_hurwitz.py`.
- No. 88 (Proposition 5.14):
  `generate_no88_collinear_certificate.py`
  `generate_no88_fiber_certificate.py`
  `generate_no88_vertical_certificate.py`
  `generate_no88_smooth_certificate.py`
  `generate_no88_zero_star_certificate.py`
  `verify_no88_root_norm_identity.py`
  `verify_no88_CM_edge_quadratic.py`.
- No. 93/98 (Propositions 5.16 and 5.21):
  `generate_no93_interval_certificate.py`
  `generate_no98_interval_certificate.py`.
- No. 94 (Proposition 5.20):
  `generate_no94_structure_certificate.py`
  `generate_no94_projective_spoke_cover.py`
  `generate_no94_equal_spoke_certificate.py`
  `generate_no94_local_certificates.py`
  `derive_no94_finite_fiber.py`
  `verify_no94_distinct_spoke_fiber.py`
  `verify_no94_h3_fiber.py`. The affine resultant, elimination and guard-saturation computations are replayed exactly at the displayed finite-field witnesses. The projective Puiseux chart analysis is supplied in the proof; the scripts replay its stated algebraic boundary certificates but do not enumerate all formal arcs.

Acknowledgments

The present paper constitutes only part of our work, and the remainder will be presented in [3].

This manuscript was prepared with AI assistance and preliminary automated verification. Rigorous manual verification by the second author is currently underway and will be incorporated into a forthcoming version, alongside a complete formal proof in Lean.

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